International Labour Organization

The psychosocial working environment



Global developments and pathways for action

Global report

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Executive summary

Work-related psychosocial risks represent a major and growing threat to workers' safety and health, organizational productivity and broader economic performance.

- According to the latest estimates[1] released by the International Labour Organization (ILO) for the first time in this report, psychosocial risk factors are responsible for more than **840,000 deaths** annually due to associated cardiovascular diseases and mental disorders.[2]
- These risks also lead to nearly **45 million disability-adjusted life years (DALYs) lost** each year. The combined impact of cardiovascular disease and mental disorders associated with psychosocial risk factors is estimated to result in **1.37 per cent of global GDP lost annually**.
- In relation to this, it is important to recall that, **long working hours,** a critical psychosocial risk factor associated with increased risk of cardiovascular disease and stroke, remain widespread. The ILO estimates that globally, **35 per cent of workers work more than 48 hours per week** (ILO 2022b).
- Exposure to bullying and other forms of violence and harassment is another major concern. The ILO estimates that **23 per cent of workers globally** have experienced at least one form of violence or harassment in their working life, with psychological violence being the most prevalent at **18 per cent** (ILO and Lloyd's Register Foundation 2022).
- In response to persistent gaps in global data availability and concerns regarding the quality and comparability of occupational safety and health (OSH) statistics, the ILO launched a targeted questionnaire to national OSH statistics focal points in 2025. Results indicate that **37 per cent of responding institutions** (41 of 111) have concrete plans to **strengthen statistics** on psychosocial risks and mental health at work within the **next five years** (ILO 2025d).
- Integration into transnational social dialogue frameworks remains modest. Only **18 per cent** of the 338 **cross-border agreements** recorded between 2000 and 2025 in the ILO Cross-Border Social Dialogue (CBSD) Repository explicitly address **mental health or psychosocial factors within OSH-related provisions**.

1 These statistics are based on the prevalence and population attributable fraction (PAF) to five work-related psychosocial risk factors— job strain, effort–reward imbalance, job insecurity, long working hours (≥55 hours/week) and bullying — using a comparative risk assessment approach based on 2021 data on mortality and DALYs (Neupane, Takala, and Descatha 2026).

2 Cardiovascular diseases include stroke and ischemic heart diseases, while mental disorders include depression.

What is the psychosocial working environment?

The psychosocial working environment encompasses the elements of work and interactions at work related to how jobs are designed, how work is organized and managed, and the broader policies, practices and procedures that govern work, as well as the ways in which these elements interrelate, all of which can influence workers' health and well-being as well as organizational performance.

The report takes note of the changing world of work and its implications for the psychosocial work environment. Digitalization and the use of artificial intelligence (AI) are transforming how tasks are coordinated, monitored and assessed. New forms of employment, including platform work, varied contractual arrangements and the expansion of remote and hybrid work, are redefining supervision, expectations and working time. Meanwhile, broader external factors, including geopolitical uncertainty, are also reshaping work organization. While these developments may create opportunities to strengthen the psychosocial working environment, they may also exacerbate psychosocial risks, underscoring the need for proactive management.

To support preventive action, the report proposes a multi-level perspective focused on identifiable and modifiable features of the working environment. It emphasizes aspects that, so far as is reasonably practicable, can be managed to minimize risks and foster healthy and productive work. In doing so, three interrelated levels of the psychosocial working environment are identified:

The job

The inherent characteristics of tasks and responsibilities, including job demands, alignment with workers' skills, access to resources, and task design features such as meaning, variety and skill use.

How work is managed and organized

How work is structured and experienced in practice, including role clarity, autonomy, workload and pace, supervision, social support and the quality of workplace interactions.

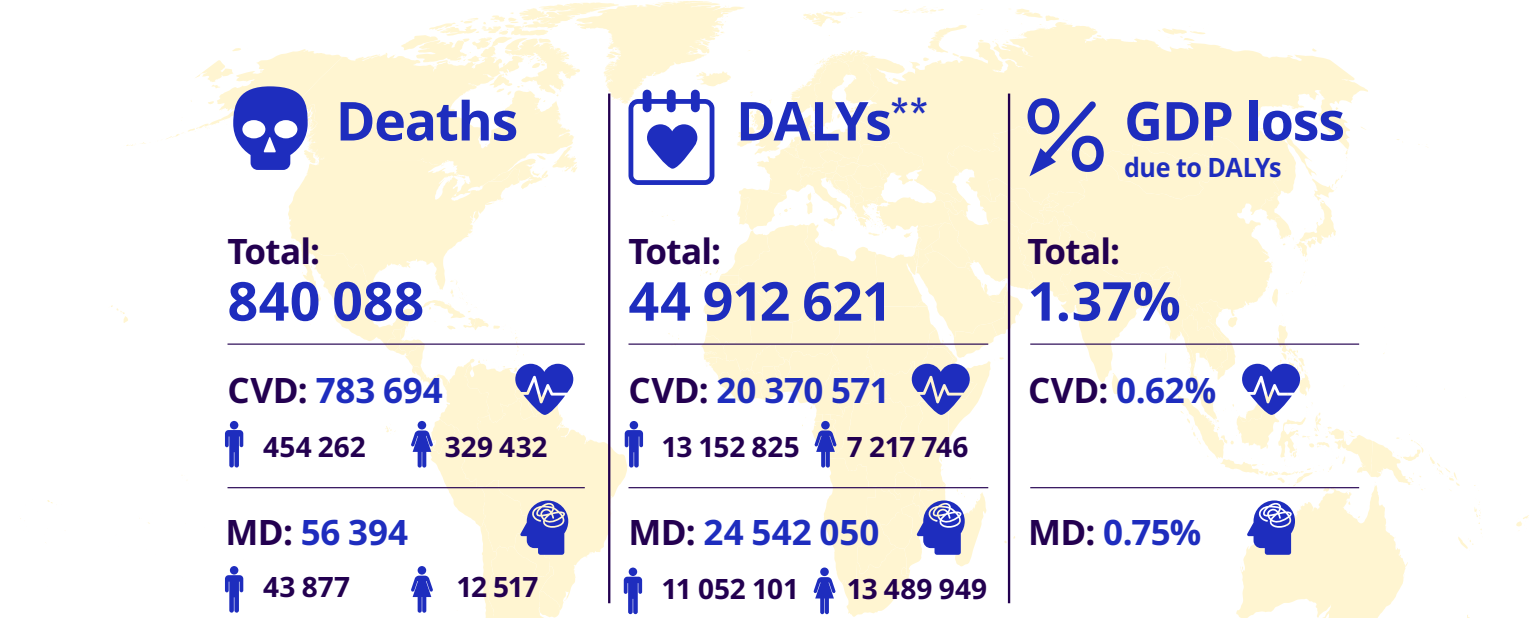
Broader policies, practices and procedures that govern work

The wider organizational and institutional systems that shape employment conditions, working time arrangements, organizational change, digital monitoring, performance and reward processes, OSH policy and management systems, procedures to prevent violence and harassment, and mechanisms for worker consultation and participation.

These levels encompass a range of psychosocial factors that reflect core features of the working environment across all forms of work and settings. How they operate and interact in specific contexts determines whether they promote health and effective performance or give rise to psychosocial hazards and adverse outcomes.

This understanding is supported by evidence spanning epidemiology, organizational psychology, labour economics and occupational health research. Although available data are subject to methodological limitations and uneven geographical coverage, they indicate that psychosocial risks are widespread.

Global Deaths, DALYs and GDP loss attributable to psychosocial risk factors*



* job strain, effort–reward imbalance, job insecurity, long working hours, and workplace bullying
** Disability-adjusted life years
CVD: Cardiovascular diseases
MD: Mental disorders

Certain work settings and diverse forms of employment, including the informal economy and self-employment, as well as specific sectoral characteristics, may increase exposure. Despite these limitations, research consistently shows that unfavourable psychosocial working environments are associated with adverse health outcomes and negative organizational impacts.

A compilation of regulatory frameworks and policies to address psychosocial risks

ILO OSH standards provide a strong normative foundation for addressing psychosocial risks and protecting workers' physical and mental health. While the fundamental OSH instruments do not explicitly refer to psychosocial risks, they provide the essential basis for comprehensive and preventive OSH policies and systems at national and enterprise levels. The Violence and Harassment Convention, 2019 (No. 190) is the first ILO instrument to explicitly refer to psychosocial risks through comprehensive framework of preventive and protective obligations. Other standards further reinforce the framework for psychosocial risk prevention and management.

The review undertaken in this report shows that **regional frameworks** increasingly incorporate psychosocial risks within broader OSH systems, although the extent and level of regulatory detail vary across regions. Some instruments explicitly identify risk factors such as violence and harassment, while others address psychosocial risks through broader references to mental health, work organization and working conditions. Across regions, these frameworks emphasize prevention and highlight organizational aspects of work as key determinants of psychosocial risk. Recent initiatives also reflect growing attention to emerging contexts, including artificial intelligence and the platform economy, alongside the development of guidance and practical tools.

The review of **national OSH policies and programmes** shows growing recognition of the prevention of psychosocial hazards as a national priority across regions. These policies promote measures to prevent and mitigate psychosocial risks, including through monitoring, training, and the use of validated tools for assessment and interventions. In many cases, they are aligned with broader national strategies promoting health and well-being, particularly those emphasizing mental health. In some countries, prevention efforts are strengthened through coordination between health and labour authorities, cross-sectoral approaches and social partners engagement.

National legislation suggests a gradual shift from framing employer duties primarily in terms of protecting "mental health", which may encourage individualized responses, towards preventive frameworks that explicitly recognize psychosocial risks. Many countries now define these risks and regulate them through provisions on work organization, risk assessment and evaluation of preventive measures, although the scope and specificity vary considerably. Monitoring of psychosocial risks is typically integrated into general OSH risk-assessment and prevention requirements and supported by grievance and enforcement mechanisms that enable preventive action. However, evidentiary requirements — particularly in establishing work-related causality — continue to influence how complaints are handled and how legislative duties are applied in practice. Where legislative frameworks are more developed, risk assessment is more clearly linked to intervention and corrective measures, although implementation remains uneven. Similar variation is observed in the recognition of work-related disorders, with post-traumatic stress disorder (PTSD) more widely recognized than other stress-related and mental health outcomes, which are often addressed through case-by-case procedures.

Collective bargaining and social dialogue play an important role in translating legal provisions into sector- and workplace-level measures. Analysis of cross-border agreements indicates that explicit attention to mental health and psychosocial factors within OSH provisions remains modest: only 18 per cent of the 338 agreements recorded between 2000 and 2025 in the ILO Cross-Border Social Dialogue (CBSD) Repository include such references. Engagement has nevertheless increased since the mid-2010s, with recent agreements more frequently addressing stress, mental health and well-being, harassment, work-life balance, digitalization and telework, while social partner engagement has supported the development of sectoral guidance tools and informed subsequent policy and regulatory developments.

Voluntary standards, guidance tools and awareness-raising campaigns are helping translate prevention frameworks into practical, action-oriented approaches. Across regions, labour inspectorates are adapting traditional inspection models to address psychosocial hazards by integrating compliance checks with preventive guidance, developing specialized tools, and adopting risk-based or sector-focused strategies. Awareness-raising campaigns have also become an important complementary instrument for strengthening psychosocial risk prevention by improving understanding of risks and encouraging preventive action at the workplace level.

Workplace level responses

The report draws on the *ILO Guidelines on Occupational Safety and Health Management Systems (ILO-OSH 2001)* as a flexible framework for managing psychosocial risks. Although not specific to these risks, the Guidelines provide a structured approach — covering policy, organization, planning and implementation, and evaluation — ensuring that the psychosocial working environment is treated as an integral component of effective OSH management rather than as a separate or parallel process.

The OSH policy sets out the organization's commitment to protecting workers' safety and health and should recognize that psychosocial risks arise from job design, work organization and broader organizational processes, aligning with relevant functions such as human resources and performance management. Effective organization requires clear responsibilities, competent leadership, cross-functional coordination and meaningful worker participation to ensure risks are identified, assessed and controlled at all levels. Planning and implementation translate this commitment into action through clear objectives, defined responsibilities and resource allocation based on a review of existing conditions.

Risk assessment is a core element of prevention within an OSH management system. It involves **identifying hazards, assessing associated risks**, and **implementing preventive or protective measures to eliminate or control them**. Identifying psychosocial hazards requires examining job design, work organization and management, and broader organizational processes shaping daily work. Consideration of interactions across these levels, together with power relations and discriminatory practices that may create or intensify psychosocial risks, is also required.

As psychosocial hazards are not always directly observable, assessment should draw on multiple sources of evidence, such as organizational data, human resources records and worker surveys. While surveys may contain subjective elements, the use of validated tools and safeguards for anonymity can improve reliability. Risk evaluation should consider the duration, frequency and cumulative nature of exposure, as many psychosocial risks develop gradually.

Prevention is the central objective. In line with the hierarchy of control, priority should be given to organizational and collective measures that address root causes, including workload management, role clarity, communication, participation and leadership practices. Immediate intervention may be required for acute hazards such as violence and harassment, while longer-term strategies may involve redesigning roles or revising systems. Individual measures that protect and promote health can support workers in coping with demands and accessing assistance but should complement, not replace, action on organizational conditions.

The report presents **examples of preventive measures** across the three interrelated levels of the psychosocial working environment, helping stakeholders identify priority areas for action and address hazards at their source. In practice, these measures should be considered collectively, as they often address multiple interacting psychosocial hazards simultaneously, reinforcing the need for a comprehensive approach.

Organizations should periodically review the implementation and effectiveness of psychosocial risk management using indicators and participatory feedback. This helps identify unintended consequences or emerging or persistent problems, and ensures continued relevance as work evolves.

The way forward

To sustain progress in psychosocial risk prevention and promote psychosocial working environments that support workers' safety and health and organizational performance, the report identifies three interrelated areas for further action.

- **Research** – More consistent data based on harmonized measurement tools and wider geographical coverage are needed to address gaps and strengthen understanding of prevalence and trends. Integrating psychosocial indicators into OSH monitoring systems and national statistics would improve tracking of exposures and outcomes. While several countries indicate plans to strengthen statistics on psychosocial risks and work-related mental health, further efforts are required to ensure routine, harmonized and internationally comparable data. Disaggregated data by worker characteristics and diverse forms of work, together with stronger evaluation of policies and workplace initiatives, would support the identification and wider adoption of effective approaches.

- **Policy and regulatory frameworks** – Greater coherence, clarity and consistency across policy and regulatory approaches are needed, as practices vary in scope, terminology and level of protection. To strengthen prevention, OSH policies and laws should explicitly address psychosocial risks, avoiding a primary focus on outcomes such as mental health at work. Alongside continued attention to acute and visible issues, including violence and harassment, greater emphasis is needed on structural and organizational aspects of work. Effective implementation requires a combination of enforcement, practical guidance, tools and capacity building, particularly for small and medium-sized enterprises, as well as clearer communication and stronger institutional coordination. Enhanced cooperation among OSH authorities, public health institutions and social partners can further support prevention and knowledge-sharing.

- **Workplace-level action** – Although not all psychosocial hazards can be fully eliminated, the associated risks can often be reduced by improving work design, organization and management. This includes reviewing workload, task allocation, supervision and other elements of the working environment, while strengthening workers' capacity to recognize and respond to risks. Leadership responsibility is central to embedding prevention in everyday management decisions and across organizational functions. Participatory approaches involving workers and their representatives, together with cooperation between OSH professionals, human resources and management, help ensure that measures are practical, context-specific and sustainable.



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Introduction

Work[3] plays a central role in people's lives. For most workers, it occupies a substantial share of waking hours, and it shapes identity, social connection and economic security (Fryers 2006). When work is well designed and well managed, it provides structure and purpose, supports financial stability, enhances health and well-being, and contributes to organizational performance and productivity. Whether work produces these positive outcomes depends largely on the psychosocial working environment, understood as the aspects of work and interactions related to how jobs are designed, how work is organized and managed, and the broader policies, practices and procedures that govern work, and the ways in which these elements interrelate.

Across the world, the psychosocial working environment is undergoing a profound transformation. Rapid technological change, including digitalisation and the introduction of AI-supported tools, has altered how work is coordinated, monitored and evaluated (Schulte et al. 2020). New forms of work, such as platform-based and other diverse forms of work, are reshaping traditional employment relationships, redefining roles, schedules and expectations (Countouris and De Stefano 2019). The expansion of remote and hybrid work following the COVID-19 pandemic has further changed where and how work is performed and supervised (WHO and ILO 2022). At the same time, supply chain disruptions, demographic shifts, climate-related pressures and geopolitical instability contribute to growing uncertainty in work organization (ILO 2025b). As work continues to evolve, workers may face complex combinations of factors affecting work content, working conditions and work organization.

The relevance of the psychosocial working environment extends far beyond office-based work or standard employment relationships. Many workers operate in the informal economy, in micro, small and medium-sized enterprises (MSMEs), or in self-employment, where exposures and protections differ substantially. In some of these contexts, workers face multiple unfavourable conditions, limited social protection and weak enforcement of labour standards, and overlapping physical and organisational constraints (ADP Research 2025). Sector-specific patterns are also significant. Public service workers, health and care professionals, transport and retail workers, and those in education and emergency services often navigate high workloads, time pressure and emotionally demanding situations (Eurofound and ILO 2019). Recognizing the particularities of different sectors and forms of work is therefore essential for understanding how psychosocial factors are experienced and how they can be improved in diverse settings.

Over the past decade, a growing body of scientific evidence spanning epidemiology, organisational psychology, labour economics and occupational health has deepened understanding of how the psychosocial working environment influence

3 The ILO understands "work" broadly as encompassing the activities, tasks and social relationships through which labour is organised and performed. This relates to all forms of work, including paid and unpaid work, own-use production work, and other diverse forms of work within economic and organisational arrangements (ILO 2018).

workers' health, well-being and work-related outcomes. This expanding evidence base provides a stronger empirical foundation for assessing and addressing aspects of the psychosocial working environment and for informing policy and preventive action. It also reinforces the understanding that these aspects are shaped by the way work is designed and organized, and are therefore modifiable, allowing them to be managed in ways that promote positive outcomes, such as work engagement, and prevent harm.

In light of these developments, this report adopts an approach that focuses on modifiable aspects of the psychosocial working environment. It supports stakeholders in identifying relevant aspects of work and workplace organization, and in understanding how they interact, in order to inform targeted preventive responses. The proposed approach is flexible in its application and can be adapted to enterprises of all sizes and forms of work, to the extent reasonably practicable. The report brings together international, regional and national policy and regulatory frameworks, as well as workplace-level preventive approaches to improve the psychosocial working environment.

Part 1 introduces the psychosocial working environment, proposing a multi-level approach to better understand how work-related psychosocial hazards arise across different elements of work.

Part 2 presents new ILO global estimates on mortality, disability-adjusted life years (DALYs) and GDP losses associated with major psychosocial risk factors, and synthesizes existing evidence on their health, organizational and economic impacts, the benefits of healthy psychosocial working environments, and patterns of exposure.

Part 3 reviews international, regional and national policy and regulatory approaches that address the psychosocial working environment, highlighting recent developments in standards, legislation and institutional practices.

Part 4 turns to workplace-level action, outlining how developing a healthy psychosocial working environment can be integrated into occupational safety and health (OSH) management systems and discussing approaches for assessment, prevention and improvement.

The **concluding section** identifies key challenges, opportunities and priorities for the way forward.

The **annexes** present selected resources, tools and guidance to support practical action on psychosocial risks across different work contexts.


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Part 1

The psychosocial working environment

The **psychosocial working environment** encompasses the elements of work and interactions at work related to how jobs are designed, how work is organized and managed, and the broader policies, practices and procedures that govern work, as well as the ways in which these elements interrelate, all of which can influence workers' health and well-being as well as organizational performance. When this environment is well structured, supportive and inclusive, it can promote motivation, engagement, productivity, job satisfaction, and overall well-being. When it is poorly designed, organized or managed, psychosocial hazards may emerge and give rise to psychosocial risks. If these risks are not effectively managed, they can have serious consequences for workers' physical and mental health and well-being, as well as for organizational outcomes, including increased absenteeism, reduced performance and higher turnover (Cox and Griffiths 2010).

Work-related psychosocial factors,[4] hereafter referred to as psychosocial factors, are aspects of the psychosocial working environment that primarily relate to how work is designed, organized and managed, and to the interactions between the work environment, job content and organizational conditions that shape how work is carried out. This includes, for example, the design and content of tasks (such as their variety, meaning, scope and repetitiveness), workload and work pace, working time arrangements, role clarity, autonomy, supervision and management practices, and organizational culture (ILO 1986; ILO 2012).[5]

4 Throughout the report, the terms work-related psychosocial factors and psychosocial factors are used interchangeably.

5 In 1984, the Joint ILO/WHO Committee on Occupational Health defined "psychosocial factors at work" as: "Interactions between and among work environment, job content, organisational conditions and workers' capacities, needs, culture, personal extra-job considerations that may, through perceptions and experience, influence health, work performance and job satisfaction" (ILO 1986). This definition illustrates the complex nature of psychosocial factors at work, encompassing aspects related to the job and the work environment as well as interactions with individual workers' abilities, needs and perceptions.

In themselves, these factors are not intrinsically positive or negative; their effects depend on how they combine and are managed within specific work settings (Leka and Jain 2024). Although their effects are experienced through the interaction between the working environment and individual characteristics — including workers' abilities, needs and perceptions — the organization and management of work play a central role in shaping their nature and impact (Cox and Griffiths 2010; Siegrist 1996). For this reason, this report places particular emphasis on identifiable work-related psychosocial factors that fall within the control, responsibility and/or influence of employers (Leka, Jain, and Lerouge 2017; Lerouge 2025).[6]

For the purpose of this report, work-related **psychosocial hazards**, hereafter referred to as psychosocial hazards, are understood as harmful aspects of work that arise from elements in the design, organization and management of work, as well as the wider organizational context, and that have the potential to harm workers' safety and health (Cobb 2022). Such hazards may be present across all sectors and forms of work and can arise from job tasks, work organization, management practices and employment arrangements (Cox et al. 2000; ILO 2016a). Furthermore, these hazards often occur together and interact with one another (Cox 2000; LaMontagne et al. 2014).

Exposure to psychosocial hazards may lead to **psychosocial risks**, which refer to the combination of the likelihood of exposure and the severity of potential injury or ill-health (ILO 2016b). Assessing these risks requires consideration of the severity, duration, and frequency of exposure, as harm may arise not only from acute events but also gradually through repeated or prolonged exposure (Pavlo Saik et al. 2024).

Over the past decades, several theoretical models have been developed to explain how the psychosocial working environment influences workers' safety and health, as well as organizational outcomes (Demerouti et al. 2001; Dollard and Bakker 2010; Karasek and Theorell 1990; Siegrist 1996; Warr 1987). These models reflect an evolution in thinking, from early approaches focusing on the fit of the individual to the working environment to later perspectives that emphasize work design, job demands and resources, fairness, and organizational context.

Many of these models have originate from efforts to understand work-related stress and its precursors. Work-related stress is often understood as an intermediate response to psychosocial hazards and an important pathway through which these hazards affect workers' safety and health outcomes (Dollard, Dormann, and Idris 2019). It arises when perceived demands exceed an individual's capacity to cope (WHO 2020). While not a disorder, prolonged or intense stress may hinder recovery and contribute to longer-term physiological, psychological, and behavioural consequences (Alalhareth et al. 2024; Maulik 2017).[7]

Evolution of key theoretical models of the psychosocial work environment

Theoretical models offer complementary perspectives on how characteristics of work function as demands, resources or stressors, and how the interaction of these characteristics shapes both individual and organizational outcomes.

Early approaches emphasize the fit between the individual and the work environment:

- The **person-environment fit theory** proposes that stress arises when there is a mismatch between workers' abilities and job demands, or when personal needs are not met by the work environment (Edwards, Caplan, and Harrison 2025; French and Caplan 1972). Similarly, Warr's **Vitamin Model** suggests that some features of work are beneficial up to a point, whilst others can be harmful when they are either lacking or excessive (Warr 1987).

Subsequent models increasingly focus on the structural characteristics of work:

- The **job demand-control model,** and its expanded version that incorporates social support, propose that job strain (an outcome of exposure to a combination of high demands and low control at work) is highest when psychological demands are high and workers' control over

6 In practice, laws and regulations, together with standards and guidance, typically address work-related psychosocial factors through psychosocial risk assessment and prevention duties that fall within employers' OSH responsibilities.

7 This mechanism is further discussed in Part 2.1.

their work is low, while social support can act as a buffer (Johnson and Hall 1988; Karasek 1979; Karasek and Theorell 1990). The more recent **job demands-resources model** groups work characteristics into job demands, which require sustained effort and carry psychological or physiological costs, and job resources, which help workers achieve their goals, support learning and development, and reduce the impact of demands. It explains how excessive demands can lead to burnout through a health impairment process, while the availability of resources promotes work engagement through a motivational process (Demerouti et al. 2001).

Other models emphasize that not all demands operate in the same way:

- The **challenge-hindrance stressor framework** differentiates demands experienced as opportunities for learning and achievement (challenges), which can stimulate effort and motivation, versus demands experienced as unnecessary obstacles (hindrances), which are associated with withdrawal and reduced engagement (Cavanaugh et al. 2000; LePine et al. 2005). In parallel, job design perspectives, including the **job characteristics model**, highlight how features such as skill variety, task identity and significance, autonomy and feedback shape workers' psychological states and motivation (Hackman 1980).

Other perspectives take a more sociological view, emphasizing reciprocity, fairness and power relations:

- The **effort-reward imbalance model** conceptualizes stress as arising from a poor exchange between the efforts workers invest and the rewards they receive, such as pay, esteem, job security and career prospects (Siegrist 1996). **Organizational justice approaches** similarly underline that workers' well-being and behaviour depend on perceived fairness in resource distribution, decision-making and interpersonal treatment (Elovainio, Kivimäki, and Vahtera 2002; Greenberg 1987; Peiró and Rodríguez 2008). These perspectives also draw attention to power asymmetries in the organization of work, as employers' authority shapes job demands, autonomy, access to resources and social support (Jespersen, Hasle, and Nielsen 2016).

More recently, attention has increasingly turned to organizational-level determinants of psychosocial risks:

- **Psychosocial safety climate theory** focuses on organizational values, policies, practices and priorities for protecting workers' psychological health, including management commitment, communication and participation, and the balance between productivity and psychological health concerns (Dollard and Bakker 2010).

Understanding how these models conceptualize the psychosocial working environment helps clarify what healthy workplaces look like. A positive psychosocial working environment combines manageable demands with sufficient autonomy, support and resources, while providing opportunities to learn and use skills that promote workers' safety, health and performance. By contrast, a poorly designed working environment – for example, one characterised by excessive demands and limited control or resources – increases the likelihood of adverse outcomes for both workers and organisations (Parker and Jorritsma 2020).

These theoretical models have been operationalized through assessment instruments that support systematic psychosocial risk management. Several frameworks have sought to harmonize key elements for prevention and intervention. Although psychosocial hazards can be conceptualized in different ways depending on purpose and policy context, the taxonomy developed by Cox (1993) and Cox and Cox (1993),[8] later refined to include macro-level factors (Leka and Jain 2024),[9] remains one of the most widely applied.

8 This taxonomy provided one of the earliest frameworks for identifying psychosocial hazards. It outlined six main domains of the psychosocial work environment: job content, workload and work pace, work schedule, control, work environment and equipment, and organizational culture and function.

9 The taxonomy was subsequently refined in Cox (2000) and expanded through the PRIMA-EF framework (Leka and Cox 2008; Leka et al. 2011) into a ten-domain model later reflected in EU-OSHA and WHO guidance. Later work (Leka, Jain, and Lerouge 2017) further elaborated features of a healthy psychosocial work environment, including related constructs such as Psychosocial Safety Climate.

1.1 A multi-level perspective on the psychosocial working environment

Building on this taxonomy, the report adopts a multi-level perspective that prioritizes modifiable features of the psychosocial working environment, from how work is designed, managed and organized to broader workplace policies and practices. This approach moves beyond individual-centred perspectives that focus primarily on workers' perceptions or personal characteristics and emphasize adaptation to existing working conditions. Instead, it highlights elements of work organization and management that fall within employers' sphere of influence. The perspective therefore supports a proactive and systemic approach to the design and management of work, focusing on psychosocial risks that, so far as is reasonably practicable, can be managed within the workplace (Wiegand et al. 2012).

Within this perspective, psychosocial factors are grouped across three broad and interrelated levels. While these levels overlap and interact, and should not be understood as independent silos, this structure helps clarify where preventive and corrective actions can be targeted to promote positive change.[10]

1. **The job** – characteristics related to the nature of the job itself, including job demands and the design and content of tasks;
2. **How work is managed and organized** – how work is planned, supervised and supported, including the quality of interactions at work;
3. **Broader policies, practices and procedures that govern work**– the wider workplace policies, systems and arrangements that shape the working environment.

10 There are multiple ways of organizing or conceptualizing psychosocial hazards; the grouping applied in this report represents one such approach.

An indicative set of examples of some psychosocial factors across the three levels

The Job

Job demands
- Cognitive, emotional and physical demands
- Quantitative and qualitative demands

Job resources and skill alignment

Level of responsibility

Task design
- Use of skills, range and variety of tasks, and contribution to work processes

How work is managed and organized

Role clarity and consistency in expectations

Predictability of tasks
- Stability of work processes
- Transparency and trustworthiness of interactions

Job control or autonomy

Assigned workload and required work pace

Supervision
- Provision of job resources and opportunities for development

Support at work from supervisors or co-workers

Broader policies, practices and procedures that govern work

Employment arrangements

Working time arrangements (including flexible work)

Management of organizational change and restructuring

Surveillance and digital monitoring practices

Rewards, performance management, opportunities for development and recruitment processes

An OSH policy and management system

Policies and procedures related to workplace violence and harassment

Consultation and worker participation mechanisms

The job

This level concerns the inherent characteristics of the job and its tasks – including the demands they impose, the level of responsibility involved, how tasks are designed, and the resources and skills required to perform them. These intrinsic features shape potential exposure to psychosocial hazards through the nature, complexity and intensity of work activities. Such elements are present across all forms of work, including formal and informal employment, and apply to diverse working arrangements, including self-employment, apprenticeships, unpaid domestic and care work, and other forms of work.

Every job imposes various types of **demands** that require sustained effort to be performed effectively (Demerouti et al. 2001). These job demands can be considered in both quantitative and qualitative terms (Kop et al. 2016). At the level of the job itself, quantitative demands refer to the inherent amount of work (i.e. workload), while qualitative demands concern the complexity and difficulty of the work. However, how workers experience workload also depends on how it is distributed and assigned in practice, an issue addressed further at the level of work management and organization.

Demands can be cognitive, emotional and physical. Cognitive demands involve the need for intellectual effort, including processing information, sustaining concentration, and using memory, reasoning and problem-solving to complete work tasks (INSHT 2012). Emotional demands require workers to regulate their feelings in response to emotionally demanding situations at work, [11] or adhere to organizationally expected emotions, such as showing empathy, understanding or compassion during interpersonal interactions (Reh and Scheibe 2025). Cognitive and emotional demands can interact with physical demands, which refer to the degree of physical effort required in the job, such as standing, walking, carrying or lifting heavy loads, among other tasks (Tanjung et al. 2025).

In order to meet job demands, workers need access to adequate resources that enable effective functioning. These encompass a broad range of positively valued organizational, social and physical elements, such as access to information, time and support, as well as appropriate physical resources, including ergonomic tools (Leka and Jain 2024). Such resources may be embedded in the design of the working environment or made available through supervisory practices and support from co-workers, as discussed further at the next level. Demands should also be aligned with workers' skills and experience,[12] so that they have the appropriate knowledge and capability to perform work competently (Kalleberg 2008; Korunka 2017).

In addition to demands, jobs entail varying **levels of responsibility**. Responsibility may relate to specific work outputs, to other people (including their safety and health[13]), or to equipment and material assets (ILO 1986; Kop et al. 2016). When responsibility is high, workers are accountable for the outcomes of their work and their decisions may carry significant consequences, including financial or material losses, or impacts on the safety and well-being of colleagues, customers or others (Schmitt, Den Hartog, and Belschak 2015). While high responsibility can enhance motivation by fostering a sense of ownership and impact, the possibility that mistakes may result in serious consequences can also be a source of strain (Hackman and Oldham 1976; Schmitt, Den Hartog, and Belschak 2015).

Task design is a central dimension of the job. It encompasses the way tasks and work processes are structured, including their range, variety and the extent to which they make use of different skills (Burr et al. 2019; Morgeson and Humphrey 2006). Jobs characterized by a broader range of tasks and effective use of workers' skills tend to be more engaging and supportive of meaningful work (Albrecht, Green, and Marty 2021). When tasks are structured in ways that allow workers to understand the purpose and impact of their work, workers are more likely to experience their work as meaningful and motivating (Allan 2017; Hackman and Oldham 1976; Morgeson and Humphrey 2006).

11 This may also involve situations where workers may witness, investigate, or be exposed to traumatic events, particularly in occupations such as and social care workers, emergency service workers, military and security personnel (Billings et al. 2023).

12 Required skills vary across occupations and relate both to skill level, which reflects the range and complexity of tasks to be performed, and to skill specialisation, which refers to the specific field of knowledge, tools and equipment used, materials worked with, and goods or services produced (ILO 2023b).

13 This may include threat-avoidant vigilant work, where individuals must remain constantly alert to prevent catastrophic outcomes, including potential loss of life (e.g. bus, taxi and truck drivers, air traffic controllers, and sea pilots) (Landsbergis et al. 2001).

Examples of psychosocial hazards related to the nature of the job

Job demands

- Sustained levels of concentration or vigilance, particularly when accuracy is required or workers are looking for infrequent events (e.g. long-distance driving or security monitoring)
- The need to show empathy, compassion, respectful service and politeness even when customers are being abusive
- Exposure to events or situations that can cause trauma
- Physically demanding, challenging or tiring work (e.g. undertaking hazardous manual tasks or strenuous physical tasks)
- Misalignment between job demands and available resources, or failure to account for individual workers' skills or capacities

Level of responsibility

- Responsibility for high-risk work where errors may have serious reputational, legal, safety, or financial consequences (e.g. air traffic control, medical care, or decisions affecting many others)

Task design

- Limited task variety, including highly repetitive or monotonous tasks
- Fragmented or meaningless work tasks

How work is managed and organized

This level concerns how work is structured, coordinated and supervised in practice. It includes the definition of roles and expectations; the allocation of tasks and workload, the degree of autonomy afforded to workers; the provision of supervision and feedback and social support, and the quality of interactions that underpin these processes. These elements shape how the inherent characteristics of the job are experienced and can either amplify or mitigate exposure to psychosocial hazards. Although they vary across sectors and settings, they are fundamental features of work organization across occupations, sectors, workplaces and diverse forms of employment.

Within the organization of work, **roles** structure how tasks, expectations and authority are defined and carried out in practice. The way roles are communicated and managed influences workers' understanding of what is expected of them, the objectives to be achieved and how their work relates to others. Role clarity refers to the extent to which these expectations and responsibilities are clearly articulated (Burr et al. 2019).[14] Role conflict may arise where expectations are incompatible or poorly coordinated. Such conflicts can stem from contradictory instructions; competing demands from multiple actors (e.g. supervisors, colleagues or clients); inconsistencies between assigned tasks and formal job descriptions; or tensions between organizational expectations and personal values (Karkkola, Kuittinen, and Hintsa 2019).

Role clarity and role conflict are closely linked to predictability at work, which describes workers' ability to anticipate tasks, processes and expectations in their daily activities (Schoellbauer et al. 2021). Predictability depends not only on stable work processes but also on timely information and consultation regarding changes and

14 A concept related to role clarity is role ambiguity. It refers to uncertainty about the actions and behaviours required to fulfil the expectations associated with a given work role (Beehr 1995).

decisions (Lindstrom et al. 2000). The quality of workplace interactions therefore plays a central role, as transparent and trustworthy exchanges facilitate the timely communication of reliable information (Burr et al. 2019).

Job control or **autonomy** concerns the degree of discretion afforded to workers in the organization of their work. It includes the freedom to schedule tasks, make decisions and choose the methods for performing them, as well as the ability to exercise initiative and judgement in daily activities (Burr et al. 2019; Hackman and Oldham 1976; Kubicek, Paškvan, and Bunner 2017; Morgeson and Humphrey 2006). As a feature shaped by managerial practices, autonomy reflects how authority and decision-making are delegated in practice (Leach, Wall, and Jackson 2003; Nie et al. 2023). However, the degree of autonomy afforded may not always be neutral, as discriminatory practices can limit decision-making freedom for certain groups of workers, even when they possess the relevant skills (Lissitsa and Chachashvili-Bolotin 2020; Meyer 2013). The degree of autonomy available also depends on how work processes are structured, including task interdependence, externally determined pace, interruptions, or requirements for close coordination (Kop et al. 2016). This highlights that autonomy is not exercised in isolation but is often shaped by how workers interact and collaborate within shared work processes.

Workload refers to both the amount and the difficulty of tasks and is largely determined by managerial decisions about how work is allocated (Bowling and Kirkendall 2012). Depending on the time available to complete assigned tasks, workers may experience quantitative overload when they are required to do more than can reasonably be completed, or quantitative underload when they are given considerably less work than they are capable of completing. Qualitative overload arises when tasks exceed workers' skills or capabilities, whereas qualitative underload occurs when tasks fall well below workers' abilities (Sales 1970; Shaw and Weekley 1985). Workload allocations may also reflect discriminatory assumptions or stereotypes, leading some workers to be disproportionately assigned routine or lower-status tasks which impact career progression and retention (Misra et al. 2021; Tiwari, Mathur, and Awasthi 2018).

The **pace of work** refers to the speed at which tasks must be performed (Burr et al. 2019) and is closely linked to workload. It may be influenced by performance targets, the speed of automated machines or systems, as well as direct demands from supervisors, colleagues or clients (Eurofound and ILO 2019). When a high workload must be completed within a limited time, the required pace of work increases and is often experienced as time pressure.[15] In this context, work intensity describes the overall level of required work demands or the mismatch between work quantity, time and task complexity (Hünefeld et al. 2025).

Supervision concerns the coordination, guidance and oversight of work activities (Heery and Noon 2008). Through planning, task allocation, feedback and day-to-day communication, supervisors shape how job demands are experienced and how support is provided, thereby influencing workers' capacity to perform effectively and maintain well-being (Dollard and Bakker 2010; Jimmieson et al. 2021; Gilbreath 2005). In smaller enterprises, these functions are often combined in owner-managers and exercised through more informal arrangements (ILO 2019). In self-employment, supervision is typically internalized and shaped by client expectations or contractual obligations, while in platform-based work coordination and monitoring may be embedded in contractual or digital management systems rather than traditional supervisory roles (Baiocco et al. 2022; Eurofound 2017a).

The way supervisors carry out these functions is closely linked to their leadership style. Supportive and participative leadership approaches are associated with clearer expectations, constructive feedback and greater access to resources, all of which can buffer the impact of high job demands (Tummers and Bakker 2021). In contrast, authoritarian or disengaged approaches may contribute to role ambiguity, excessive monitoring or limited support, which can increase the difficulty of achieving work goals (Al-Malki and Juan 2018).

Support from supervisors constitutes a key organizational resource. This may include practical guidance and problem-solving assistance and responsiveness to emerging difficulties, including recognizing concerns early and facilitating reasonable adjustments where needed (WHO 2022b).[16] Effective supervision also supports ongoing competence and development by providing recognition, constructive feedback, identifying skill gaps and enabling learning opportunities (Casalini 2023; Williams et al. 2014). The manner

15 Time pressure has been defined as "the extent to which employees feel they have insufficient time to finish their work tasks" (Ohly and Fritz 2009, 544).

16 Further information on manager training, identified as a key organizational intervention, can be found in the *WHO guidelines on mental health at work* (2022).

in which supervisory authority is exercised further shapes whether power is applied fairly and inclusively, influencing access to opportunities and evaluation processes (Adams, Meyers, and Sekaja 2020).

Support from co-workers complements supervisory support. Co-worker support, expressed through collaboration, mutual assistance and respect, provides an additional resource that can mitigate work-related stress (Arora and Kamalanabhan 2013; Attiq et al. 2017). Beyond their supportive function, everyday interactions between colleagues and supervisors shape how formal structures of work organization are enacted and experienced in practice.

Examples of psychosocial hazards related to how work is managed and organized

Roles and expectations

- Scenarios where workers do not have clear guidelines on the tasks they are expected to do (and not do)
- Expectations within a role that undermine one another (e.g. being expected to provide good customer service, but also to not spend a long time with customers)
- Conflicting, uncertain, or frequently changing expectations and work standards (e.g. workers are given conflicting deadlines or instructions)
- Insufficient, unclear or contradictory information (e.g. necessary information to complete tasks is not passed on)

Job control or autonomy

- Prescriptive processes and not allowing workers to apply their skills or judgment (e.g. work is tightly scripted, and workers cannot adapt to the specific situation)
- Low levels of influence and independence (e.g. inability to influence the speed, order or schedule of tasks or workload, sometimes resulting in long idle periods where workers cannot perform other tasks)
- Discriminatory work organization practices that result in unequal freedom in decision-making or unequal opportunities to apply new competencies, skills and knowledge

Workload and work pace

- Distribution of workload leading to overload (e.g. having too much to do) or underload (e.g. running out of work), including where these arise from biased or discriminatory allocation
- Time pressures or fast paced work (e.g. unreasonable deadlines or unrealistic pace of work)

Supervision and support

- Lack of support, resources, information or training to facilitate performance
- Limited support or unempathetic leadership (e.g. supervisors do not notice when workers are struggling, do not take issues seriously or provide a safe space to raise issues)
- Receiving insufficient feedback or recognition (e.g. workers do not receive feedback on their work or are not given information to help them improve; workers are not acknowledged or rewarded for high effort or supporting colleagues), which may also be influenced by discriminatory practices
- Workers not being able to ask for help when needed, including in uncooperative or uncollaborative workplaces that discourage co-worker support (e.g. competitive environment)

Broader policies, practices and procedures that govern work

Organizational policies and practices shape the framework within which work is structured and managed. They define employment arrangements, working time systems, performance and reward processes, and the procedures in place to prevent and address unacceptable behaviour. These governance arrangements operate across different enterprise sizes and forms of work, although their degree of formalization may vary (ILO 2019; ILO 2023a).

Employment arrangements are defined by the contractual terms governing the duration of employment, the stability of working hours, wages and employment rights, and access to social protection (ILO 2018). Although labour legislation establishes standards affecting working conditions,[17] such as minimum wages and maximum working hours and equal opportunities and treatment in employment, how these protections operate in practice depends on whether workers are effectively covered through their employment arrangements and workplace practices.

Different forms of work, including full-time and part-time contracts, temporary and agency work, dependent self-employment and self-employment, as well as work in the informal economy, are associated with differences in working conditions and levels of protection (ILO 2018). These differences are reflected in effective coverage by statutory social insurance; access to paid leave and other employment benefits; employment income relative to defined benchmarks (such as minimum wages); the type and duration of employment agreements; actual hours worked and exposure to time-related underemployment; and affiliation to trade unions or coverage by collective agreements (ILO 2023a). Together, these factors illustrate how employment arrangements shape decent work outcomes and overall quality of employment.

For example, both wage levels and the duration of employment influence the adequacy and predictability of earnings. Contracts of shorter or uncertain duration may reduce income stability and limit effective access to non-monetary benefits, including entitlement to and use of paid sick leave and paid annual leave (Benach et al. 2014). These differences are particularly relevant for workers with care responsibilities, as eligibility for and take-up of leave and care support often vary by employment status and workplace arrangements (Dobrotić and Blum 2019). Comprehensive maternity protection[18] and related leave policies, including paternity, parental and long-term care leave, are essential to enabling continued participation in employment while preventing discrimination linked to pregnancy or caregiving responsibilities (ILO 2022c; ILO 2024a). Care-related provisions and services, such as childcare and breastfeeding breaks, further support the work-home interface (ILO 2022c). Effective access to these measures depends not only on social protection systems but also on workplace accommodations that allow workers to exercise their rights in practice.

Working time arrangements, which define the number, timing and scheduling (stability or flexibility) of work hours, are another important aspect of the quality of working conditions (ILO 2008). Standard schedules typically involve regular daytime hours on weekdays, whereas flexible arrangements may include shift work, night and weekend work and rotating schedules (Soltanzadeh et al. 2024). While flexibility may create opportunities, such arrangements can also result in unsocial hours[19] when work takes place during socially valued times and rest occurs during less valued periods (Arlinghaus and Nachreiner 2016). Unstable or unpredictable patterns in working hours can be more prevalent in certain forms of work, including casual, platform and gig work, as well as contracts with very short or unpredictable hours, such

17 Working conditions is used here as an umbrella term covering multiple aspects of employment quality, aligned with the ILO's Decent Work Agenda (employment creation, rights at work, social protection and social dialogue). The ILO Decent Work Indicators framework comprises 75 statistical and 21 legal framework indicators structured around ten elements: employment opportunities; adequate earnings and productive work; decent working time; combining work, family and personal life; work that should be abolished; stability and security of work; equal opportunity and treatment in employment; safe work environment; social security; and social dialogue (Gammarano 2020).

18 ILO standards set a minimum maternity leave period of 14 weeks under the Maternity Protection Convention, 2000 (No. 183), while the accompanying Maternity Protection Recommendation, 2000 (No. 191) encourages to extend leave to at least 18 weeks to further protect the health of the mother and child.

19 According to the European Working Conditions Survey (EWCS), "unsocial working hours" refers to long hours, night work, working at short notice, and working in free time (Eurofound 2022).

as on-call or zero-hours arrangements.[20] In addition, excessive overtime, especially when unscheduled or without advance warning, can disrupt workers' personal and family lives, generate out-of-pocket costs (e.g. additional childcare, etc.) if not suitably compensated, and reduce opportunities for rest and recovery (ILO 2013; ILO 2019). Contractual provisions and workplace practices should therefore specify when overtime may be required, establish rules on notice and compensation, and respect limits consistent with working time legislation to prevent long working hours.[21]

In addition to the number and timing of working hours, compatibility between work schedules and workers' personal requirements, including family-related activities and the balance between working time, rest and social life, is a key aspect of employment quality (ILO 2019). Access to childcare and flexible working arrangements, including flexitime and working from home or hybrid work, can support the work-home interface by helping workers manage time and energy to meet demands both at work and outside work (Herrera-Ballesteros et al. 2025; Thompson, Payne, and Taylor 2014). However, flexible arrangements can also blur boundaries between professional and personal life, allowing work demands to intrude into private time (Palumbo et al. 2021). Digital communication may reinforce expectations of constant availability and intensify work, underscoring the importance of protecting rest time through accompanying measures (ILO 2025c).[22]

Practices for managing organizational change influence how restructuring, changes in organizational culture, and the introduction of new technologies are experienced by workers. Whilst these may be triggered by external pressures, organizations remain responsible for how change is planned, communicated and implemented.[23] Choices about the timing and scope of change, the redistribution of tasks and responsibilities, and the support provided during the transition can positively or negatively affect workloads, clarity of roles, social support, trust in management and, in some cases, job insecurity (Landsbergis, Grzywacz, and LaMontagne 2012; Quinlan and Bohle 2009). Change processes characterized by lack of transparency, poor communication, limited consultation and low predictability tend to heighten uncertainty and stress, whereas those marked by clear information, visible leadership and meaningful worker involvement can buffer negative effects and promote more sustainable adjustment (Egan et al. 2007; Kieselbach et al. 2010).

Both employment arrangements and organizational change processes shape workers' perceptions of employment stability and predictability. In this context, job insecurity, understood as the anticipation or concern about involuntary job loss (De Witte 2005; Vander Elst, De Witte, and De Cuyper 2014), may arise across diverse forms of work and sectors (Ferrie et al. 2008; Landsbergis, Grzywacz, and LaMontagne 2012). Beyond economic conditions, job insecurity is shaped by organizational decisions on contracts, staffing, working time and communication (Llorens et al. 2009).

Surveillance and digital monitoring refer to management's capacity to monitor, record and track workers' performance, behaviour and personal characteristics (Ball 2010). Although surveillance has long existed, digital technologies, remote work and platform-based labour present new means for expanding its scope and frequency.[24] In algorithmic management systems, work is allocated, monitored and evaluated through continuous data collection, real-time metrics and automated decision-making (Masoodi et al. 2021; Mateescu and Nguyen 2019). Such practices can support coordination, fairness, productivity and performance management, thereby benefiting both employers and workers. However, they become problematic when they are intrusive, undermine autonomy and trust, or exceed what is necessary for work organization (Ball 2021). Risks also arise when surveillance systems collect sensitive personal data or generate biased or unfair assessments that disproportionately affect specific groups of workers.

20 The online platform economy is characterized by algorithmic management, customer rating systems, income instability and unpredictable working hours, which may heighten exposure to psychosocial risks such as work intensification, job insecurity and reduced control over working conditions.

21 Long working hours generally refer to working hours exceeding 48 hours per week, in line with the limits established in the Hours of Work (Industry) Convention, 1919 (No. 1) and the Hours of Work (Commerce and Offices) Convention, 1930 (No. 30). In occupational health research, long working hours are often defined as 55 hours or more per week.

22 This can include the right to "disconnect from work", a regulatory development pioneered by European countries (Lerouge and Trujillo Pons 2022).

23 Please refer to *Health in Restructuring (HIRES): Recommendations, national responses and policy issues in the EU* by Kieselbach et al. (2010) for an in-depth analysis on the implications of restructuring and policy recommendations.

24 For example, electronic performance monitoring typically involves email monitoring, phone tapping, tracking computer content and usage times, video monitoring and GPS tracking (Moore 2017).

Rewards, performance management, opportunities for development and recruitment processes influence perceptions of fairness, recognition and motivation. Rewards may include financial remuneration, job security, promotion opportunities and access to training and development. Remuneration systems should be grounded in objective job evaluation and equal remuneration principles, without discrimination on grounds such as sex[25] and applied consistently across comparable roles (Mujtaba and Shuaib 2010).

Performance management systems often inform decisions on pay progression, promotion, development and, in some cases, disciplinary action (Lawler, Benson, and McDermott 2012; Mercer 2019). To be effective and fair, these systems should rely on job-relevant, transparent and consistently applied criteria, and be supported by constructive feedback (Ikramullah et al. 2016).[26] Effective supervision plays a critical role in this process, as regular, objective and clear feedback helps guide development efforts. Furthermore, systems that identify skill gaps should be matched with development activities that support both current and future performance (Chiang and Birtch 2012).

Fair and valid procedures for hiring and promotion are equally essential. Selection criteria should clearly map onto job requirements and be applied consistently to ensure equality of opportunity (Cropanzano, Bowen, and Gilliland 2007). Beyond fairness, these processes help ensure that roles are matched to the skills and competencies needed to meet organizational demands (De Crom and Rothmann 2018). While hiring and promotion opportunities depend on the availability of suitable positions and organizational needs (Cappelli and Conyon 2018), decisions should be based on objective, job-related criteria and applied without discrimination.

Increasingly, artificial intelligence and algorithmic tools are used in performance assessment, hiring and promotion. While these technologies may enhance efficiency and consistency, they raise concerns regarding transparency, bias, privacy and limited human oversight (Gupta and Tembhurnekar 2024; Pero, Wyckoff, and Vourc'h 2022). When evaluation and reward systems lack transparency, consistency or equity, they may undermine organizational justice and increase work-related stress.

An OSH policy and management system provides a structured approach for identifying, preventing and controlling workplace hazards to protect workers' safety, health and wellbeing. It helps organizations ensure compliance with regulations, reduce accidents, illnesses and costs, and foster a culture of continuous improvement and accountability.

Procedures to prevent and address violence and harassment at work are a core element of organizational responsibility. Violence and harassment encompass unacceptable behaviours and practices, or threats thereof, whether a single occurrence or repeated, that aim at, result in, or are likely to result in physical, psychological, sexual or economic harm.[27] Violence and harassment may also be gender-based when directed at persons because of their sex or gender, or when they disproportionately affect persons of a particular sex or gender.

Violence and harassment are closely linked to broader features of work organization and workplace culture, including power dynamics, workload pressures and employment insecurity (Nielsen and Einarsen 2018; Quinlan and Bohle 2009). Harmful behaviours and actions may occur vertically in supervisory relationships, horizontally among colleagues, or through interactions with third parties such as clients or members of the public. In doing so, they can influence task allocation, access to resources and cooperation among colleagues, while certain job contexts, particularly client-facing roles or work involving valuables, may increase exposure to risks. In this way, they both shape and are shaped by other psychosocial risk factors (WHO and ILO 2022).

These forms of abuse often co-occur or escalate if left unaddressed (Einarsen et al. 2017; Leake, Amankwaa, and Elisabeth 2025). Whether such behaviours are prevented, normalized or effectively addressed depends on the presence, clarity and credibility of organizational policies and procedures (Einarsen et al. 2017; Park

25 The Equal Remuneration Convention, 1951 (No. 100) establishes the principle of equal remuneration for men and women workers for work of equal value.

26 Effective implementation of performance management systems requires clear documentation of ratings and decisions; appraisers who communicate performance expectations, set specific targets, and regularly monitor progress; and the provision of timely and supportive feedback to help workers improve (Ikramullah et al. 2016).

27 As stated in the Violence and Harassment Convention, 2019 (No. 190), violence and harassment can take place in any work-related context, including during commuting, work-related travel, training or in digital work environments.

2023).[28] Workplace governance arrangements therefore play a central role in shaping accountability and supporting prevention.

Consultation and worker participation are core organizational practices and central pillars of OSH risk management, particularly for psychosocial hazards, which are often less identifiable. Through consultation and participation mechanisms, including joint OSH committees and the designation of workers' OSH representatives, workers can provide valuable insights and influence decisions that affect their working conditions, safety and health. Meaningful participation supports early identification of problems, strengthens trust in management decisions and enhances the legitimacy and effectiveness of organizational interventions.



28 Organizational measures may include, inter alia, clearly articulated codes of conduct; accessible reporting and complaint-handling procedures; training and awareness-raising; monitoring and oversight mechanisms; fair and confidential investigation processes; protection against retaliation for complainants, witnesses and whistle-blowers; and proportionate disciplinary measures where allegations are substantiated. Prevention should be integrated into everyday operations and risk controls, including work processes and the physical environment, and may be complemented by workplace facilities that reduce exposure to discrimination, violence and harassment (Einarsen et al. 2017; ILO 2019; Park 2023).

Examples of psychosocial hazards related to practices at the level of the organization

Employment arrangements

- Work that involves uncertainty on conditions of employment, including possibility of redundancy or temporary loss of work with reduced pay
- Low-paid or insecure employment
- Working in situations that are not properly covered or protected by labour law or social protection

Working time arrangements

- Working time arrangements that are not adapted to operational demands and workers' needs
- Excessive or misaligned working time arrangements, including insufficient breaks and rest periods in relation to workload
- Lack of organizational policies that make reasonable adjustments in working hours (e.g. flexitime) and practices on the location of work (e.g. telework) that support work-home interface

Organizational change management

- Poorly planned changes (e.g. changes are disorganized, do not have a clear goal, poor communication that does not provide timely or clear information)
- Poor consideration of OSH risks (e.g. not considering OSH risks when undergoing structural shifts or introducing changes, including new technology)
- Providing insufficient support to workers throughout the change process (e.g. not providing adequate training on how to perform new role or use a new process)

Surveillance

- Unreasonable levels of digital supervision for the purpose of work performance monitoring

Rewards, performance management, opportunities for development and recruitment processes

- Unfair, biased, opaque, or inequitable distribution of recognition and rewards (e.g. workers being rewarded for the efforts of others, promotions based on favouritism)
- Career stagnation and uncertainty, under-promotion or over-promotion, lack of opportunity for skill development
- Over-reliance on untransparent use of algorithms for management and evaluation purposes

An OSH policy and management system

- Lack of integration of psychosocial risks into OSH policies and management systems
- Unclear or inadequate procedures for assessing and controlling psychosocial risks
- Failure to account for interactions between the psychosocial working environment and the physical working environment

Violence and harassment at work

- Lack of procedures to report incidents and processes that lack transparency of how unacceptable behaviour will be managed

- Failing to appropriately address (actual or alleged) inappropriate or harmful behaviour, or misconduct (e.g. not investigating allegations of sexual harassment or not taking necessary disciplinary sanctions)
- Lack of procedures dealing with personal information, failing to treat workers' information sensitively or maintain their privacy

Consultation and participation

- Not consulting workers on changes in the workplace (e.g. not talking to workers or genuinely considering their views) and how these can affect them
- Leadership and management practices that do not visibly prioritize psychological health at work, including non-transparent or unethical practices that that undermine fairness and trust

Beyond formal policies and procedures, the effectiveness of organizational arrangements depends on how they are enacted and experienced in practice, which is shaped by organizational culture and climate.

Organizational culture encompasses shared values, norms and expectations that influence how rules are interpreted, how behaviour is regulated and whether psychological safety and health are prioritized alongside performance objectives (Guldenmund 2000; Schein 2010). Within this broader context, **psychosocial safety climate** reflects shared perceptions of the extent to which organizational policies, practices and procedures protect and promote psychological safety and health (Dollard and Bakker 2010; Dollard et al. 2017). A strong psychosocial safety climate signals that psychological health is not subordinate to productivity pressures and that concerns can be raised without fear of negative consequences.

Closely related is **organizational justice**,[29] understood as the perceived fairness of decision-making processes, resource distribution and interpersonal treatment within the organization (Baldwin 2006). Perceptions of fairness and transparency influence trust in organizational systems, the legitimacy of managerial decisions and the willingness of workers to engage in reporting and participation processes.

Coherence between these principles and everyday organizational practices is equally important. Inclusive organizational arrangements, including accessible environments, reasonable accommodation, equitable leave and flexibility provisions, and policies that recognize diverse working situations, shape whether workers can participate safely and with dignity. Such environments help mitigate psychosocial risks by reducing exclusion, unequal exposure and structural disadvantage across different groups of workers (ILO 2016b; Jack, Baird, and Hill 2025).

29 Organizational justice is commonly described as comprising distributive justice (fairness of outcomes such as pay, promotion and access to opportunities), procedural justice (fairness of decision-making processes, including consistency, lack of bias and opportunities to voice concerns or appeal), and interactional justice (fairness in the quality of interpersonal treatment and explanations for decisions, including dignity, respect and clear communication) (Baldwin 2006).

Interactions with the physical environment and work equipment

The overall working environment includes not only physical, biological and chemical hazards but also the design and condition of workspaces, including layout, ergonomics, hygiene and the availability, suitability and maintenance of tools, equipment and other resources needed to carry out tasks safely (Kop et al. 2016; Kouvonen et al. 2016). These elements interact closely with psychosocial factors and can either exacerbate or mitigate impacts on workers' safety, health and organizational performance.

When workers face dangerous tasks, poorly maintained equipment, limited space, insufficient lighting or excessive noise, they are exposed not only to direct physical risks but may also become mentally preoccupied with potential harm, increasing stress and reducing attentional capacity (Kouvonen et al. 2016; Seo, Hyun, and Yoon 2024). Working alone, whether remotely or in isolated locations, similarly raises physical and psychological risks, particularly where access to assistance from colleagues or emergency support is limited.[30] Risks also arise in traumatic, unstable or conflict- and disaster-affected environments,[31] where physical danger is combined with high emotional demands (Bierens de Haan 2020; Guisolan et al. 2022).

Workers' perceptions of risk are shaped not only by the frequency, duration and severity of hazards, but also by their confidence in managing them, including trust in safety safeguards such as alarms, communication systems, workplace layout and timely supervisory support (Leiter, Zanaletti, and Argentero 2009). For example, the physical layout can influence exposure to violence and harassment, particularly in settings involving contact with risk groups (ILO 2020).[32] Measures such as secure entry systems, surveillance, locks and alarms can reduce incidents and strengthen feelings of safety (ILO 2020).

Well-designed physical environments and equipment therefore act as protective resources that enhance safety, control and effective task performance, while poor design can amplify psychosocial risks (Leiter, Zanaletti, and Argentero 2009). This highlights the importance of considering how physical conditions interact with job design, work organization and organizational policies to support both safety and well-being.

Examples of psychosocial hazards arising from interactions with the physical work environment and equipment

- Work in remote or isolated settings with poor access to resources and difficult communication
- Poor workplace conditions such as lack of space, poor lighting, excessive noise
- Inadequate availability, suitability or maintenance of equipment or other resources needed to complete work tasks
- Working in extreme conditions or situations (e.g. extreme temperatures, at height etc.)
- Working in unstable working environments such as conflict zones and natural disaster zones
- Working environments that limit protections against violence from clients or outsiders (e.g. work in isolated or remote locations, at evening and/or night; working in contact with the public)

30 Examples include security staff working alone in large or poorly lit facilities, technicians maintaining infrastructure in remote locations, home-care workers travelling alone between clients' homes, or workers closing retail outlets or petrol stations on their own.

31 In contexts of armed conflict, health systems and organizational supports often collapse, which exposes workers to extreme physical danger, repeated traumatic events and chronic insecurity. Contemporary conflicts have also normalized violence against medical and humanitarian staff, further undermining the structures needed to manage psychosocial risks (The Lancet 2026).

32 Examples include special risk groups such as psychiatric patients, customers who have been consuming drugs or excessive amounts of alcohol, or criminals.



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Part 2

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The global health and economic burden of psychosocial hazards

Psychosocial hazards are associated with substantial health, organizational and economic consequences. This part presents new ILO global estimates on mortality, disability-adjusted life years (DALYs) and GDP losses associated with major psychosocial risk factors and situates these findings within the broader available evidence on adverse health effects on workers, the organizational costs of psychosocial risks, and patterns of exposure to work-related psychosocial hazards.

2.1. New ILO global estimates on mortality, DALYs and GDP losses associated with psychosocial risks

New global estimates provide a quantitative assessment of the burden attributable to five major work-related psychosocial risk factors: job strain, effort–reward imbalance, job insecurity, long working hours (55 hours or more per week) and workplace bullying. Using prevalence and population attributable fraction estimates applied to global mortality and disability-adjusted life years (DALYs) data, the analysis quantifies the contribution of these exposures to work-related disease and mortality.[33]

The findings indicate that work-related psychosocial risk factors are associated with an annual estimated **840,088 deaths annually worldwide** (783, 964 from cardiovascular diseases — ischaemic heart disease and stroke — and 56, 394 from mental disorders). These exposures account for approximately **44.9 million DALYs lost each year**, including 20.4 million attributable to cardiovascular diseases[34] and 24.5 million to mental disorders. While cardiovascular diseases account for the majority of attributable deaths, the overall loss of healthy life years is greater for mental disorders, reflecting their chronic and disabling nature. At the macroeconomic level, the combined burden corresponds to an estimated loss of **1.37 per cent of global gross domestic product (GDP)**, with regional variation ranging from 1.12 per cent in the ILO Americas Region to 1.72 per cent in the Africa Region.

Although causal pathways are complex and exposures are often multifactorial and cumulative, a substantial body of longitudinal research and systematic reviews shows consistent associations between adverse psychosocial work exposures — particularly job strain, effort–reward imbalance, job insecurity and violence and harassment — and mental and cardiovascular health outcomes (Cox 1993; Kivimäki and Steptoe 2018; Niedhammer, Bertrais, and Witt 2021; Rugulies et al. 2023).[35] Evidence remains more limited for low- and middle-income countries, MSMEs, informal work settings and some sectors, underscoring the need for continued research and improved data systems.


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33 These estimates are based on a comparative risk assessment approach combining prevalence data for five work-related psychosocial risk factors with population attributable fractions derived from systematic reviews and meta-analyses. Prevalence inputs draw on international survey sources, including the European Working Conditions Survey, while health outcome estimates are based on WHO Global Health Estimates and Global Burden of Disease data for 2021. This translational methodology aligns with approaches previously applied in ILO and WHO work on work-related disease burden.

34 Trend analysis shows that DALYs attributable to psychosocial risk factors increased by almost 14 per cent, rising from 17.9 million in 2019 to 20.4 million in 2021.

35 This recognition is also reflected in the WHO International Classification of Diseases (ICD-11), which includes health problems associated with employment conditions — such as stressful work schedules, uncongenial work and other work-related factors — within the category of factors affecting health status.

Global Deaths, DALYs and GDP loss attributable to psychosocial risk factors*

Global

Deaths

Total:
840 088

CVD: 783 694
454 262 329 432

MD: 56 394
43 877 12 517

DALYs**

Total:
44 912 621

CVD: 20 370 571
13 152 825 7 217 746

MD: 24 542 050
11 052 101 13 489 949

GDP loss
due to DALYs

Total:
1.37%

CVD: 0.62%

MD: 0.75%

* Job strain, effort–reward imbalance, job insecurity, long working hours, and workplace bullying

** Disability-adjusted life years

CVD: Cardiovascular diseases

MD: Mental disorders

Regional deaths, DALYs and GDP loss attributable to psychosocial risk factors*

Africa

Deaths

Total:
174 226

CVD: 162 844
88 656 | 74 188

MD: 11 382
8563 | 2819

DALYs**

Total:
8 191 936

CVD: 2 997 019
2 114 111 | 882 908

MD: 5 194 917
2 156 981 | 3 037 936

GDP loss due to DALYs

Total:
1.72%

CVD: 0.63%

MD: 1.09%

The Americas

Deaths

Total:
79 134

CVD: 73 871
41 921 | 31 950

MD: 5263
4049 | 1214

DALYs

Total:
5 135 499

CVD: 2 807 244
1 602 383 | 1 204 861

MD: 2 328 255
1 019 920 | 1 308 335

GDP loss due to DALYs

Total:
1.12%

CVD: 0.61%

MD: 0.51%

* Job strain, effort–reward imbalance, job insecurity, long working hours, and workplace bullying

** Disability-adjusted life years

CVD: Cardiovascular diseases

MD: Mental disorders

Arab States

Deaths

Total:
11 082

CVD: 10 206
8319 1887

MD: 876
804 72

DALYs

Total:
625 296

CVD: 345 623
285 173 60 450

MD: 279 673
202 407 77 266

GDP loss due to DALYs

Total:
1.16%

CVD: 0.64%

MD: 0.52%

Asia and the Pacific

Deaths

Total:
463 313

CVD: 431 701
259 575 172 126

MD: 31 612
25 072 6540

DALYs

Total:
25 075 720

CVD: 11 711 937
7 818 501 3 893 436

MD: 13 363 783
6 315 400 7 048 383

GDP loss due to DALYs

Total:
1.33%

CVD: 0.62%

MD: 0.71%

Europe and Central Asia

Deaths

Total:
112 333

CVD: 105 072
55 791 49 281

MD: 7 261
5389 1872

DALYs

Total:
5 884 170

CVD: 2 508 748
1 332 657 1 176 091

MD: 3 375 422
1 357 393 2 018 029

GDP loss due to DALYs

Total:
1.43%

CVD: 0.61%

MD: 0.82%

Impact of psychosocial risks on workers and enterprises

Workers



- Depression
- Anxiety disorders
- Burnout
- Sleep disturbances
- Fatigue
- Cognitive decline and dementia
- Suicidal behaviours
- Cardiovascular diseases
- Acquired metabolic disorders
- Digestive and immune function
- Cancer
- Musculoskeletal disorders
- Reproductive and perinatal outcomes
- Occupational injuries
- Unhealthy coping behaviours

Enterprises



- Absenteeism, presenteeism and leaveism
- Work engagement, performance and productivity
- Return-to-work outcomes
- Early exit from work, disability retirement and job quitting
- Job satisfaction and staff retention
- Organizational reputation, trust and collaboration

2.2. A review of the adverse health effects on workers

Psychosocial risks affect workers' health through multiple mechanisms, with stress constituting a primary pathway linking poor psychosocial working environments to a range of physical and mental health outcomes (Cox 1993; Rugulies et al. 2023; WHO and ILO 2022). Stress occurs when individuals perceive an imbalance between demands and their capacity to cope (Leka, Griffiths, and Cox 2003). Although short-term stress may enhance alertness and performance, prolonged, recurrent or intense stress without adequate recovery can result in cumulative physiological strain and increase the risk of adverse physical and mental health outcomes (APA 2024; Leka, Griffiths, and Cox 2003; WHO and ILO 2022).[36] Exposure to severe or traumatic events may also contribute to trauma-related conditions such as PTSD.

36 Physiologically, stress activates the autonomic nervous system and the hypothalamic–pituitary–adrenal (HPA) axis, leading to the release of stress hormones such as adrenaline, noradrenaline and cortisol. While these responses support short-term adaptation, chronic or repeated activation can result in neuroendocrine dysregulation, metabolic disturbance and chronic low-grade inflammation. This cumulative physiological burden is often described as *allostatic load* and represents a key biological pathway through which sustained psychosocial stress contributes to disease (Eddy et al. 2023; Eguchi et al. 2023; Kivimäki and Steptoe 2018; McEwen and Stellar 1993).

Sustained work-related stress may further influence behaviour and workplace relationships. In certain organizational contexts, elevated stress is associated with increased interpersonal tensions and conflict, potentially heightening the risk of violence, harassment and other forms of unacceptable behaviour, including bullying – particularly where excessive job demands, low control, organizational injustice and weak leadership are present (Einarsen et al. 2020; Nielsen and Einarsen 2018; WHO and ILO 2022).

Although many health effects of psychosocial risks operate through stress-related mechanisms, evidence shows that psychosocial risks may also affect health through other pathways. These operate through health-related behaviours (such as smoking, alcohol use or physical inactivity); biological and circadian disruption linked to long or irregular working hours; and social and material conditions, whereby employment arrangements influence living conditions, access to healthcare and health risks over time (Bell 2017; Benach et al. 2014).

Mental health conditions represent a substantial share of the global burden of disease among working-age adults. An estimated 15 per cent live with a mental disorder, and depression and anxiety account for approximately 12 billion lost workdays annually (WHO 2022a; WHO 2022b). A large body of research links psychosocial working environments to these outcomes, with depression, anxiety, burnout, sleep disturbances and trauma-related conditions among the most frequently examined consequences of adverse psychosocial conditions at work. New global estimates indicate that major psychosocial risk factors account for approximately 24.5 million DALYs related to mental disorders worldwide, confirming their substantial contribution to this burden (Neupane, Takala, and Descatha 2026). [37] Survey findings further show that such exposures are widespread: in Europe, nearly one in three workers reports job-related stress, depression or anxiety, with women reporting these problems more frequently than men (EU-OSHA 2025c).

Stigma: a barrier to addressing mental health at work

Stigma around mental health remains a major global barrier to prevention and support at work. Despite growing awareness, negative perceptions and discrimination persist in most regions, discouraging workers from reporting psychosocial hazards, from disclosing difficulties or seeking help. Cultural and organizational norms play a strong role: in some settings, mental health conditions are still viewed as signs of personal weakness rather than legitimate health concerns (WHO 2022a).

In Europe, the EU-OSHA OSH Pulse 2025 survey shows how such attitudes vary widely. Workers are evenly split on whether disclosing a mental health condition would harm their career (48 per cent agree, 48 per cent disagree). In Greece (69 per cent), Cyprus (67 per cent), and *France* and Italy (63 per cent), most fear negative consequences, whilst in the Nordic countries, about 80 per cent of workers feel comfortable discussing mental health with their manager.

Depression[38] and related outcomes are the most extensively documented mental health consequences associated with psychosocial risks (Leka and Jain 2024; Niedhammer, Bertrais, and Witt 2021; Rugulies et al. 2023). Studies report associations between depressive symptoms and psychosocial risks such as organizational injustice, job insecurity, and workplace violence and harassment, although findings vary across contexts and study designs (Niedhammer, Bertrais, and Witt 2021; Nielsen and Einarsen 2018; Rugulies et al. 2023). While evidence remains insufficient to establish causality for specific exposures, prospective cohort studies indicate that poor psychosocial safety climates are associated with an increased risk of incident depressive symptoms (Zadow et al. 2021).

Anxiety disorders[39] have also been linked to psychosocial risks at work, although the evidence base is more limited and heterogeneous (Rugulies et al. 2023; Stansfeld and Candy 2006). Longitudinal and

37 See the global estimates presented earlier in this chapter, which quantify the burden attributable to five major work-related psychosocial risk factors — job strain, effort–reward imbalance, job insecurity, long working hours and workplace bullying — using a comparative risk assessment approach based on 2021 data on mortality and DALYs (Neupane, Takala, and Descatha 2026).

38 See International Classification of Diseases, 11th Revision — ICD-11, code 6A70-6A7Z (Depressive disorders).

39 See International Classification of Diseases, 11th Revision — ICD-11, codes 6B00–6B0Z (Anxiety or fear-related disorders).

meta-analytic research identifies associations between anxiety symptoms and psychosocial risks including high job demands, low control and support, job insecurity, and workplace violence and harassment, with variation across contexts and study designs (Niedhammer, Bertrais, and Witt 2021; Nielsen and Einarsen 2018; Wieclaw et al. 2008). Some studies suggest these associations may be moderated by individual and contextual factors, including coping resources and psychological resilience (Stansfeld and Candy 2006; Vollrath and Torgersen 2008).

Burnout[40] is increasingly recognized as a significant occupational phenomenon characterized by exhaustion, cynicism, and reduced professional efficacy (WHO 2019). It commonly arises in work environments marked by sustained high job demands combined with low control and limited support (Maslach, Schaufeli, and Leiter 2001). Although conceptually distinct from depression and anxiety, it frequently co-occurs with other mental health conditions. In the EU, 13 per cent of workers report physical and emotional exhaustion (a risk factor for burnout), with substantial variations among occupations, reflecting differences in workplace culture and organizational support (Eurofound 2022). In the United States, 19 per cent of workers reported feeling very burned out in 2022 – an increase of more than 50 per cent since 2018 (Nigam et al. 2023).

Sleep disturbances[41] and **fatigue**[42] have been associated with psychosocial risks at work, including high job demands, shift work and work-family conflict, and can contribute to adverse health outcomes, as persistent sleep disruption and insufficient recovery increase the risk of anxiety, depression and burnout (Harvey et al. 2017; Salvagioni et al. 2017; Virtanen et al. 2009). Insufficient or poor sleep is widely reported across regions, ranging from around one-third of industrial workers in parts of South and East Asia (Yu, Chang, and Chang 2024), to approximately one-fifth of European workers (Eurofound 2025b), and affecting about 19 per cent of U.S. employees overall and over 30 per cent among night-shift workers (Yong, Li, and Calvert 2017). Rates are particularly high in health care (one-third to two-thirds), reflecting long hours, night shift work and emotional demands (Nilsson et al. 2025; Rahman et al. 2025).

Cognitive decline[43] and possible increases in **dementia**[44] **risk** have also been observed in relation to long-term exposure to high-strain and low-control work. These associations suggest that adverse psychosocial conditions may accelerate cognitive ageing through vascular and neuroendocrine processes (Andel et al. 2011; Wang et al. 2012).

Suicidal behaviours[45] are a serious but less extensively studied outcome in relation to psychosocial risks at work. Evidence indicates psychosocial risks may increase suicide risk,[46] particularly in interaction with pre-existing mental health conditions, social disadvantage or cumulative stressors (LaMontagne et al. 2024; Milner et al. 2018). Elevated risk has been observed in construction, agriculture and emergency response occupations, where psychosocial risks may intersect with traumatic exposure, irregular schedules, norms discouraging help-seeking and, in some sectors, access to lethal means (Greiner and Arensman 2022; Milner et al. 2018; Tiesman et al. 2015). In New Zealand, national records show that around 12 per cent of suicides were assessed as work-related (Rowan 2024). However, data remain limited and methodologically uneven, requiring cautious interpretation and consideration of work as one contributing factor within a wider social and health context (Greiner and Arensman 2022).

40 See International Classification of Diseases, 11th Revision — ICD-11, code QD85 (Burnout).

41 See International Classification of Diseases, 11th Revision — ICD-11, codes MG41 (Sleep disturbance, not elsewhere classified) and sleep–wake disorders, corresponds to codes 7A00–7A8Z.

42 See International Classification of Diseases, 11th Revision — ICD-11, code MG22 (Fatigue).

43 See International Classification of Diseases, 11th Revision — ICD-11, codes 6D70–6D72 (Neurocognitive disorders).

44 See International Classification of Diseases, 11th Revision — ICD-11, codes 6D80–6D86, 6D8Y and 6D8Z (Dementia).

45 See International Classification of Diseases, 11th Revision — ICD-11, code MB23.S (Suicidal behaviour). "Work-related suicide" has recently been defined from an OSH perspective as "death by suicide that is wholly or partly caused by work or working condition" (LaMontagne et al. 2024).

46 Associations have been reported between suicidal behaviours and psychosocial risks such as job insecurity, low control, poor support, long or irregular working hours, shift work, and exposure to violence or harassment, although findings vary across contexts and study designs (Greiner and Arensman 2022; Kim et al. 2024).

Moral injury: an emerging dimension of psychosocial risk

Moral injury refers to the psychological, emotional, and moral distress that occurs when individuals perpetrate, witness, or fail to prevent acts that conflict with their deeply held moral or ethical values. Originally identified in military contexts, the term has since expanded to include a wide range of occupations, especially those involving high emotional and ethical demands – such as healthcare, social care, emergency response, and humanitarian work.

Research shows that moral injury can have serious mental health consequences, including depression, anxiety, PTSD, and suicidal ideation (Hall et al. 2021; Jamieson et al. 2023; Williamson et al. 2018). During the COVID-19 pandemic, studies of healthcare workers and public safety personnel documented significant levels of moral injury related to ethical conflicts, perceived injustice, and chronic exposure to trauma (Thibodeau et al. 2023; Xue et al. 2022).

Unhealthy coping behaviours are another path through which psychosocial exposures affect health. Chronic work stress may lead to increased smoking, alcohol consumption, overeating and physical inactivity (Azagba and Sharaf 2011; Heikkilä, Fransson, et al. 2013). Although often adopted to manage tension or fatigue, these behaviours contribute over time to obesity, hypertension and other chronic diseases. Stress-related fatigue and time pressure may also discourage preventive habits such as sufficient sleep, physical activity and medical check-ups (Rugulies et al. 2023). Evidence also suggests that health-related behaviours and psychosocial risks interact over time, with unhealthy behaviours reinforcing and amplifying the adverse health effects associated with psychosocial stress exposure (Rodgers et al. 2021).

Psychosocial risks at work are associated with a range of **physical health outcomes**. Whilst these outcomes are multifactorial and shaped by biological, behavioural and social determinants, evidence indicates that chronic exposure to poorly managed psychosocial risks may contribute to physiological dysregulation and increased risk of diseases. The strength of evidence varies across outcomes, with the most consistent findings observed for cardiovascular and metabolic diseases, and more heterogeneous evidence for other conditions (Kivimäki and Steptoe 2018; Niedhammer, Bertrais, and Witt 2021; Rugulies et al. 2023).

Cardiovascular diseases[47], particularly ischaemic heart disease (IHD) and stroke, are among the most consistently documented physical health outcomes associated with adverse psychosocial exposures at work. Epidemiological studies and systematic reviews link exposures such as high demands combined with low control, effort–reward imbalance and long working hours (55 hours or more per week) with increased cardiovascular risk (Fishta and Backé 2015; Niedhammer, Bertrais, and Witt 2021; Nielsen and Einarsen 2018; Rugulies et al. 2023). Meta-analyses show that working 55 hours or more per week is associated with elevated risk of IHD and stroke (Li et al. 2020; Descatha et al. 2020),[48] and WHO–ILO estimates attribute around 745,000 deaths annually to long working hours, largely from these conditions (Pega et al. 2021). New global burden estimates further indicate that major work-related psychosocial risk factors account for approximately 20.4 million DALYs lost annually due to cardiovascular diseases worldwide, underscoring their substantial contribution to the occupational cardiovascular disease burden.[49]

47 See International Classification of Diseases, 11th Revision — ICD-11, codes 11A00-11Z99 (Diseases of the circulatory system).

48 While associations between psychosocial risks, long working hours and cardiovascular disease are consistently observed at the population level, effect sizes are generally modest and influenced by contextual and individual factors. Evidence on specific biological and behavioural mechanisms remains limited, and many studies rely on single baseline exposure measures. Further research using repeated exposure assessments and refined causal approaches may help clarify pathways and timing of risk (Virtanen and Kivimäki 2018).

49 These figures derive from the global burden estimates presented earlier in this chapter, which quantify mortality and disability attributable to five major work-related psychosocial risk factors using a comparative risk assessment approach based on 2021 data (Neupane, Takala, and Descatha 2026).

Karoshi and *Karo-jisatsu* in Japan: Death and Suicide from Overwork

In Japan, deaths and suicides linked to overwork have long been recognized as distinct yet related occupational health phenomena. ***Karōshi*** [sudden death from overwork] most commonly results from cardiovascular events such as stroke or myocardial infarction following prolonged excessive working hours. ***Karō-jisatsu*** [suicide from overwork] denotes suicide associated with overwork or severe job stress, often occurring in the context of chronic exhaustion, depression or intense workplace pressure (Bittle et al. 2025; LaMontagne et al. 2024).

The phenomena have drawn sustained public and policy attention since the 1980s, reflecting the intense demands of Japan's work culture and long-hour norms. According to a 2025 report by Japan's Ministry of Health, Labour and Welfare, there were 1,304 officially recognized cases of overwork-related deaths and health disorders in 2024, up from 1,108 the previous year. Of those 1,304 cases, 247 involved strokes or heart conditions (*karōshi*) and 1,057 involved mental health disorders, including 89 suicides or attempted suicides (MHLW 2025).

Whilst policy reforms in Japan have begun to address excessive overtime and work style (such as the 2018 *Work Style Reform*), the data suggests that overwork-related health disorders and suicide remain an urgent occupational and public health challenge.

Acquired **metabolic disorders**,[50] including obesity, metabolic syndrome and type 2 diabetes, have been associated with work-related stress and specific psychosocial exposures. Evidence indicates that long and irregular working hours, shift work, high job demands combined with low control, effort-reward imbalance and workplace bullying are linked to disrupted circadian and metabolic regulation (Boini et al. 2022; Kivimäki and Steptoe 2018; Kivimäki et al. 2015; Li et al. 2014; Peña-Gralle et al. 2022; Proper et al. 2016; Xu et al. 2017).

Digestive and immune function may be influenced by chronic work-related stress. Evidence suggests links between stress and functional gastrointestinal disorders, including irritable bowel syndrome and dyspepsia (Fond et al. 2014; Konturek, Brzozowski, and Konturek 2011). Chronic stress is also associated with altered immune functioning and increased susceptibility to infections, partly through neuroendocrine and inflammatory pathways (Glaser and Kiecolt-Glaser 2005).

Cancer risk has been examined in relation to sustained work-related stress,[51] but evidence remains limited and mixed. Some studies report associations between chronic stress or low job control and certain cancers, including colorectal, lung and oesophageal cancer (Heikkilä et al. 2013; Lillberg et al. 2003; Yang et al. 2018). Proposed mechanisms include immune and inflammatory dysregulation, hormonal changes and stress-related behaviours, although findings are heterogeneous and causal relationships remain uncertain (Antoni and Dhabhar 2019; Chida et al. 2008).

Musculoskeletal disorders (MSDs),[52] including back, neck and shoulder pain, are well-established consequences exposure to psychosocial risks. High job demands, low control, poor social support and effort-reward imbalance are consistently associated with MSDs (Bernal et al. 2015; Bezzina et al. 2023; Buruck et al. 2019; Taibi et al. 2021). Psychosocial hazards interact with physical load, amplifying both onset and chronicity (Taibi et al. 2021).

Reproductive and perinatal outcomes have gained increasing attention. Irregular hours, shift work and work-related stress are associated with menstrual disturbances, reduced fertility and adverse pregnancy outcomes, including preterm birth and low birth weight (Cai et al. 2019; Lee, Moawad, and Pien 2025). These effects appear to operate through biological and circadian disruption.

50 See International Classification of Diseases, 11th Revision — ICD-11, codes 5C50–5D2Z for an indicative list of acquired metabolic disorders (Metabolic disorders).

51 See International Classification of Diseases, 11th Revision — ICD-11, chapter 2 for a definition of the cellular dysregulation that leads to cancer (Neoplasms).

52 See International Classification of Diseases, 11th Revision — ICD-11, chapter 15 for an indicative list of musculoskeletal disorders (Diseases of the musculoskeletal system or connective tissue).

Occupational injuries and **workplace safety** can be influenced by psychosocial risks. Workers in high-hazard sectors such as construction, agriculture, forestry and mining are particularly vulnerable, with evidence linking high demands, low control, bullying, job insecurity and organizational change to increased accident rates and unsafe behaviours (Derdowski and Mathisen 2023; Sun et al. 2023). Fatigue, cognitive overload and reduced attention commonly mediate this relationship, as sustained pressure may lead workers to skip safety procedures, take shortcuts or delay reporting near misses. Unhealthy coping behaviours – including alcohol use, physical inactivity and poor sleep – can further impair alertness and reaction time (Taylor and Dorn 2006), while prolonged stress may affect cognitive control and situational awareness, increasing accident risk (Lewine 2024). Workers in informal employment remain especially exposed (Koranyi et al. 2018), and fear of reprisal or job loss may discourage incident reporting, contributing to underestimation of workplace harm (Kyung et al. 2023).

2.3. The escalating costs of psychosocial risks to organizations

Work-related stress creates a substantial economic burden through productivity losses, healthcare expenditure and social protection costs. As previously noted, the annual global economic losses associated with exposure to psychosocial risks at work are equivalent to 1.37 per cent of GDP through their contribution to ischaemic heart disease, stroke and mental disorders (Neupane, Takala, and Descatha 2026). Reviews of national studies estimate that the cost of work-related stress ranges from approximately US$221 million to US$187 billion, largely driven by productivity losses (Hassard et al. 2018). In the EU, annual costs of work-related depression are estimated to exceed €100 billion, with employers bearing more than 80 per cent of the burden (Sultan-Taïeb et al. 2024).

Epidemiological modelling estimates that workplace bullying and harassment are associated with global productivity losses of approximately US$1.7 trillion annually through their link with depression and work impairment (Dhanani, LaPalme, and Joseph 2021). This complements reviews of national cost studies showing that violence and harassment contribute to psychosocial risk costs ranging from over US$100 million to more than US$35 billion per year, depending on country and methodology (Hassard et al. 2018).

Absenteeism[53] and **presenteeism**[54] are among the most visible organizational effects of exposure to psychosocial hazards. Job strain, low organizational justice, poor psychosocial safety climate and inadequate leadership have been consistently associated with higher sickness absence (Amiri and Behnezhad 2020; Taibi et al. 2021; White et al. 2013), while mental-health-related absences have been particularly associated with effort-reward imbalance and high psychological demands (Duchaine et al. 2020). Across 62 countries, epidemiological modelling based on pooled survey data suggests that workers exposed to bullying and harassment may be absent between two and seven additional days per year compared to other workers (Dhanani, LaPalme, and Joseph 2021). Consistent patterns are observed at national level. For example, in the United States, workers reporting fair or poor mental health experience nearly 12 days of unplanned absence per year, around four times more than those reporting good or excellent mental health (Gallup 2022). Similar associations are observed for presenteeism, which has been closely linked to burnout, exhaustion and excessive job demands, and may be exacerbated by discrimination or poor social relations at work (Brborović et al. 2017; Miraglia and Johns 2016). Comparable patterns have been observed in Europe, where exposure to job strain has been associated with significantly higher risks of both sickness absence and presenteeism (Eurofound 2017b). A more recently described phenomenon, leaveism,[55] has

53 Absenteeism refers to absence from work due to illness or other health-related reasons and is commonly used as an indicator of work-related health outcomes in OSH research (WHO and ILO 2022).

54 Presenteeism refers to attending work while unwell and therefore functioning below full capacity (Miraglia and Johns 2016).

55 Leaveism refers to the use of leave or rest time to continue working or to recover from excessive workload (Hesketh and Cooper 2014).

also been associated with sustained organizational pressure and limited opportunities for recovery where psychosocial hazards are poorly managed (Hesketh and Cooper 2014).

Work engagement, performance and **productivity** are reduced under conditions of high stress, insecurity and unfair treatment (Arends, Prinz, and Abma 2017; Kotera and Vione 2020; Mauno et al. 2023; Sverke et al. 2019). Psychological disengagement at work (commonly labelled quiet quitting) refers to withdrawal from work roles whilst remaining formally employed. Survey evidence indicates that disengagement is more prevalent in contexts characterized by high demands, low recognition and limited organizational support, reflecting unmet psychosocial needs rather than individual motivation deficits (Gallup 2023). This pattern is consistent with longstanding research on work disengagement and withdrawal behaviours in response to adverse psychosocial working environments (Kahn 1990; Schaufeli, Bakker, and Salanova 2006). Poor psychosocial working environments additionally undermine organizational safety performance, increasing error rates and procedural violations, particularly in safety-critical sectors such as healthcare, transport and manufacturing (Christian et al. 2009; Nahrgang, Morgeson, and Hofmann 2011). A 2023 multi-country survey in Asia found that workers widely reported impaired productivity linked to mental health concerns, with country-level rates ranging from 31 per cent in Japan to over 60 per cent in Malaysia.[56] The region also recorded the lowest overall productivity-impact score globally (Aon and TELUS Health 2023). Workplace violence and harassment, social exclusion and interpersonal conflict have also been linked with counter-productive work behaviour.[57]

Return-to-work outcomes after illness or injury are also influenced by the psychosocial work environment. Workers facing high demands, low control or lack of supervisor support return to work more slowly and are at greater risk of relapse (Haveraaen, Skarpaas, and Aas 2017; Nielsen and Yarker 2022). An Australian study using workers' compensation data found that workers from organizations with poor psychosocial safety climates had markedly longer work absence durations and substantially higher compensation costs than those from organizations with stronger psychosocial safety climates (Dollard et al. 2024).

Early exit from work, disability retirement and **job quitting** are among the long-term organizational consequences of exposure to an adverse psychosocial working environment. Psychosocial hazards such as low job control, effort-reward imbalance and monotonous work are consistently linked to earlier or disability retirement (Dragano and Schneider 2011; Hintsa et al. 2015; Knardahl et al. 2017). Cohort analyses from Australia indicate that, although age and health remain primary retirement drivers, dissatisfaction and job-related stress meaningfully contribute to early departure from the workforce (Butterworth, Kiely, and Timmins 2017). National statistics from Finland highlight that mental and behavioural disorders accounted for disability pensions for more than half of recipients in 2024, and that depression was the single most common diagnosis, affecting 36 per cent of women and 20 per cent of men (Finnish Centre for Pensions 2025). Growing evidence also shows that psychosocial risks may contribute to voluntary labour-market exit. For example, a large multinational workforce survey across multiple countries reported that 44 per cent of respondents had left a job in the previous year due to a toxic work environment, with higher proportions among women and younger workers (Randstad 2025).

Job satisfaction and **staff retention** are strongly affected by psychosocial risks. Burnout, stress, poor organizational justice and exposure to harassment, incivility and bullying significantly increase turnover and reduce morale (Gerhardt et al. 2021; Li and Yao 2022; Özkan 2022; Penconek et al. 2021)

Organizational reputation, trust and **collaboration** deteriorate under sustained exposure to psychosocial risks, undermining innovation, learning capacity, and adaptability to change (Dediu, Leka, and Jain 2018; Leka and Jain 2024; Montani, Courcy, and Vandenberghe 2017).

In service and care sectors, high levels of work stress and burnout due to psychosocial risk exposure are further associated with reduced service quality and lower client or patient satisfaction, thereby reinforcing the erosion of organizational reputation and public trust (Adriaenssens, De Gucht, and Maes 2015; Hakanen and Bakker 2017).

56 According to the Aon and TELUS Health (2023) survey, country-level proportions of workers reporting that their mental health negatively affected their productivity were highest in Malaysia (62 per cent) and the Philippines (60 per cent), followed by Viet Nam (53 per cent) and Thailand (50 per cent). Substantial proportions were likewise reported in China and Indonesia (both 46 per cent), Singapore (43 per cent) and Japan (31 per cent).

57 Counterproductive work behaviour refers to behaviours that negatively affect the productivity of an organization or its workers (e.g. withdrawing effort, leaving early, or engaging in acts of sabotage) (Bruk-Lee and Spector 2006).

Research evidence on how healthy psychosocial working environments can support workers and organizations

When the psychosocial working environment is characterized by fairness, trust, and support, it enhances health and well-being, boosts motivation and productivity, and reduces work-related stress and its negative impacts. Positive psychosocial factors act as "resources" that both buffer job demands and stimulate engagement, consistent with the Job Demands-Resources model (Bakker and Demerouti, 2017; Schaufeli 2017). Beyond protecting health, such environments foster learning, creativity, and innovation, enhancing organizational adaptability and resilience (Edmondson and Lei 2014; Montani, Courcy, and Vandenberghe 2017).

- **Supportive and participatory leadership** plays a central role in shaping positive outcomes. Leaders who communicate openly, recognize effort, and encourage participation foster trust, a positive psychosocial safety climate, and stronger organizational commitment, contributing to a healthier and more productive workplace culture (Efimov, Harth, and Mache 2020; Skakon et al. 2010). Survey-based evidence further suggests that variation in leadership and management practices explains a substantial share of differences in team engagement and well-being, highlighting the influence of day-to-day leadership behaviour alongside formal organizational policies (Gallup 2025b). Leadership quality is therefore a key determinant of a healthy organizational culture and psychosocial working environment.
- **Fairness and organizational justice** are equally important. When workers perceive fair treatment and respect, they experience lower stress, higher job satisfaction, and greater engagement (Bronkhorst et al. 2015; Colquitt et al. 2013). A climate of justice and trust enhances cooperation and reduces conflict, strengthening both individual well-being and team performance. According to ILO evidence, when diversity and inclusion actions are embedded in enterprise strategy and organizational culture, workers are more likely to report high levels of personal well-being, reinforcing the link between fairness, inclusion and positive psychosocial outcomes (ILO 2022d).
- **Autonomy and social support** act as protective resources that mitigate stress and promote resilience. Opportunities to influence work decisions, combined with strong peer and supervisory support, buffer the impact of high demands and help prevent burnout (Halbesleben 2010; Van der Doef and Maes 1999). Such environments also promote learning and collaboration, essential for healthy team dynamics. Evidence shows that belonging and social cohesion predict higher engagement, performance and retention (Huettermann and Bruch 2019; Walton 2014). Consistent with this, recent findings from a multinational survey indicate that a large majority of workers perceive workplace community and social support as contributing positively to their performance (Randstad 2025).
- **Work-life balance** contributes to long-term health and engagement. Flexible and family-friendly work arrangements, predictable schedules, and respect for recovery time reduce emotional exhaustion and support sustained productivity (Kossek et al. 2011). Reflecting this priority, recent global survey data show that work-life balance is consistently rated by workers as one of the most important aspects of current and future jobs (Randstad 2025).
- **Psychosocial safety climate** represents a cornerstone of positive organizational culture. High psychosocial safety climate is consistently linked to lower stress, better relationships, and greater motivation and engagement across different work settings, including remote and hybrid contexts (Amoadu, Ansah, and Sarfo 2023; Juutinen et al. 2023). In sectors where creativity is a key source of competitive advantage, studies indicate that a strong psychosocial safety climate is fosters higher work engagement, job performance, and creative behaviour by enabling collaboration and knowledge sharing (Li and Wareewanich 2024; Zadow et al. 2023).

2.4. A review of the evidence on the prevalence of exposure to work-related psychosocial hazards

Understanding the prevalence of psychosocial hazards requires reliable and comparable measurements.[58] Although attention to the study of psychosocial hazards has increased, global data remain fragmented and uneven. A key challenge is that psychosocial hazards differ in how easily they can be measured. Some relate to observable features of work organization, such as working hours, which can be directly quantified and with widely used measurement approaches. Others depend on individual workers' experiences and are therefore commonly measured through self-report.[59] Responses in such cases may be shaped by social norms, workers' awareness and workplace cultures, making the comparison of trends between workers more complex when these modifiers are varyingly present.[60] Nevertheless, aggregated results can help to control for this variation and therefore remain important for providing insights on the prevalence of psychosocial hazards.

International organizations, including the ILO, have produced global estimates for a limited set of more readily measurable indicators of psychosocial risks and have conducted global survey initiatives. However, for many other psychosocial hazards, evidence comes mainly from national and regional working conditions surveys, which are often concentrated in high-income settings. These surveys typically involve structural or administrative indicators or self-reported measures of exposure, capturing different psychosocial hazards and their dimensions. They provide valuable insights into sectoral and demographic patterns and, when repeated over time, enable the analysis of trends.

Differences in definitions, measurement tools and composite indices constrain cross-country comparability, particularly for hazards that rely on more subjective assessment. In addition, in many low- and middle-income countries, and particularly in the informal economy, such surveys are infrequent or absent, further limiting global coverage.[61] Industry specific and independent surveys also examine selected psychosocial factors, but their representativeness is limited and they further illustrate the lack of harmonized measurement approaches across sources.

In this context, the following section draws on available surveys and studies to provide an overview of those psychosocial hazards for which documentation and statistical coverage are relatively more developed, while noting important data gaps and ongoing challenges in measurement harmonization.

Long working hours are amongst the most well-documented psychosocial hazards. In 2019,[62] per ILO estimates, 35 per cent of workers globally worked more than 48 hours per week. The Asia-Pacific had the highest burden of all studied regions (47 per cent), whilst wholesale and retail trade (49 per cent), transport and communications (45 per cent) and manufacturing (45 per cent) sectors were most impacted (ILO 2022b). Long working hours were also more common in informal employment compared to formal employment (affecting 41 versus 28 per cent of workers, respectively)[63] (ILO 2022b). However, evidence from 35 countries in Europe[64] suggests that the occurrence of working for 48 or more hours weekly is declining, having decreased from 19 to 11 per cent between 2005 to 2024 (Eurofound 2025b).

58 Prevalence estimates are not only descriptive; they are a key input for quantifying the preventable burden of disease linked to psychosocial risks. Epidemiological studies combine exposure prevalence with relative risk estimates to calculate population attributable fractions (PAFs), indicating the proportion of disease cases that could be avoided if harmful exposures were reduced, as illustrated by the WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury (2000–2016) (WHO and ILO 2021).

59 Studies indicate that self-report data produce results comparable to other measurement methods, such as job exposure matrices, supporting the validity of self-reports for assessing psychosocial risk perception (Madsen et al. 2018).

60 When surveys are independent from workplaces, such as national surveys that are not linked to employers, these concerns are less likely to affect data collection.

61 It is worth noting that little global research has examined gendered effects and their intersection with psychosocial factors at work, including access to employment, work quality and exposure to risk factors.

62 These statistics employ data of 2019 or latest available year as of 2022 and are limited to paid work.

63 The ILO estimates that in 2025, almost 58 per cent of workers globally were in the informal economy (ILO 2026).

64 Includes over 36,500 workers across the 27 EU Member States, as well as Norway, Switzerland, Albania, Bosnia and Herzegovina, Kosovo, Montenegro, North Macedonia and Serbia.

Working arrangements such as shift, night and weekend work, are often documented in labour surveys. In the European Union (EU), 21 per cent of workers worked shifts in 2024, with higher shares in health (34 per cent) and transport (32 per cent) sectors (Eurofound 2025a).[65] Across Europe in 2021,[66] more than one in five workers undertook night work, including security staff (65 per cent), drivers (46 per cent) and healthcare professionals (39 per cent) (Eurofound 2022). In the United States of America, in 2024, 30 per cent of workers worked on an average Saturday, Sunday or holiday, rising to 42 per cent in transport and material moving occupations and 41 per cent in service occupations (U.S. Bureau of Labor Statistics 2025).

Conversely, **flexible arrangements** in the form of working entirely remotely (3 per cent), in hybrid arrangements (9 per cent) or occasional remote work (16 per cent) were also practised in Europe in 2024 (Eurofound 2025b). In the United States, 33 per cent of workers (29 per cent of men and 36 per cent of women) worked from home in 2024 (U.S. Bureau of Labor Statistics 2025). Despite the benefits of these arrangements, the psychosocial working environment can be negatively impacted by domestic violence when working from home. Moreover, adverse psychosocial working environments may themselves increase vulnerability or limit access to protection and support, highlighting a bidirectional relationship between these issues.[67]

The growing use of digital tools has translated into the implementation of **digital** and **algorithmic management** in workplaces. Enterprise level evidence indicates that monitoring via algorithmic tools is reported by 90 per cent of enterprises in the United States, 67 per cent in Europe, and 31 per cent in Japan. Automated evaluations are also reported by 90 per cent of enterprises in the United States, and 35 per cent in Europe and 11 per cent in Japan (Milanez, Lemmens, and Ruggiu 2025).[68] In the EU, worker surveys show that in 2025 around one in four workers were exposed to technologies that automatically allocate tasks or shifts, provide third party performance ratings, generate automated instructions, or operate monitoring systems (EU-OSHA 2025c).[69]

Different aspects of **job control and autonomy** have been captured in working conditions surveys. In Europe, half of men report having some control over their working hours, compared with 43 per cent of women (Eurofound 2025b). Meanwhile, 17 per cent of workers in the EU report a lack of autonomy or influence over work pace and processes (EU-OSHA 2025c). Links with digital technologies have also been also explored. Technology sets the work pace for 48 per cent of workers, reduces opportunities to use their knowledge and skills for 19 per cent, and limits decision -making for 16 per cent (EU-OSHA 2025c). In Colombia, national data for 2021 demonstrates variation in workers' autonomy of decisions by employment status.[70] The ability to act on one's own ideas is reported as always possible by 81 per cent of self-employed workers, 78 per cent of employees and 61 per cent of informal workers (Ministro del Trabajo and OISS 2022). In the United States, a 2025 survey indicates 28 per cent of workers strongly agree that their opinion counts at work (Gallup 2025a).[71]

Across Europe, the share of workers reporting **monotonous work** tasks increased from 39 per cent in 1995 to 48 per cent in 2024 (Eurofound 2025b). Sectoral results at the EU level indicate particularly high exposure in agriculture (60 per cent), transport (56 per cent) and commerce and hospitality (53 per cent) (Eurofound 2025a). In Colombia (2021), 63 per cent of workers reported performing monotonous or **repetitive work**, rising to 89 per cent in mining and quarrying and 73 per cent in manufacturing (Ministro del Trabajo and OISS 2022).

65 Whilst the EWCS is administered across the EU and non-EU countries, this statistic is derived from the data explorer, which does not provide an aggregate of all countries surveyed.

66 Data collected by the European Working Conditions Telephone Survey (EWCTS), conducted in the 27 EU Member States, Albania, Bosnia and Herzegovina, Kosovo, Montenegro, North Macedonia, Norway, Serbia, Switzerland and the United Kingdom.

67 Convention No. 190 recognizes the impact of domestic violence on the world of work and calls for measures to mitigate its effects.

68 This OECD survey covers over 6 000 firms in six countries (France, Germany, Italy, Japan, Spain and the United States).

69 While the OSH Pulse interviewed 28,220 workers across the EU-27, Iceland, Norway and Switzerland, the reported statistics in this section refer only to the EU.

70 In this survey, "employees" refers to workers with a formal employment contract; "self-employed" refers to individuals contributing independently to the social security system; and "informal workers" refers to those without formal contracts or social security coverage.

71 These survey questions are drawn from Gallup's 12-item employee engagement instrument (Q12). Results are based on self-administered web surveys conducted with a random sample of U.S. adults aged 18 and older who work full-time or part-time for an organization and are members of the Gallup Panel.

Regional and national worker surveys have documented aspects related to **workload** and **time pressure.** In the EU, 44 per cent of workers report exposure to severe time pressure or overload, and 43 per cent of enterprises[72] identify time pressure as a risk (EU-OSHA 2025a; EU-OSHA 2025c). An increase in workload linked to the use of technologies is also reported by 28 per cent of workers (EU-OSHA 2025c). The transport sector reports the highest exposure to tight deadlines (46 per cent) and working at high speed for most of the time (43 per cent). [73] Rarely or never having enough time to do the job is most common in education (12 per cent) and health (11 per cent) (Eurofound 2025b). National findings also highlight workload concerns that vary by sector. In Canada, 32 per cent of workers in health care and social assistance report heavy workloads, compared with 24 per cent overall (Statistics Canada 2023). In Chile, 23 per cent of establishments report risk of high workload, rising to 69 per cent in social services (Superintendencia de Seguridad Social 2024). An industry study of 2025 on desk-based workers across 31 countries[74] indicates eight in ten office-based workers reported having insufficient time to complete their work (Microsoft 2025).

Emotional demands are monitored in several working conditions surveys, though indicators differ. EU data from 2025 show that women consistently report higher exposure than men across multiple metrics: 29 per cent compared with 22 per cent for hiding emotions, 18 per cent compared with 12 per cent for handling angry clients, and 13 per cent compared with 8 per cent for emotionally disturbing situation (Eurofound 2025a).[75] This finding is consistent with previous survey editions, and the pattern is closely linked to structural gender inequalities in labour markets, including occupational segregation and the concentration of women in service, care and people-facing roles that often involve high levels of emotional labour. In Chile,[76] emotional demands (considering exposure to upsetting situations, dealing with clients' personal problems and hiding emotions at work) rank among the three top psychosocial risks assessed as high risk, affecting an estimated 21 per cent of all workplaces in 2024 (Superintendencia de Seguridad Social 2024). **Cognitive demands** are also assessed across a range of metrics. In the EU, 64 per cent of workers report undertaking complex tasks. Their prevalence varies by sector, reaching 81 per cent in financial services, compared to 47 per cent for commerce and hospitality (Eurofound 2025a). In Colombia, around half of workers across all employment arrangements report always having to maintain high or very high attention at work.[77] The proportion reporting that they always carry out complex or difficult tasks is 16 per cent among the self-employed-employed, 9 per cent among informal workers and 8 per cent among employees (Ministro del Trabajo and OISS 2022).

Workplace support is measured through indicators of relationships with colleagues and supervisors, as well as access to resources. In Europe, 73 per cent of workers report receiving support from colleagues always or most of the time, and between 64 to 65 per cent report support from managers, whilst 11 per cent report requesting training but not receiving it (Eurofound 2025b). In the United Kingdom of Great Britain, a 2025 survey found that 79 per cent of workers viewed their manager as supportive when they had a problem, whilst 64 per cent said their manager supported their learning and development and 62 per cent reported receiving helpful feedback (CIPD 2025).[78] In the United States, 39 per cent of workers reported feeling cared for by a supervisor or someone at work in 2025, whilst only 36 per cent reported having the materials and equipment needed to do their job properly (Gallup 2025a).

72 41,458 establishments across the EU-27, Iceland, Norway and Switzerland were surveyed in the Fourth European Survey of Enterprises on New and Emerging Risks (ESENER 2024); however, the statistics reported here refer only to the EU-27.

73 These two survey questions refer to situations occurring three quarters of the time or more.

74 This is an online survey of full-time employed or self-employed knowledge workers, defined as people who typically work at a desk in an office or at home, either fully in-person or with some remote work. It covers over 31,000 workers, collecting around 1,000 respondents per country (Argentina, Australia, Brazil, Canada, China, Colombia, Czechia, Finland, France, Germany, Hong Kong, India, Indonesia, Italy, Japan, Malaysia, Mexico, Netherlands, New Zealand, Philippines, Poland, Singapore, Republic of Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, the United Kingdom, the United States, and Vietnam), with results reported as aggregated global averages.

75 While the survey question on hiding emotions refers to doing so most or all of the time, the other two indicators capture situations occurring three quarters of the time or more.

76 This survey is applied every 2 years in organisations with more than 10 workers. In 2024, 20,451 workplaces were evaluated, 16,262 private and 4,189 public, covering a total of 1,016,765 workers. It reports percentage of workplaces risk status based on the results of a questionnaire with 12 dimensions of psychosocial risks (see manual at Superintendencia de Seguridad Social, 2022).

77 Specifically, 47 per cent for employees, and 51 per cent for self-employed and informal workers.

78 This survey collects responses on 5,000 workers from different sectors.

Statistics on access to development, as well as other forms of **rewards on work performance**, have started to emerge. In 2024, a global survey found that 27 per cent of workers felt paid unfairly (28 per cent of women versus 23 per cent of men) (ADP Research 2025).[79] In the EU, around one in three workers report a lack of rewards (e.g. pay, career opportunities, recognition) for their efforts (EU-OSHA 2025c). Additionally, 27 per cent of EU workers disagree that they are paid appropriately for their efforts and achievements, with the highest shares in agriculture at 37 per cent and in health at 34 per cent (Eurofound 2025a).

Data on **role clarity** are available in some countries, although measurement approaches vary. In a 2024 national survey in Japan, 20 per cent of highly stressed workers identified changes in their role or role status as one of their top three sources of stress (MHLW 2025). In Colombia, more than one in four informal workers reported insufficient information to perform their roles, compared with around one in five self-employed workers and employees in 2021 (Ministro del Trabajo and OISS 2022). Canadian data of 2021 show variation by sector and age. Overall, 62 per cent of workers report often knowing what is expected of them at work, falling to 58 per cent in transportation. Reported role clarity is also lower among workers aged 18-34 (63 per cent) than among those aged 55 or older (82 per cent) (MHRC 2021).[80] In the United States, fewer than half of workers (48 per cent) reported being clear about their work expectations in 2025, down from 61 per cent in 2015 (Gallup 2025a).

Data on **violence and harassment at work** have historically been scarce and only available in a few countries. The first global source, an ILO survey conducted in 2021, found that more than one in five (23 per cent) workers had experienced at least one form of violence and harassment in their working life, most commonly psychological (18 per cent), followed by physical (9 per cent) and sexual (6 per cent)[81], with higher reported exposure to sexual violence and harassment among women (8 per cent vs 5 per cent) (ILO and Lloyd's Register Foundation 2022). Efforts to improve internationally comparable data are ongoing. In 2024, the ILO piloted a new survey methodology in Burkina Faso, Côte d'Ivoire and Senegal. Results show that 65 per cent of respondents in Senegal, 58 per cent in Côte d'Ivoire and 43 per cent in Burkina Faso had experienced violence and harassment at some point in their lives. Psychosocial forms were the most common, and women were more likely to experience sexual violence and harassment. Workers in market services also reported markedly higher exposure, with 45 per cent in Côte d'Ivoire affected in the past 12 months compared with 23 per cent of agricultural workers (ILO 2024b).

Available data suggest that **consultation** practices vary by topic. In Europe, 45 per cent of workers report being consulted on stressful aspects of work, compared with 18 per cent of workplaces consulting on working from home in 2024. When digital technologies are introduced, 25 per cent of enterprises report consulting workers on OSH impacts, with higher shares in education (42 per cent) and ICT (41 per cent) (EU-OSHA 2025a; EU-OSHA 2025c). Consulting workers when implementing algorithmic management tools is reported by 63 per cent of managers across the US, Europe and Japan (Milanez, Lemmens, and Ruggiu 2025).[82] Evidence on **psychosocial safety culture and climate** focuses on stigma and workplace action. In the EU, workers are evenly split on whether disclosing a mental health condition would harm their career (48 per cent agree; 48 per cent disagree) (EU-OSHA 2025c). A survey carried out in Asia in 2023 shows that 54 per cent of workers believe that disclosing mental health concerns would limit their career options (Aon and TELUS Health 2023).[83] At the same time, 53 per cent of EU workers report that awareness-raising, information or training on well-being and coping with stress is available at their workplace (EU-OSHA 2025c). In Colombia, only 5 per cent of workplaces reported carrying out promotion or prevention activities on psychosocial risks in 2020, compared with much higher uptake in other OSH areas (Ministro del Trabajo and OISS 2022).

79 Survey covers 38,000 workers across 34 markets,

80 This survey collected over 5,500 responses from employed Canadians.

81 The study defines psychological violence and harassment as including insults, threats, bullying or intimidation; physical violence as including hitting, restraining or spitting; and sexual violence and harassment as including unwanted sexual touching, comments, pictures, emails or sexual requests.

82 Surveyed European countries include France, Germany, Italy and Spain.

83 The survey collected data from 13,000 workers and revealed substantial differences across countries: The Philippines (75 per cent), Malaysia (71 per cent), India (66 per cent) and Singapore (64 per cent) displayed the highest burdens while Japan, had the lowest (28 per cent). Other countries including Viet Nam (58 per cent), Thailand (51 per cent), Indonesia (49 per cent), China (45 per cent), and the Republic of Korea (39 per cent) fell between these points.



Part 3

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OSH policy frameworks addressing psychosocial risks in the world of work

The expanding evidence base on psychosocial hazards and impacts on both workers and organizations has contributed to their increasing recognition within OSH policy frameworks. Furthermore, the affirmation of a safe and healthy working environment as a fundamental principle and right at work has reinforced the understanding that protecting safety and health requires attention to the full range of conditions under which work is performed, including both physical and psychosocial aspects of the working environment.

Across countries and regions, OSH policy and regulatory frameworks have incorporated psychosocial risks in different ways, reflecting diverse legal traditions, institutional arrangements and policy priorities (Leka et al. 2011). As a result, approaches to psychosocial risk prevention vary in scope, specificity and mechanisms for implementation and enforcement. Differences in legal coverage may also affect certain categories of workers, particularly where employment relationships fall outside standard regulatory protections.

Research indicates that coherent policy and regulatory frameworks play a key role in shaping workplace prevention by clarifying duties and embedding psychosocial hazards within established risk management systems (ILO 2016b; Leka et al. 2015). The way OSH policy frameworks are designed and implemented therefore influences how psychosocial risks are addressed in practice.

3.1. The ILO framework on psychosocial risks and mental health at work

Since its foundation in 1919, the ILO has recognized that the protection of workers' safety and health is central to social justice and decent work.

The **ILO Constitution** (1919), in its Preamble, establishes safety and health as a fundamental objective of the Organization by calling for the improvement of working conditions, including "the protection of the worker against sickness, disease and injury arising out of employment", in order to address conditions of labour involving injustice, hardship and privation that endanger social peace.[84]

The **Declaration of Philadelphia** (1944) includes the solemn obligation of the Organization "to further among the nations of the world programmes which will achieve adequate protection for the life and health of workers in all occupations" (Para. III(g)). It also affirms that "all human beings, irrespective of race, creed or sex, have the right to pursue both their material well-being and their spiritual development in conditions of freedom and dignity, of economic security and equal opportunity" (Para. II(a)).

These foundational principles were explicitly reaffirmed in 2022 when the International Labour Conference recognized a safe and healthy working environment as a **fundamental principle and right at work**,[85] confirming that protecting workers' safety and health is a necessary condition for decent work. Within this framework, the fundamental OSH Conventions (the Occupational Safety and Health Convention, 1981 (No. 155) and the Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187)) provide the essential basis for comprehensive and preventive OSH systems at national and enterprise levels. Whilst these instruments do not refer explicitly to psychosocial risks, their provisions are sufficiently broad to encompass the organizational and human factors that influence workers' physical and mental health.

Convention No. 155 requires each Member to formulate, implement and periodically review a coherent national policy on OSH and the working environment, with the aim to protect the physical and mental health of workers. It defines "health" in relation to work as "not merely the absence of disease or infirmity" but also including "the physical and mental elements affecting health which are directly related to safety and hygiene at work" (Art. 3(e)). The Convention calls for measures to prevent occupational accidents and diseases by minimizing, as far as reasonably practicable, the causes of hazards inherent in the working environment, and emphasizes the adaptation of work, equipment and organization to the physical and mental capacities of workers (Art. 5). In defining employers' responsibilities, Convention No. 155 explicitly refers not only to workplaces, machinery and equipment, but also to "processes under their control", which should be safe and without risk to health, "as far as is reasonably practicable" (Art. 16(1)). The accompanying **Recommendation No. 164** reinforces these principles by urging measures to prevent "harmful physical or mental stress due to conditions of work" (Para. 3(e)) and by requiring that "work organization, particularly with respect to hours of work and rest breaks, does not adversely affect occupational safety and health" (Para 10(f)). It further provides that employers take all reasonably practicable measures to eliminate "excessive physical and mental fatigue" (Para. 10(g)). Taken together, Convention No. 155 and Recommendation No. 164 provide a strong normative basis for addressing psychosocial risks and promoting workers' physical and mental health, within preventive OSH policies and measures.

84 In calling for the improvement of working conditions, the Preamble to the ILO Constitution refers, inter alia, to the regulation of working hours and labour supply; the prevention of unemployment and the provision of an adequate living wage; the protection of children, young persons and women; provision for old age and in case of injury; protection of the interests of workers employed outside their country of origin; recognition of the principles of equal remuneration for work of equal value and freedom of association; and the organization of vocational and technical education.

85 In 2022, the International Labour Conference adopted the Resolution on the Inclusion of a Safe and Healthy Working Environment in the ILO's Framework of Fundamental Principles and Rights at Work, thereby amending the ILO Declaration on Fundamental Principles and Rights at Work (1998) to recognize a safe and healthy working environment as a fifth fundamental principle and right at work (ILO 2022e). The fundamental principles and rights at work now include: (i) freedom of association and the effective recognition of the right to collective bargaining; (ii) the elimination of all forms of forced or compulsory labour; (iii) the effective abolition of child labour; (iv) the elimination of discrimination in respect of employment and occupation; and (v) a safe and healthy working environment.

ILO Convention No. 187 complements **ILO Convention No. 155** by establishing a promotional framework for the continuous improvement of national OSH systems. It calls for the promotion of a national preventive safety and health culture, in which the right to a safe and healthy working environment is respected at all relevant levels, and in which governments, employers and workers actively participate in securing OSH through a system of defined rights, responsibilities and duties, with the principle of prevention accorded the highest priority.

In addition to the fundamental Conventions, other OSH instruments further elaborate provisions relevant to the management of psychosocial risks and mental health at work.

The **Occupational Health Services Convention, 1985 (No. 161)** defines occupational health services as preventive and advisory bodies responsible for assisting employers, workers and their representatives in maintaining a safe and healthy working environment and promoting workers' physical and mental health. It provides that these services shall advise on "the requirements for establishing and maintaining a safe and healthy working environment which will facilitate optimal physical and mental health in relation to work," and on "the adaptation of work to the capabilities of workers in the light of their state of physical and mental health" (Art. 1(a)). Among their core functions, occupational health services must identify and assess workplace risks, monitor working conditions, and provide guidance on the organization of work and work design (Art. 5). The accompanying **Recommendation No. 171** further calls for the surveillance of the working environment to include the assessment of factors in work organization that may give rise to risks for workers' health (Para. 5(1)(b)), and for participation in job and work-method analysis to ensure a better adaptation of work to workers (Para. 8(e)).

The **List of Occupational Diseases Recommendation, 2002 (No. 194)** provides for the periodic review and updating of the ILO list of occupational diseases, which serves as an international reference for their identification and notification. The 2010 revision expanded this list to include mental and behavioural disorders, such as PTSD and "other mental or behavioural disorders [...] where a direct link is established scientifically, or determined by methods appropriate to national conditions and practice, between the exposure to risk factors arising from work activities and the mental or behavioural disorder(s) contracted by the worker."

The **Violence and Harassment Convention, 2019 (No. 190)** is the first ILO instrument to refer explicitly to *psychosocial risks* through a comprehensive approach to violence and harassment in the world of work.[86] It recognizes "the right of everyone to a world of work free from violence and harassment, including gender-based violence and harassment" and defines such conduct as "a range of unacceptable behaviours and practices, or threats thereof, whether a single occurrence or repeated, that aim at, result in, or are likely to result in physical, psychological, sexual or economic harm" (Art. 1). The Convention sets out preventive and protective obligations, including the requirement for employers, as appropriate and taking into account their degree of control, to "take into account violence and harassment and associated psychosocial risks in the management of occupational safety and health" (Art. 9(b)). **Recommendation No. 206** further specifies that risk assessments should take into account factors that increase the likelihood of violence and harassment, including psychosocial hazards and risks, arising from working conditions and arrangements, work organization and human resource management (Paragraph 8(a)); involving third parties, such as clients, customers, service providers, users, patients and members of the public (Para. 8(b)); and arising from discrimination, abuse of power relations, and gender, cultural and social norms that support violence and harassment (Para. 8(c)).

Other International Labour Standards that are pertinent to the prevention and management of psychosocial risks include those addressing equality of opportunity and treatment, non-discrimination, working time and night work.[87]

86 The concept of the "world of work" adopted in the Convention aligns with OSH standards, notably the definition of "workplace" in Convention No. 155 as encompassing all places where workers need to be or go by reason of their work and which are under the direct or indirect control of the employer (Art. 3(c)).

87 For example, the Discrimination (Employment and Occupation) Convention, 1958 (No. 111); Equal Remuneration Convention, 1951 (No. 100); Workers with Family Responsibilities Convention, 1981 (No. 156) and its Recommendation No. 165; Hours of Work (Industry) Convention, 1919 (No. 1); Forty-Hour Week Convention, 1935 (No. 47); Night Work Convention, 1990 (No. 171) and its Recommendation, 1990 (No. 178).

ILO normative framework relevant for addressing psychosocial risks

Foundational mandate

ILO Constitution (1919)

- Protection of the worker against sickness, disease and injury arising out of his employment (Preamble)

Declaration of Philadelphia (1944)

- Adequate protection for the life and health of workers in all occupations (Para. III(g))
- Right to pursue both their material well-being and their spiritual development in conditions of freedom and dignity, of economic security and equal opportunity (Para. II(a))

Declaration on Fundamental Principle and Right at Work (1998, as amended in 2022)

- Safe and healthy working environment as a fundamental principle and right at work

Fundamental OSH Conventions

Occupational Safety and Health Convention, 1981 (No. 155)

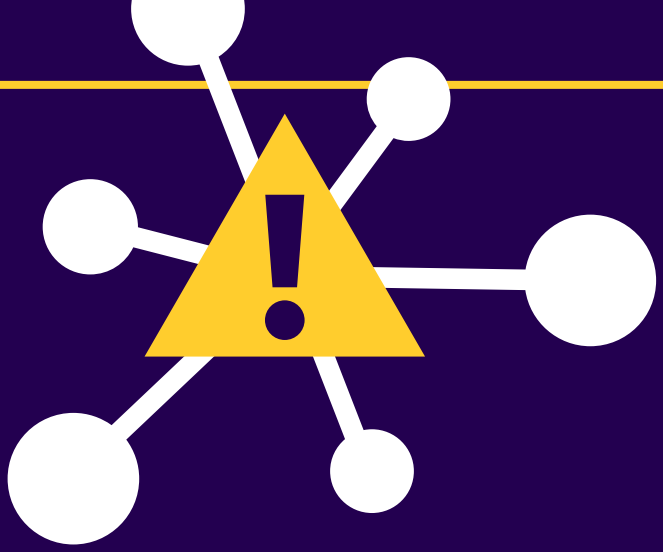
- **Physical and mental** elements affecting health (Art. 3(e))
- Adaptation of machinery, equipment, **working time, organisation of work and work processes** to the physical and mental capacities of the workers (Art. 5(b))
- Workplaces, machinery, equipment and **processes** under employers' control to be safe and without risk to health, so far as is reasonably practicable (Art. 16(1))

Occupational Safety and Health Recommendation, 1981 (No. 164)

- Prevention of harmful **physical or mental stress** due to conditions of work (Para. 3(e))
- **Work organisation**, particularly with respect to **hours of work and rest breaks**, to not adversely affect OSH (Para. 10(f))
- Measures to eliminate excessive **physical and mental fatigue** (Para. 10(g))

Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187)

- Continuous improvement of OSH & promotion of a preventive safety and health culture (Art.2)



Related International Labour Standards

Occupational Health Services Convention, 1985 (No. 161) & Recommendation (No. 171)

- Preventive functions and advisory role for establishing and maintaining a safe and healthy working environment which will facilitate optimal **physical and mental health** in relation to work; the adaptation of work to the capabilities of workers in the light of their state of physical and mental health (Art. 1)

List of Occupational Diseases Recommendation, 2002 (No. 194)

- Recognition of **PTSD** and **work-related mental disorders** (Annex 2.4)

Violence and Harassment Convention, 2019 (No. 190) & Recommendation (No. 206)

- Explicit reference to **psychosocial risks** (Art. 9(b))
- Right of everyone to a world of work free from violence and harassment (Art. 4(1))
- Preventive and protective obligations, including take into account **violence and harassment and associated psychosocial risks** in the OSH management (Art. 9)
- Risk assessments to consider **psychosocial hazards and risks arising from working conditions and arrangements, work organization and human resource management**; and arising from **discrimination, abuse of power relations, and gender, cultural and social norms** that support violence and harassment (Art. 9 and Para. 8)

Labour Inspection Convention, 1947 (No. 81)

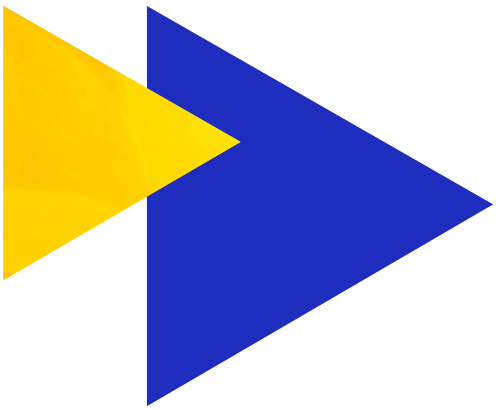
- Enforcement of legal provisions relating to conditions of work and the protection of workers; technical information and advice to employers and workers (Art. 3(1))

The **Labour Inspection Convention, 1947 (No. 81)** also forms an essential element of this framework. Pursuant to Article 3, the main functions of the labour inspection system are to secure the enforcement of legal provisions relating to conditions of work and the protection of workers whilst engaged in their work;[88] to supply technical information and advice to employers and workers on the most effective means of complying with these provisions; and to bring to the notice of the competent authority any defects or abuses not covered by existing legislation. The accompanying **Recommendation No. 81** reinforces these functions by providing guidance on the organization, methods and resources of labour inspection services, as well as the training and competence required of inspectors.

Within its normative mandate, the ILO is engaged in ongoing **standard-setting discussions**, including deliberations at the International Labour Conference in 2025 and 2026 on possible international labour standards concerning decent work in the platform economy. These deliberations concern conditions of work for platform workers, including the role of digital and algorithmic management of work, subject to the outcome of the standard-setting process.[89]

Standard-setting activities also extend to the statistical domain. The ILO has launched a consultative revision of the 1998 OSH statistical standard, with a view to adoption at the 22nd International Conference of Labour Statisticians (ICLS) in 2028. This work seeks to improve the availability, quality and comparability of OSH statistics, which is particularly important for understanding psychosocial hazards and work-related mental health, where definitions, reporting practices and data sources remain uneven across countries (ILO 2025d). Furthermore, the development and piloting of methodological guidelines and survey tools on statistics concerning Work-related Violence and Harassment are also expected to be presented for endorsement at the ICLS in 2028 (ILO 2025d; ILO 2024b).

In addition to its normative instruments, the ILO has developed a range of guidance and awareness-raising materials addressing organizational and psychosocial factors that may impact both the physical and mental health and well-being of workers. These materials support the implementation of OSH policies and practices at both national and workplace levels by governments, employers, workers' representatives and occupational health services.[90]



88 Conditions of work under the Convention No. 81, are not limited to OSH and may, depending on national legislation, encompass a range of other employment conditions. Article 3(1)(a) provides that labour inspection systems are responsible for securing the enforcement of legal provisions relating to conditions of work and the protection of workers while engaged in their work, including, inter alia, provisions relating to hours of work, wages, safety, health and welfare, the employment of children and young persons, and other connected matters, insofar as such provisions are enforceable by labour inspectors. The scope and institutional organization of labour inspection functions are determined in accordance with national laws and practice.

89 A compendium of resources regarding the platform economy from the ILO includes recent publications to support upcoming standard-setting procedures. These resources are available at: https://www.ilo.org/digital-labour-platforms#-publications.

90 A list of selected ILO publications and resources addressing psychosocial risks at work, including guidance, tools and awareness-raising materials, is provided in the Annex 1.

New Global Policy Review on work-related psychosocial hazards and future directions

In 2025, a global survey conducted in consultation with the ILO gathered inputs from national experts in 70 countries across diverse regions[91] and income groups (Potter et al. forthcoming). The survey assessed whether countries have measures addressing work-related psychosocial hazards and examined the consistency of expert responses within countries as an indication of awareness and shared understanding.

Presence of OSH law(s)

Overall, 61 countries (87 per cent) had at least one expert reporting that OSH legislation addressing work-related psychosocial hazards exists (including partial provisions). At the same time, 21 countries (30 per cent) had at least one expert reporting that such legislation does not exist and 13 countries (19 per cent) showed mixed responses (both "yes" and "no" within the same country). This pattern suggests differences in awareness or interpretation of the scope of existing legislation.

Presence of nation-wide and/or sector-oriented initiatives

There are 56 countries (80 per cent) who had at least one expert reporting the existence of national or sectoral initiatives (programmes, strategies, frameworks, or campaigns) addressing work-related psychosocial risks. However, in 51 countries (73 per cent), at least one expert did not identify such initiatives, and in 38 countries (54 per cent) responses were mixed, indicating uneven awareness or recognition of such initiatives within countries.

Looking specifically at nation-wide initiatives, 52 countries (74 per cent) had at least one expert reporting national-level programmes or strategies. At the same time, in 35 countries (50 per cent) nº such initiatives were identified by at least one expert, and in 20 countries (29 per cent) expert views differed, indicating scope for stronger coordination, visibility and communication of existing measures.

Presence of national surveys or studies

In 43 countries (61 per cent), at least one expert reported the existence of national surveys or studies assessing the prevalence and impact of work-related psychosocial hazards or risks. However, in 31 countries (44 per cent) these were not reported by at least one expert, while 14 countries (20 per cent) showed mixed responses. These findings suggest that national evidence systems remain uneven and not consistently recognised. Clearer definitions, improved dissemination of existing data sources and strengthened capacity for monitoring psychosocial risks are therefore needed.

The need for actionable evidence

To support the ongoing revision of the OSH statistical standard and inform deliberations at the 22nd ICLS in 2028, a questionnaire on national OSH statistical practices was circulated to national OSH statistics focal points in 2025.

Among 111 responding institutions, 41 (37 per cent) reported concrete plans to improve OSH statistics over the next five years in the area of psychosocial risks and work-related mental health conditions (ILO 2025d). This points to growing efforts to develop more robust, comparable and policy-relevant data to guide prevention and monitor progress.

91 Countries across all ILO regions included as follows: Africa: Egypt, Eswatini, Ghana, Kenya, Liberia, Mauritius, Morocco, Mozambique, Namibia, Nigeria, South Africa, Sudan, Tunisia, Uganda and Zimbabwe; Americas: Argentina, Barbados, Brazil, British Virgin Islands, Canada, Chile, Colombia, Costa Rica, Jamaica, Mexico, Paraguay, Peru, Saint Lucia, Trinidad and Tobago, United States and Bolivarian Republic of Venezuela; Arab States: Jordan, Qatar, Saudi Arabia and United Arab Emirates; Asia and the Pacific: Australia, Bangladesh, Brunei Darussalam, China, Fiji, India, Indonesia, Japan, Kazakhstan, Malaysia, Nepal, New Zealand, Philippines, Republic of Korea, Sri Lanka and Thailand; and Europe and Central Asia: Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Cyprus, Denmark, Finland, France, Germany, Greece, Italy, Ireland, Lithuania, Luxembourg, Kingdom of the Netherlands, North Macedonia, Norway, Poland, Portugal, Russian Federation, Slovenia, Spain, Sweden, Switzerland, Türkiye and United Kingdom and Northern Ireland.

3.2. Regional frameworks

Regional institutions have long supported cooperation and policy alignment in OSH, providing platforms through which Member States exchange approaches, develop common principles and strengthen national systems. Within this wider OSH role, several regional instruments and guidance documents refer to psychosocial risks and mental health, typically by incorporating them into general OSH duties, sector-specific strategies or practical tools. These contributions help promote shared understandings of issues such as work organization, stress, violence and harassment, and offer reference points for countries seeking to reinforce the preventive dimensions of their national OSH frameworks.

In **the EU**,[92] references to psychosocial risks appear within the broader legal and institutional landscape governing OSH. These are underpinned by articles three and 31 of the *Charter of Fundamental Rights of the European Union,* which provide for the right to mental integrity and protection from psychosocial hazards, including excessive working hours. Article 151 of the *Treaty on the Functioning of the European Union* commits Member States to improving working conditions, and article five of the Framework Directive 89/391/EEC requires employers "to ensure the safety and health of workers in every aspect related to the work", including those linked to task design, organization of work and social relations. Complementary directives also address psychosocial dimensions and include reference to various issues. For example, reference to "mental stress" is found in Directive 90/270/EEC on the minimum safety and health requirements for work with display screen equipment, and in Directive 92/85/EEC on the introduction of measures to encourage improvements in the safety and health at work of pregnant workers and workers who have recently given birth or are breastfeeding. Reference to psychosocial risks is found in Directive 2010/32/EU implementing the Framework Agreement on prevention from sharp injuries in the hospital and healthcare sector. More recently, reference to psychological harm is found in Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonized rules on artificial intelligence. Whilst Directive (EU) 2024/2831 on improving working conditions in platform work includes reference to psychosocial risks, mental health, stress, anxiety, violence and harassment. Several other complementary directives address psychosocial hazards, such as long or irregular working hours[93] and job insecurity.[94] Additionally, the resolution 2022/C 347/10 adopted by the European Parliament has called on the European Commission to adopt a new directive on psychosocial risks and well-being at work.[95] Social partner instruments, notably the framework agreements on work-related stress (2004) and on harassment and violence at work (2007), offer further guidance implemented through national social-dialogue processes. EU-OSHA initiatives, including tools for psychosocial risk assessment and sector-specific guidance, support employers – particularly small and medium-sized enterprises – in addressing these issues within the risk-prevention cycle.

In **the Americas**, the 1998 Social and Labour Declaration adopted by the Southern Common Market (MERCOSUR) recognizes workers' right to protection of their physical and mental health and calls on Member States to establish and update OSH policies aimed at preventing occupational accidents and diseases.[96] Building on this, the *Prevención de Riesgos Psicosociales en el Trabajo* recommendation Nº 04/23 urges Member States to incorporate psychosocial risk prevention into national OSH policies and programmes; develop assessment and management tools; incorporate gender considerations into the eval-

92 The EU-27 Member States are Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and Sweden.

93 Council Directives 2002/15/EC, 2003/88/EC, and 2000/79/EC.

94 Council Directive 2008/94/ECV and 98/59/EC. Additional issues considered in further complementary directives include workplace stress during pregnancy (Council Directive 92/85/EEC), mental stress from screen use (Council Directive 90/270/EEC), young workers' vulnerabilities (Council Directive 94/33/EC), and equal treatment and non-discrimination (Council Directive 2000/78/EC).

95 A study by the European Parliament Employment Committee specifies that a new directive on psychosocial risks and well-being at work should: distinguish between "psychosocial risks" and "mental health"; refer to psychosocial risks in concrete and specific terms; address psychosocial risks related to telework and digitalization of workplaces; and recognize the right to disconnect (Makarevičienė et al. 2023).

96 Current MERCOSUR Member States are Argentina, Bolivia (Plurinational State of), Brazil, Paraguay, Uruguay, and Venezuela (Bolivarian Republic of) (suspended at the time of writing).

uation of psychosocial risks and related actions; strengthen labour inspector training; promote employer action; and disseminate related rights and obligations. Measures that address psychosocial risks were also included in the *Plan Regional de Salud y Seguridad de las Trabajadoras y los Trabajadores en el MERCOSUR (2023-2024)*: specifically, the development of common guidelines; comparison of national legislations on the prevention of psychosocial risks; provision of workplace rights and obligations; and training to integrate psychosocial risks prevention in OSH systems. In the Andean Community,[97] the Andean Instrument on Safety and Health at Work (Decision No. 584) defines occupational health as encompassing physical, mental and social well-being and identifies organizational, ergonomic and psychosocial factors of the work environment as relevant to risk management. In addition, regional public-health strategies developed by the Pan American Health Organization identify the workplace as a priority setting and encourage Member States to address psychosocial risks, such as violence and poor organizational conditions, within national workers' health and mental health policies. The *Guía de Prevención de Riesgos Psicosociales en el Trabajo de Plataformas Digitales de Reparto (2025)* builds on these regional frameworks by focussing on emergent psychosocial risk within the growing platform economy of Latin America. The guide was produced by the Alianza del Pacífico (Chile, Colombia, Mexico, Peru) with Argentina and Uruguay, and identifies risk factors related to the organization of work, dependence on algorithms, job insecurity, social isolation, and exposure to violence. It also recommends counter measures including legal recognition of platform work, regulation of working hours, access to social protection, and promotion of mental health.

In the **Asia-Pacific region**, the Asia-Pacific Economic Cooperation (APEC)[98] Digital Hub for Mental Health's *White Paper on Workplace Mental Health and Safety* sets out shared concepts for promoting mental well-being at work, including attention to organizational factors, early identification of concerns and supportive workplace practices. The APEC *Roadmap to Promote Mental Wellness in a Healthy Asia Pacific (2021-2030)* similarly highlights the workplace as an important setting for mental health promotion and encourages economies to exchange good practices and develop guidance that supports preventive approaches. More broadly, the *Guidelines on Occupational Safety and Health (OSH) Risk Management for Small and Medium Enterprises in ASEAN (2020)* Member States[99] generally offer only a high-level reference point, with psychosocial hazards indicated as one possible element of workplace risk assessment.

In **Africa**, psychosocial risks have been addressed primarily through sector-specific strategies and OSH guidance. The African Union's[100] COVID-19 related guidance for health workers recommends including psychosocial hazards in risk assessments and addressing them through national and workplace programmes.[101] In the education sector, the *Continental Strategy on Mental Health and Psychosocial Support for Teachers in Africa (2025)* provides a regional framework for creating enabling school environments, strengthening mental health literacy, improving early identification and referral, and supporting teachers' return to work following mental health difficulties.

97 Current Andean Community Member States are Bolivia (Plurinational State of), Colombia, Ecuador and Peru.

98 Current APEC Member Economies are Australia, Brunei Darussalam, Canada, Chile, China, Chinese Taipei, Hong Kong (China), Indonesia, Japan, Malaysia, Mexico, New Zealand, Papua New Guinea, Peru, Philippines, Republic of Korea, Russian Federation, Singapore, Thailand, United States, Viet Nam.

99 Current Association of Southeast Asian Nations (ASEAN) Member States are Brunei Darussalam, Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Viet Nam.

100 Current African Union Member States are Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Democratic Republic of the Congo, Djibouti, Egypt, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Côte d'Ivoire, Kenya, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sahrawi Arab Democratic Republic, São Tomé and Príncipe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, Sudan, United Republic of Tanzania, Togo, Tunisia, Uganda, Zambia and Zimbabwe.

101 African Union Development Agency (AUDA-NEPAD), Health and Wellness of Health Workers, Volume 3, AU COVID-19 OSH Guidelines Series, 2020.

3.3. National frameworks

National OSH policies and programmes

An increasing number of countries now address issues related to the psychosocial working environment and mental health concerns within their **OSH policies and programmes**. Supported by a substantial international evidence base (Boot, LaMontagne, and Madsen 2024; Leka and Jain 2024), this policy evolution marks a shift from a traditional focus on physical hazards toward a more integrated approach to working conditions and overall well-being at work.

In Asia, Japan's *Fourteenth Occupational Safety and Health Program (2023-2027)* identifies the prevention of psychosocial hazards including overworking and mental health problems as national priorities, supported by workplace stress assessments and employer guidance. In the Republic of Korea, the *Employment and Labour Policy (2024)* calls for measures to prevent work-related mental stress, including the mandatory establishment of workplace rest facilities while providing specialized support via nationwide Workers' Health Centres and Occupational Trauma Centres, including counselling for workers exposed to traumatic incidents. Preventive approaches to sexual harassment and gender-based discrimination are also promoted through monitoring and training.

In Latin America and the Caribbean, psychosocial hazards are frequently included in national OSH polices. Chile's most recent OSH policy promotes protective psychosocial factors, [102] and builds on its previous policy which formally recognized psychosocial risks and encouraged the use of validated tools and organizational interventions.[103] Uruguay's OSH policy recognizes safe and healthy work environments as a fundamental right at work, which must guarantee physical and psychosocial integrity.[104] Across the region, a range of other national policies promote efforts to prevent and mitigate psychosocial hazards, often alongside measures to raise mental health awareness, including those of Colombia,[105] Ecuador,[106] Grenada,[107] Guyana,[108] Peru[109] and Saint Lucia.[110] Costa Rica's 2021-2026 plan further aims to strengthen the evidence base for action through a comprehensive national survey on working conditions that includes psychosocial factors. [111]

Emerging developments in Africa point in the same direction, with several countries beginning to recognize psychosocial hazards as elements of national OSH priorities. Namibia's national OSH policy highlights workplace health-promotion efforts that address psychosocial issues, [112] whilst Nigeria's national OSH policy explicitly prioritizes the prevention and management of psychosocial problems such as mental stress, violence and substance abuse. [113] The revised *national Occupational Safety and Health Policy of Mauritius (2015)* also incorporates, for the first time, provisions for identifying and managing workplace psychosocial hazards.[114] Similarly, South Sudan's national OSH policy[115] defines psychosocial risk and recognizes stress-related and mental health disorders as occupational outcomes.

In Europe, this trend has been reinforced by the *European Union Strategic Framework on Health and Safety at Work 2021-2027*, which calls on Member States to prioritize mental health and the prevention of psycho-

102 Política Nacional de Seguridad y Salud en el Trabajo, 2024.
103 Protocolo de Vigilancia de Riesgos Psicosociales en el Trabajo, 2013.
104 Decreto N°246/024 relativo a la fijación de una política nacional de seguridad y salud laboral, prevención de riesgos laborales, conforme a lo establecido por la OIT.
105 ABC del Plan Nacional de Seguridad y Salud en el Trabajo (2022–2031).
106 Política Nacional de Salud en el Trabajo (2019 – 2025).
107 National Occupational Safety and Health Policy (2023).
108 National Policy on Occupational Safety and Health.
109 Política de Seguridad y Salud en el Trabajo (2030).
110 National Occupational Safety and Health (OSH) Policy for Saint Lucia.
111 Plan de Acción de la Política Nacional de Salud Ocupacional (2021 – 2026).
112 National Occupational Safety and Health Policy (2021).
113 National Policy on Occupational Safety and Health (2020).
114 National Occupational Safety and Health policy – Mauritius (revised 2015).
115 National Occupational Safety and Health Policy (2022)..

social risks. This orientation is reflected in the strategies of several countries, including Finland,[116] where national policy emphasizes the proactive management of mental workload and work ability, and France,[117] where the prevention of psychosocial risks is identified as a key objective, as well as in the OSH strategies of Spain,[118] Belgium[119] and Denmark,[120] which set specific targets and indicators on psychosocial risk assessment, prevention and awareness-raising.

A complementary trend can be observed in **broader national policies** on mental health and well-being, which increasingly recognize the workplace as a key setting for prevention and promotion. This apg proach is evident in national mental health strategies such as *Sharing the Vision: A Mental Health Policy for Everyone (2020-2030)* in Ireland, France's *Stratégie nationale de santé mentale et de psychiatrie* (2018-2022), and *Kia Manawanui Aotearoa – Long-term Pathway to Mental Well-being* (2021) in New Zealand, each of which emphasises the role of safe, supportive and inclusive workplaces in promoting mental well-being. Comparable approaches are found elsewhere: South Africa's *National Mental Health Policy Framework and Strategic Plan (2023-2030)* and Chile's *Plan Nacional de Salud Mental* (2017-2025) both include commitments to address work-related psychosocial factors through coordination between health and labour authorities. Some national frameworks, such as Australia's *National Mental Health and Suicide Prevention Plan* (2021), extend this approach by identifying mentally healthy workplaces as a cross-sectoral priority,[121] linking preventive and early-intervention measures to employment and industry policy. Likewise, several suicide-prevention strategies – including the United Kingdom's *Suicide Prevention Strategy for England* (2023-2028) and Canada's *Federal Framework for Suicide Prevention* (2016) – explicitly recognise workplace stress and employer engagement as key components of national prevention efforts.



116 Safe and healthy working conditions and workability for everyone (2030).

117 Plan Santé au Travail n°4 2021–2025.

118 Spanish Strategy on Occupational Safety and Health (2023-2027).

119 Plan d'action national pour l'amélioration du bien-être des travailleurs lors de l'exécution de leur travail (2022-2027).

120 A strategy for working environment efforts up to 2020.

121 The Mentally Healthy Workplaces national digital hub is endorsed by the Australian government and identifies a mentally healthy workplace as one which "actively minimises risks to mental health, promotes positive mental health and well-being, is free from stigma and discrimination and supports the recovery of employees with mental health conditions" (LaMontagne et al. 2014; National Mental Health Commission, n.d.).

Strengthening data collection for evidence-based policies and action on psychosocial risks

Evidence-based OSH policies and preventive action depend on reliable data to identify exposure patterns, set priorities, target measures and monitor whether interventions reduce harm over time. For psychosocial risks in particular – where outcomes are multifactorial and often underreported – this requires combining data from multiple sources, including working-conditions surveys, establishment surveys and administrative records (such as labour inspection, occupational health services and compensation data), supported by trusted and confidential reporting arrangements.

Where routine data-collection systems are limited, psychosocial risks tend to be addressed through ad hoc studies or isolated incidents, constraining the ability of governments and social partners to assess trends, compare groups or evaluate policy impact. Strengthening regular, policy-relevant data collection is therefore a key enabler of effective psychosocial risk governance.

At the international level, as noted earlier in this report, the ILO is undertaking a multi-year consultative revision of the international OSH statistical standard, with a view to adoption at the 22nd ICLS in 2028. This process aims to improve the availability, quality and comparability of data on psychosocial hazards and work-related mental health, where definitions and reporting practices remain uneven across countries (ILO 2025d).

At regional and national levels, harmonized survey instruments play a critical role. In Europe, periodical surveys such as the European Working Conditions Survey (EWCS), the EU Labour Force Survey (EU-LFS), the European Survey of Enterprises on New and Emerging Risks (ESENER), and the OSH Pulse provide a consistent evidence base for comparing psychosocial risk exposure and prevention practices across countries. National initiatives further strengthen this foundation: for example, Spain's National Institute for Safety and Health at Work (INSST) integrates psychosocial indicators into national working-conditions surveys and enterprise-level assessments. Beyond Europe, countries such as *Costa Rica* have introduced comprehensive national surveys on working conditions that explicitly include psychosocial risks, supporting more targeted and evidence-based OSH policy development.[122]

OSH laws and regulations

Across regions, OSH legislation increasingly recognizes that workers' health encompasses both physical and mental dimensions. In many jurisdictions, labour and OSH laws explicitly define health in this way or impose a general duty on employers to protect workers' physical and mental health. Several countries make this explicit by placing a direct duty on employers to protect workers' physical and mental health as part of their OSH responsibilities. This approach appears in countries including Argentina,[123] Australia,[124] Finland,[125] New Zealand,[126] the Philippines,[127] and Togo.[128] In some systems, this link is reinforced through provisions requiring employers to organize or adapt work in line with workers' physical and mental capacities, as seen in Brazil,[129] Estonia,[130] France,[131] Finland,[132] Norway[133] and Sweden.[134]

122 The Ministry of Labour and Social Security of Costa Rica is developing a new method to evaluate psychosocial risk factors in workplaces (the MEPS-CSO tool), aimed at helping companies identify and manage risks that affect workers' well-being, health, and productivity. As part of this process, the Ministry implemented a national survey on psychosocial factors at work from 4 June to 4 July 2025, with participating workplaces receiving results and recommendations in 2026.

123 Ley de Higiene y Seguridad en el Trabajo, Law No. 19,587 of 21 April 1972.

124 Johnstone, R., and Way K., "The evolution of regulation to prevent psychosocial harm in Australia", *Australian Journal of Labour Law,* 37, No.3 (2024): 271-294.

125 Työturvallisuuslaki, Act No. 738/2002 of 23 August 2002 and Työterveyshuoltolaki, Act No. 1383/2001 of 21 December 2001.

126 Health and Safety at Work Act 2015, Public Act No. 70 of 4 September 2015.

127 Department Order No. 208, Series of 2020, Guidelines for the Implementation of Mental Health Workplace Policies and Programs for the Private Sector, 11 February 2020.

128 Code du travail, Loi du 15 juin 2021.

129 Norma Regulamentadora n.º 1 – Disposições Gerais e Gerenciamento de Riscos Ocupacionais, 27 de agosto de 2024.

130 Occupational Health and Safety Act, 16 June 1999.

131 Code du Travail.

132 Occupational Safety and Health Act, 2002 (as amended).

133 Act No. 62 of 17 June 2005 respecting working environment, working hours and employment protection (Working Environment Act).

134 Svensk författningssamling, Work Environment Act (1977:1160).

Key areas covered by national laws and regulations on psychosocial risks



General duty to protect physical and mental health

Employer duty of care, to protect both physical and mental health

Argentina, Australia, Finland, New Zealand, Philippines, Togo

Integration of psychosocial risks into OSH prevention

Identification of psychosocial hazards; risk assessment; preventive measures within work organization and management.

Belgium, Estonia, Finland, France, Germany, Norway, Spain, Sweden

Dedicated regulations on psychosocial risks

Specific legal definitions, assessment tools, and structured prevention procedures.

Belgium, Chile, Colombia, Mexico

Monitoring and health surveillance

Periodic examinations and surveillance of psychosocial risk exposure.

Belgium, Colombia, Italy, Japan, Republic of Korea

Grievance and enforcement mechanisms

Complaint procedures, burden of proof approaches, labour inspection and preventive enforcement.

Australia, Belgium, Chile, Colombia, France

Recognition and compensation of work-related mental disorders

Inclusion of PTSD and other mental disorders in occupational disease national lists, through listed, open or mixed models (including case-by-case recognition).

Australia, Canada, Chile, Italy, Japan, Kingdom of the Netherlands, Portugal, Republic of Korea

Violence and harassment within OSH frameworks

Recognition of violence and harassment as occupational risks; preventive duties and risk assessment.

Brazil, Canada, Chile, Colombia, France, Lesotho, Spain

Digitalization and telework

Regulation of psychosocial risks linked to remote work, digital overload, algorithmic management and right to disconnect.

Argentina, Belgium, Chile, France, Italy, Japan, Portugal, Spain

The growing attention to mental health within OSH legislation has underscored the importance of addressing the work-related factors that shape it. Framing workplace issues primarily in terms of mental health outcomes risks shifting attention towards individualized responses, which are often reactive or remedial, rather than preventing the underlying organizational and psychosocial hazards (Harvey et al. 2017; LaMontagne et al. 2014; Lerouge 2025).

An effective **preventive approach** within OSH regulation therefore involves addressing psychosocial hazards alongside other occupational hazards through risk assessment and preventive action. From this perspective, psychosocial risks are managed through the same preventive logic that underpins OSH regulation more broadly, including the identification of hazards, assessment of risks and implementation of preventive measures within work organization and management. As psychosocial risks are linked not only to mental health outcomes but also to a range of physical health effects, this reinforces the importance of addressing their organizational sources. Evidence indicates that when psychosocial risks are explicitly addressed in legislation, workplaces are more likely to conduct risk assessments and develop preventive action plans, strengthening the practical implementation of OSH prevention requirements (Jain et al. 2022; Lerouge 2025).

Against this background, many countries have progressively reflected psychosocial factors, hazards or risks more explicitly in their legislation, although the form and level of detail vary substantially. The scope of these legislative provisions also varies, with some categories of workers – such as the self-employed, informal economy workers and, in some cases, public sector workers – remaining partially or fully outside OSH coverage in a number of countries.

Historically, the Nordic countries were among the first to integrate the psychosocial dimension of work into OSH legislation, embedding it within general preventive duties rather than treating it as a separate regulatory domain. Sweden's *Work Environment Act of 1977* already required employers to organize work so that it did not cause undue physical or mental strain, and subsequent binding regulations progressively expanded obligations relating to the organizational and social work environment. Finland incorporated these dimensions in 2002, when its Occupational Safety and Health Act introduced explicit requirements to manage mental workload and the psychosocial work environment as part of overall risk prevention. In Iceland, amendments adopted in 2003 provide for the implementation of systematic preventive measures and to assess the psychosocial aspects of the working environment.[135] Norway followed in 2005 with a revised *Working Environment Act* that explicitly addressed the psychosocial working environment. An amendment adopted in 2026 further clarified preventive requirements in relation to psychosocial factors, including workload and time pressure, emotional demands, role clarity and support at work.[136]

Belgium was among the first countries outside the Nordic region to establish a comprehensive legal framework on psychosocial risks. Building on measures introduced in the 1990s and consolidated through the *2014 Law on Well-being at Work and its Royal Decre*e, Belgian legislation introduced a statutory definition of psychosocial risks and a structured prevention approach centred on systematic risk assessment using defined procedures.[137]

Other European countries have progressively incorporated psychosocial risks into their OSH legal frameworks, often through integration into general risk-assessment and prevention requirements.[138] Estonia OSH legislation requires the identification of psychosocial hazards alongside physical hazards, and the assessment and management of associated risks, including risks related to workload, work pace and organizational practices.[139] Germany's OSH Act includes psychological stress within mandatory risk assessment,[140] clarified through national guidance addressing factors such as time pressure, emotional demands and role conflicts. [141] In France, OSH and labour law provisions require the assessment of risks arising

135 Act on Working Environment, Health and Safety in Workplaces, No. 46/1980, amended 2003.

136 Section 4-3 of The Working Environment Act (LAW-2005-06-17-62), amended 2026.

137 Law of 28 February 2014 and Law of 28 March 2014 amending the Act of 4 August 1996 on Well-being at Work, and Royal Decree of 10 April 2014 on the prevention of psychosocial risks at work.

138 The European Commission conducted extensive peer reviews in 2019 and again in 2024, examining how EU Member States address psychosocial risks through their national practices, legislative frameworks, and enforcement mechanisms. These reviews generated a robust evidence base and were undertaken with the intention of helping shape future EU-level action, including consideration of a dedicated Directive on psychosocial risks

139 Occupational Health and Safety Act, 16 June 1999.

140 Occupational Health and Safety Act (ArbSchG), Federal Law Gazette I p. 1246.

141 Consideration of psychosocial factors in risk assessment: Recommendations for implementation in business practice.

from work organization and related psychosocial factors,[142] complemented by a structured framework for preventing harassment.[143] Spain similarly requires evaluation of risks linked to monotony, mental strain and work organization.[144] Comparable approaches are found in other European countries, including Italy,[145] the Kingdom of the Netherlands[146] and Portugal,[147] where psychosocial aspects are addressed through requirements related to work organization and related factors.

Outside Europe, several countries in the Americas have adopted specific regulations or detailed provisions to address psychosocial risks as part of OSH and labour protection, including within employers' general responsibilities for risk prevention. Colombia, adopted a dedicated OSH resolution in 2008 establishing definitions, assessment procedures and preventive measures for psychosocial risks.[148] The *Resolución 2764* builds on this by adopting a validated battery of instruments[149] and detailed technical protocols for the undertaking of duties,[150] whilst *Resolución 5137 de 2024* extends psychosocial risk prevention to informal workers with new tools and mental-health actions.[151] Mexico introduced a binding standard defining psychosocial risks and addressing factors such as workload, work schedules, organizational practices and workplace violence.[152] In Brazil, legislation requires the evaluation of psychosocial risks, and the assessment of the effectiveness of preventive measures.[153] Chile requires the preventive management of occupational risks, including organizational factors,[154] and applies a specific mandatory OSH protocol on psychosocial risk assessment, based on standardized tools and preventive action plans.[155] Costa Rica integrates psychosocial factors directly into its OSH regulations, linking them to safe work organization and preventive risk management.[156] Ecuador has recently incorporated psychosocial risks into technical OSH regulations.[157] Other countries in the region address psychosocial risks and their outcomes through wider OSH or labour law obligations. For example, Argentina requires examining psychological factors influencing occupational accidents and diseases,[158] and Uruguay mandates work-system design that addresses workers' physiological, psychological and social well-being.[159]

A similar trend is visible in the Asia-Pacific region, where several countries have progressively addressed psychosocial risks in their OSH laws and regulations. In Australian jurisdictions,[160] psychosocial hazards are explicitly recognized in OSH regulations, covering factors such as high job demands, low control, role conflict, violence and harassment, organizational change and exposure to traumatic events. In the Republic of Korea, legislative reforms have strengthened preventive approaches addressing work organization, excessive working hours and job stress.[161] In Japan, the OSH framework combines preventive measures on working time, and work organization with mandatory psychosocial risk monitoring through an annual workplace stress check system that now includes workplaces with less than 50 employees.[162] Malaysia has

142 Code du travail, Article L4121-3.
143 Code du travail, Chapitre II : Harcèlement moral. (Articles L1152-1 à L1152-6).
144 Ley de Prevención de Riesgos Laborales, BOE-A-1995-24292 Ley 31/1995, de 8 de noviembre.
145 Decreto Legislativo n. 81/2008, 9 aprile 2008..
146 Working Conditions Act (Arbowet) and Working Conditions Regulation (Arbeidsomstandighedenbesluit).
147 Lei n.º 102/2009, de 10 de setembro. Diário da República, Série I, n.º 176.
148 Resolución 2646 de 2008.
149 Riesgo Psicosocial.
150 Resolución 2764 de 2022.
151 Resolución 5137 de 2024..
152 NOM-035 -STPS-2018, Factores de riesgo psicosocial en el trabajo – Identificación, análisis y prevención.
153 Portaria nº 1.419, de 27 de agosto de 2024.
154 Decreto N° 44 de 2024 que aprueba el nuevo reglamento sobre gestión preventiva de los riesgos laborales para un entorno de trabajo seguro y saludable.
155 Resolución Exenta N° 1448, 2022.
156 Código de Trabajo. Ley No. 2 de 26 de agosto de 1943, con reformas.
157 Norma Técnica en Seguridad e Higiene del Trabajo.
158 Ley N° 19.587 de Higiene y Seguridad en el Trabajo (1972).
159 Decreto 291/007.
160 Examples include, but are not limited to, Victoria, Occupational Health and Safety (Psychological Health) Regulations 2025, Statutory Rule No. 103/2025; New South Wales, Environmental Planning and Assessment (Infrastructure Contributions) Amendment Regulation 2025; and Work Health and Safety (Psychosocial Risks) Amendment Regulations 2023 (No. 92 of 2023) (South Australia), under the Work Health and Safety Act 2012.
161 Occupational Safety and Health Act (as amended up to Act No. 17433 of 9 June 2020).
162 Act on Promotion of Preventive Measures against Karoshi (2014); Industrial Safety and Health Act (ISHA); and Outline of the Act Partially Amending the Industrial Safety and Health Act and the Working Environment Measurement Act.

also amended its OSH legislation to integrate psychological aspects and work organization into general OSH risk-prevention frameworks.[163]

In Africa and the Arab States, the integration of psychosocial risks into OSH legislation remains limited. Most OSH frameworks rely on general provisions on the protection of workers' physical and mental health, without detailed or specific regulatory treatment of psychosocial risks. Nonetheless, some countries have begun to introduce more explicit references. In Ethiopia, the Labour Proclamation includes psychological hazards among the categories of occupational hazards addressed within OSH prevention frameworks, alongside physical, ergonomic, biological and chemical hazards.[164] In Jordan, ministerial instructions under the labour and OSH regulatory framework identify psychosocial hazards as one category of occupational risks to be considered within risk assessment processes.[165]

Across the world, many countries have addressed violence and harassment in the world of work through labour law, OSH legislation or dedicated legal instruments. In a growing number of cases, violence and harassment are also explicitly recognized within OSH prevention frameworks, reflecting their close links with psychosocial risks. Examples include OSH or OSH-linked provisions addressing workplace violence and harassment in Brazil,[166] Canada,[167] Chile,[168] Colombia,[169] Ecuador,[170] France,[171] Peru,[172] and Spain.[173] More recent legislative developments have further embedded violence and harassment within OSH law, such as the Occupational Safety and Health Act adopted in Lesotho in 2024.[174] In other countries, prevention relies on combined approaches in which OSH provisions interact with labour and equality legislation to support risk assessment and preventive action. [175]

While the approaches described above rely on binding OSH legislation to address psychosocial risks, including violence and harassment, some countries have adopted a different regulatory technique. In these cases, authorities have chosen to introduce soft law, voluntary standards or other legally non-binding instruments, either to complement binding regulations addressing psychosocial risks or as the primary means of addressing them. Such instruments typically take the form of management standards, codes of practice or guidelines that support the application of general OSH principles to psychosocial risks (ILO 2022a). Measures of this kind have been implemented, for example, in Austria,[176] Canada,[177] Australia (Queensland)[178] and the United Kingdom.[179]

163 Occupational Safety and Health (Amendment) Act 2022.
164 Labour Proclamation No. 1156/2019.
165 Instructions for Identifying the Types of Sources of Occupational Hazards in the Work Environment and the Necessary Preventive Precautions and Measures (2023).
166 Lei nº 14.457/2022.
167 Work Place Harassment and Violence Prevention Regulations (SOR/2020-130).
168 Ley 21.643.
169 Ley 1010 de 2006 and Ley 2365 de 2024.
170 Norma Técnica de Seguridad e Higiene del Trabajo.
171 Code du travail, Titre V – Harcèlements, Section : Harcèlement moral et sexuel.
172 Ley N.° 27942.
173 Ley Orgánica 10/2022.
174 Occupational Safety and Health Act, 2024. Act No. 4 of 2024.
175 For an in-depth analysis of instruments addressing violence and harassment, please refer to the existing ILO publications Safe and healthy working environments free from violence and harassment, 2020; Violence and Harassment in the World of Work: A Guide on Convention No. 190, 2021, and Preventing and addressing violence and harassment in the world of work through occupational safety and health measures, 2024.
176 Arbeitsplatzevaluierung psychischer Belastungen.
177 National Standard of Canada on Psychological Health and Safety in the Workplace: Prevention, Promotion and Guidance to Staged Implementation (2013).
178 Managing the risk of psychosocial hazards at work: Code of Practice 2022 (WorkSafe Queensland)..
179 Health and Safety Executive (HSE), Management Standards for work-related stress; and Guidance on the Management of Psychosocial Risks in the Workplace.

Psychosocial risks linked to telework and digitalization: examples of regulatory approaches

Digital transformation and the expansion of telework have prompted several countries to introduce legal provisions addressing psychosocial risks linked to new forms of work organization. In some jurisdictions, psychosocial considerations are integrated into telework regulations, including measures related to working time boundaries, constant availability and work organization. Examples include telework regulations or legislation in Argentina,[180] which address disconnection and psychosocial risk prevention; Chile,[181] which requires risk assessments adapted to remote work; and *Portugal*,[182] whose framework includes measures to prevent work intensification, isolation and excessive out-of-hours communication.

Recognizing risks associated with digital overload, a number of countries have introduced statutory or regulatory approaches commonly referred to as a "right to disconnect". For example, in Australia workers have the right to refuse unreasonable out-of-hours contact.[183] Belgium's recent reforms require enterprises to define digital disconnection policies,[184] while in *France* companies are required to negotiate disconnection mechanisms.[185] Spain's legislation establishes a general right to disconnect and requires the adoption of internal policies aimed at preventing fatigue and work-life boundary erosion.[186]

Telework regulations have also begun to intersect with concerns related to violence and harassment, including domestic violence, as remote work arrangements may affect exposure to risk and access to support, with implications for workers' safety and health. For example, Mexico's telework standard (NOM-037) explicitly includes special protection mechanisms related to domestic violence.[187]

Some OSH frameworks address the psychosocial impacts of digital tools and algorithmic management more broadly. In Finland,[188] the cognitive ergonomics checklist assesses to cognitive strain linked to information and communication technology use, while Japan's stress check system includes aspects related to mental workload and work pace.[189] In the context of platform work, Italy requires transparency regarding automated decision-making systems and mandates risk assessments covering algorithmic controls and work intensification, while Spain requires disclosure of algorithms affecting working conditions and reinforces OSH responsibilities in relation to organizational and psychosocial risks.[190]

Overall, these developments reflect growing recognition that digitalization and new forms of work organization shape the psychosocial work environment and require adapted, process-oriented preventive approaches. Regulatory responses vary across countries, highlighting the importance of context-specific solutions and of regularly reviewing and adjusting measures as work organization, technologies and risks continue to evolve.

Other areas of labour and employment legislation are also relevant to the psychosocial work environment. These include regulations on working time, rest periods and shift work; employment protection; and anti-discrimination and equality. However, these areas are not reviewed specifically, as they fall outside the scope of this chapter, which focuses on OSH legislation.

Surveillance and monitoring mechanisms are an essential complement to preventive OSH duties, ensuring that psychosocial risks are evaluated and controlled over time. Monitoring requirements help embed continuous evaluation into OSH management systems, strengthening the practical application of

180 Ley 27.555.
181 Ley N°21.220.
182 Law No. 83/2021.
183 Fair Work Legislation Amendment Act 2024.
184 Loi sur des dispositions diverses relatives au travail.
185 Loi n° 2016-1088.
186 Ley Orgánica 3/2018 and Ley 10/2021.
187 NORMA Oficial Mexicana NOM-037-STPS-2023, Teletrabajo-Condiciones de seguridad y salud en el trabajo.
188 Cognitive ergonomics checklist.
189 ストレスチェック等の職場におけるメンタルヘルス対策・過重労働対策等.
190 Ley 39/2015.

legal requirements. Monitoring mechanisms for psychosocial hazards and risks may be embedded within general OSH risk assessment frameworks as in Spain,[191] Germany[192] and Estonia,[193] or established through dedicated regulations on psychosocial risks, as seen in Belgium[194] and Colombia[195] In addition, several OSH systems incorporate health surveillance mechanisms that include outcomes of exposure to psychosocial risks. Some countries embed psychosocial factors within broader occupational health surveillance frameworks, requiring periodic health monitoring or screening when workplace stressors are present. For example, in Japan, the Industrial Safety and Health Act[196] mandates annual stress checks for workplaces with 50 or more workers, linking individual screening results to workplace risk-reduction measures. The Republic of Korea[197] applies a similar approach, integrating job stress indicators into periodic worker health examinations and follow-up procedures under the OSH Act. In Italy, the OSH legislation provides for medical surveillance where work-related hazards may pose risks to workers' health, allowing occupational physicians to monitor health effects linked to work-related stress and mental workload when clinically relevant.[198] These surveillance mechanisms help detect early signs of harm, guide targeted preventive action and complement organizational risk-assessment duties.

Effective **grievance and complaint mechanisms** are also essential to enable workers to report psychosocial risks and, where necessary, initiate formal procedures to obtain adequate support and corrective action. Most countries rely on general OSH complaint procedures, but these systems were primarily designed for physical hazards and can be difficult to apply to psychosocial risks, which often develop cumulatively and without visible traces (Chirico et al. 2019).

The burden of proof can pose a particular challenge. Comparative legal analyses show that in systems where workers must demonstrate both the harm suffered and a direct causal link with organizational conditions, psychosocial risks are rarely the subject of formal complaints, even when exposure is widespread (Leka, Jain, and Lerouge 2017).

In light of these complexities, approaches to evidentiary requirements in complaints related to psychosocial risks vary across jurisdictions. In France, case law relating to workplace harassment has introduced a modified allocation of the burden of proof. Once workers present factual elements suggesting the existence of harassment or psychosocial harm, employers must demonstrate that the conduct in question was not harmful or that appropriate preventive measures were taken, in line with the general obligation to protect workers' safety and health.[199] In some Australian jurisdictions, preventive enforcement under OSH legislation allows labour inspectors to intervene on the basis of a reasonable belief that a risk to safety and health exists, including psychosocial risks, without the need to establish that harm has already occurred. This approach applies in particular to improvement notices and other risk-based regulatory tools, rather than criminal prosecution.[200] In other countries, the examination of complaints related to psychosocial risks is closely linked to employers' compliance with documented prevention requirements, without a formal reversal of the burden of proof. In Belgium, legislation on psychosocial risks provides for dedicated internal and external intervention procedures, under which complaints are assessed against evidence of psychosocial risk assessment and the preventive or corrective measures implemented.[201] Similarly, in Colombia[202] and Chile,[203] complaints are examined in light of mandatory psychosocial risk assessments, standardized

191 Ley 31/1995 (Ley de Prevención de Riesgos Laborales).
192 Arbeitsschutzgesetz (Occupational Safety and Health Act) (1996, §5–6).
193 Occupational Health and Safety Act (RT I, 1999, 60, 616; updated 2023).
194 Act of 4 August 1996 on the Well-being of Workers in the Performance of Their Work and Royal Decree of 10 April 2014 on Prevention of Psychosocial Risks at Work.
195 Resolution 2646 of 2008.
196 Industrial Safety and Health Act (ISHA), Act No. 57 of 1972 (2015 amendment).
197 Occupational Safety and Health Act (OSHA), amended 2020.
198 Legislative Decree No. 81/2008, Consolidated Act on Safety and Health at Work.
199 See *France, Labour Code, Art. L1154-1*. Furthermore, the *Cour de cassation*, settled case law on moral harassment and employer duty of safety.
200 Model Work Health and Safety Act 2011, ss. 90–99 (Provisional improvement notices).
201 Loi relative au bien-être des travailleurs (4 August 1996, as amended) and Arrêté royal relatif à la prévention des risques psychosociaux au travail (10 April 2014).
202 Resolución 2646 de 2008 Ministerio de la Protección Social.
203 Protocolo de Vigilancia de Riesgos Psicosociales en el Trabajo (MINSAL, 2013).

measurement tools and action plans required by binding regulations, making documentation of preventive processes central to grievance handling.

Another important dimension of legal OSH frameworks concerns the **recognition and compensation** of work-related disorders linked to psychosocial risks. Countries differ widely in how occupational diseases are defined and classified, but an increasing number have begun to include mental and stress-related conditions, either through explicit inclusion in occupational disease schedules or through mechanisms allowing recognition on a case-by-case basis (ILO 2016b; Leka, Jain, and Lerouge 2017).

Across regions, disorders caused by traumatic events, most commonly PTSD, are among the earliest and most widely recognized conditions. This reflects the fact that PTSD is typically associated with clearly identifiable incidents, such as violence, serious accidents or other traumatic events at work, which tend to satisfy evidentiary requirements more readily than conditions arising from prolonged exposure to organizational stressors (ILO 2016b).

Beyond PTSD, some jurisdictions explicitly recognize other mental health conditions within their occupational disease frameworks. Examples include Italy,[204] which recognizes chronic adjustment disorders such as anxiety and depression in addition to PTSD; Lithuania,[205] which includes occupational diseases caused by stress; the Kingdom of the Netherlands,[206] which recognizes stress-related disorders such as burnout and job-related depression; and Chile,[207] which lists disabling neurosis and other stress-related mental disorders. In some countries, mental health disorders arising from violence and harassment at work are also explicitly recognized. For example, in Portugal,[208] damages resulting from occupational illnesses arising from harassment are compensable, while in the Republic of Korea,[209] illness caused by work-related mental distress due to workplace harassment is included within the definition of occupational diseases. In a limited number of systems, statutory recognition extends beyond mental disorders to severe physical outcomes associated with prolonged psychosocial exposure. For example, in Japan,[210] the occupational disease and compensation framework includes overwork-related cardiovascular diseases, such as strokes and heart attacks linked to excessive working hours.

Beyond these forms of explicit statutory recognition, many jurisdictions rely on case-by-case recognition of health outcomes arising from exposure to psychosocial risks at work. In these systems, compensation may be granted when medical and occupational evidence shows that the disorder is predominantly or mainly caused by work. For example, whilst some jurisdictional variation exists within Australia,[211] claims are generally accepted for a medically diagnosed psychological injury when employment is the main contributing factor, and where the injury was not caused by reasonable management action; many Canadian provinces[212] recognize work-related mental disorders when work is the predominant cause; and Brazil[213] and Argentina[214] allow the recognition of stress-related disorders, including burnout or reactive depression, on a case-by-case basis when a work-related origin can be demonstrated.

Finally, many countries operate open or mixed systems that combine explicit recognition for certain conditions with open clauses which, in principle, permit the recognition of disorders whether listed or not, provided the worker can demonstrate a direct and essential causal link with work. While these systems theoretically offer broad coverage, comparative analyses indicate that the evidentiary thresholds they

204 Decreto del Presidente della Repubblica 30 giugno 1965, n. 1124 – Testo Unico per l'assicurazione obbligatoria contro gli infortuni sul lavoro e le malattie professionali, including the Tabelle delle Malattie Professionali (as periodically updated by INAIL).

205 Lietuvos Respublikos Vyriausybės nutarimas Nr. 2012-09-12 – Dėl Profesinių ligų sąrašo patvirtinimo.

206 Nederlands Centrum voor Beroepsziekten (NCvB), Registratierichtlijnen voor beroepsziekten.

207 Decreto Supremo N° 109, de 1996, del Ministerio de Salud – Reglamento sobre Enfermedades Profesionales.

208 Labour Code (Código do Trabalho, as amended in 2017), Art. 283(8-9).

209 Industrial Accident Compensation Insurance Act (as amended in 2018), Article 5; as well as Enforcement Decree and official recognition criteria for occupational diseases.

210 Industrial Accident Compensation Insurance Act (IACIA), the Recognition Criteria for Mental Disorders (MHLW, 2011; revised 2015), and List of Occupational Diseases (under IACIA).

211 Safety, Rehabilitation and Compensation Act 1988 (Commonwealth), as well as State/Territory workers' compensation acts, for example the Workers Compensation Act 1987 (New South Wales), and the Work Health and Safety Act 2011.

212 Workplace Safety and Insurance Act, 1997 (Ontario); Workers Compensation Act (British Columbia); Workers' Compensation Act (Alberta); Act Respecting Industrial Accidents and Occupational Diseases (Quebec).

213 Decreto n° 3.048, de 06 de maio de 1999 – Regulamento da Previdência Social.

214 Ley 24.557 sobre Riesgos del Trabajo (1995), and Decreto 658/1996 – Listado de Enfermedades Profesionales.

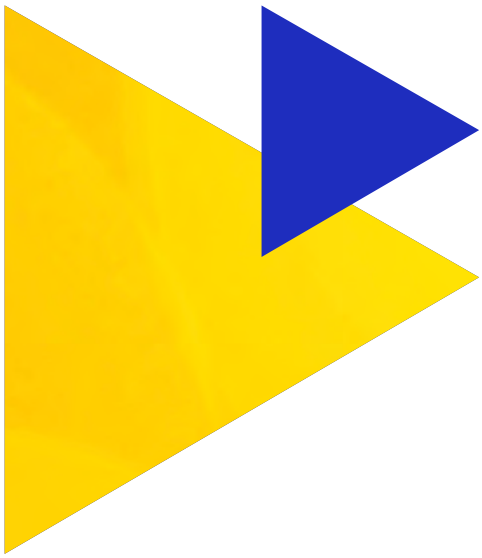
impose are often difficult to meet for multifactorial mental and stress-related disorders, as recognition typically requires proof not only of the medical diagnosis but also of its occupational origin (Lerouge 2025).

To mitigate these challenges, some countries complement case-by-case or open systems with detailed administrative or regulatory criteria. France[215] allows recognition of non-listed disorders such as depression or anxiety when specialized committees determine that work was the "direct and essential" cause. Japan applies ministerial criteria linking long working hours, heavy workload or potentially traumatic events to mental disorders.[216] It also relies on administrative rules to guide recognition of exhaustion-related or stress-induced disorders.[217] Such criteria provide clearer pathways for recognizing conditions arising from prolonged psychosocial risk exposure and help reduce evidentiary uncertainty.

Recognition of work-related suicide in legal and compensation frameworks: some examples

In some countries, legal or compensation frameworks have begun to recognize suicide or attempted suicide as work-related under specific conditions, typically where a link with work-related mental distress, overwork or serious psychosocial hazards is established.

In Japan, suicide may be recognized within the workers' compensation system where it is causally linked to work-related mental disorders or excessive working hours, based on detailed administrative criteria applied on a case-by-case basis.[218] In *France*, courts have in several cases held employers liable where suicide or attempted suicide was connected to serious psychosocial risks that were foreseeable and insufficiently prevented, including under the concept of gross negligence.[219] In Australia, recent reforms to OSH notification frameworks require reporting of suicide, attempted suicide and certain serious psychological incidents, strengthening preventive oversight and regulatory response.[220]



215 Code de la Sécurité Sociale – Tables des Maladies Professionnelles (Livre IV) (listing PTSD) and CRRMP (Comité Régional de Reconnaissance des Maladies Professionnelles) procedure (for complementary recognition).

216 Industrial Accident Compensation Insurance Act (IACIA).

217 Bekendtgørelse om fortegnelse over erhvervssygdomme (Executive Order on the List of Occupational Diseases), and the Administrative rules of Arbejdsmarkedets Erhvervssikring (AES) for recognition of exhaustion-related and stress-induced disorders.

218 Industrial Accident Compensation Insurance Act and related administrative criteria on mental disorders and overwork-related outcomes (karōjisatsu).

219 See case law on employer liability and gross negligence in relation to work-related suicide (e.g. France Télécom and subsequent cases).

220 Safe Work Australia, Notifiable incidents, extended absences and suicides handbook.

Social dialogue and collective agreements

Social dialogue plays a central role in shaping national approaches to the psychosocial working environment.[221] Evidence shows that OSH policies and regulations tend to be more effective when employers' and workers' organizations are involved in their development, monitoring and implementation, as participation strengthens legitimacy, improves compliance and facilitates adaptation to sectoral and organizational realities (EU-OSHA 2021). In the area of psychosocial risks in particular, the knowledge and experience of workers and employers are essential for identifying organizational features, understanding sectoral risk patterns and designing responses that reflect actual work processes. Social dialogue mechanisms – whether through national tripartite bodies, bipartite committees, sectoral platforms or workplace-level structures – thus provide an important foundation for embedding psychosocial risk prevention within OSH systems (Vesper et al. 2025).

Approaches differ across countries as regards the respective roles of legislation and collective bargaining. In some contexts, trade unions have favoured keeping psychosocial risk prevention firmly anchored in binding OSH legislation, in order to ensure universal coverage and avoid making fundamental protections subject to negotiation. Collective bargaining is then used to develop sector- or enterprise-specific provisions that operationalize legal requirements, rather than to replace them. This division of roles helps preserve minimum standards while allowing negotiated solutions to reflect differences in sectors, occupations and work organization.

Collective bargaining and sectoral agreements can nonetheless play an important role in translating shared priorities into concrete measures. Sector-level agreements are often particularly relevant, as psychosocial risks frequently follow sectoral patterns linked to work organization, exposure to emotional demands or working time arrangements. In many countries, collective agreements complement legislation by specifying procedures for identifying psychosocial risks, defining indicators of stress or workload imbalance, establishing consultation mechanisms and outlining preventive actions tailored to sectoral contexts. At enterprise level, workplace agreements and cooperation mechanisms offer flexibility and adaptability to specific organizational circumstances, although their effectiveness may depend on the presence of worker representation, organizational capacity and enforcement mechanisms. Taken together, negotiated approaches at different levels present both advantages – such as contextual relevance and ownership – and limitations, including uneven coverage and variability in implementation.

Collective bargaining addressing psychosocial risks and related issues

An analysis of the **ILO Cross-Border Social Dialogue (CBSD) Repository**[222] indicates that only **18 per cent of the 338 transnational initiatives** recorded between 2000 and 2025 explicitly incorporate issues related to mental health or psychosocial factors within OSH-related provisions (61 agreements in total). This limited share highlights that, despite increasing policy attention to psychosocial risks, their integration into transnationally negotiated OSH frameworks remains partial and uneven.

Most of these initiatives take the form of collective agreements, although non-binding instruments – such as joint declarations, charters and memoranda of understanding – are also represented (ILO 2025a). The large majority were concluded after 2010, with particularly strong growth from the mid-2010s onwards. More recent texts increasingly refer to **work-related stress, mental health and well-being, harassment, work-life balance, digitalization and telework**, reflecting broader developments in international and regional OSH discourse.

The agreements vary in duration and regulatory approach, ranging from fixed-term arrangements aligned with bargaining cycles to open-ended frameworks with defined revision procedures, suggesting diverse institutional strategies in addressing psychosocial risks.

221 Social dialogue includes all types of negotiation, consultation and exchange of information between or among representatives of governments, employers and workers on issues of common interest relating to economic and social policy. More information of social dialogue can be found at https://www.ilo.org/topics-and-sectors/social-dialogue-and-tripartism.

222 The full database and links to the texts are available at: Transnational Company Agreements | CBSD.

Beyond collective bargaining, other forms of social dialogue have also contributed to psychosocial risk prevention. Advisory opinions and recommendations issued by national economic and social councils, joint frameworks developed by social partners, and collaborative initiatives such as joint surveys, pilot projects, guidance materials and awareness campaigns have played an important role in shaping policy debates and supporting implementation. These instruments can influence legislation, inform enforcement practices and promote shared understanding, even where they are not legally binding.

In Europe, social partners have been particularly active in shaping responses to psychosocial risks through a range of social dialogue mechanisms. The 2004 European Framework Agreement on Work-related Stress stimulated national and sectoral engagement not only through collective bargaining, but also through joint guidelines, recommendations, awareness-raising activities and policy-oriented initiatives, and influenced legislative developments in several countries (e.g. Belgium,[223] Czechia,[224] Hungary,[225] Italy,[226] Latvia,[227] Lithuania,[228] Portugal,[229] and Slovakia[230]) (European Social Partners 2008). In Belgium, national bipartite social dialogue has combined inter-professional agreements with non-binding recommendations and pilot initiatives on psychosocial risks and burn-out, which have informed workplace practices and supported subsequent regulatory and policy developments (*Conseil national du travail* 2023). In France, the cross-industry agreement on work-related stress sets out shared objectives to raise awareness, identify stress factors, and promote measures to prevent, reduce, or manage work-related stress.[231] Denmark has a long-standing tradition of joint initiatives, including public-sector agreements, sectoral guidance and tripartite tools requiring workplace cooperation bodies to develop stress-prevention measures that complement OSH legislation (EU-OSHA 2025b). In Romania, the European agreement was implemented directly through national collective bargaining, while in countries such as the Kingdom of the Netherlands, Hungary and Spain, psychosocial risk prevention has been integrated into broader cross-industry or sectoral frameworks combining agreements, joint guidance and sector-specific tools, including in agriculture, call centres, social services, hospitality and the public sector (European Commission 2011).

Beyond Europe, social dialogue on psychosocial risks has developed more unevenly, but several examples illustrate how collective bargaining, tripartite cooperation and joint initiatives can complement legislation and public policy. In Australia, unions and employer organizations have worked through tripartite structures under Safe Work Australia and state jurisdictions to develop model codes of practice on managing psychosocial hazards,[232] which are commonly referenced in sectoral and enterprise agreements dealing with workload, rostering, bullying, organizational change and consultation and dispute resolution.[233] In Canada, particularly within the federal public service, social dialogue has combined collective agreements with joint initiatives to implement the National Standard for Psychological Health and Safety in the Workplace (2013). This has included the establishment of joint mental health committees, shared monitoring arrangements and collaborative tools addressing organizational change and workload management.[234] In Chile, recent framework agreements in the central public administration include a dedicated workstream on mental health and psychosocial well-being, with joint commitments to strengthen preventive measures and improve working conditions affecting public employees.[235] In South Africa, bargaining councils in the health sector have negotiated access to counselling, debriefing and psychosocial support for workers exposed to traumatic events.[236]

223 Arrêté royal sur la prévention de la charge psychosociale au travail.
224 Zákon č. 262/2006.
225 évi CLXI. törvény a munkavédelemről szóló 1993. évi XCIII. törvény módosításáról.:
226 Decreto Legislativo 9 aprile 2008.
227 Ministru kabineta noteikumi Nr. 660.
228 Įsakymas dėl psichosocialinių rizikos veiksnių tyrimo metodinių nurodymų patvirtinimo.
229 Lei 102/2009.
230 Zákon č. 124/2006; Zákon č. 355/2007; and Zákon č. 542/2007.
231 Accord national interprofessionnel du 2 juillet 2008 relatif au stress au travail.
232 Safe Work Australia, Model Code of Practice: Managing Psychosocial Hazards at Work (2022).
233 The Australian Public Service Enterprise Agreement; the New South Wales Teachers Federation agreements; and the Queensland Health Certified Agreement.
234 Memorandum of Understanding on Mental Health in the Workplace between the Treasury Board of Canada Secretariat (TBS) & Public Service Alliance of Canada (PSAC), appended to several federal public-service collective agreements (first signed 2015, renewed in subsequent bargaining rounds).
235 Central Unitaria de Trabajadores (CUT). 2024. Protocolo de Acuerdo Gobierno – Mesa del Sector Público 2024–2025: Agenda de Trabajo 2025.
236 Resolution 4 of 2017: Agreement on the Payment of Special Allowance and Danger Allowance.

Promoting compliance and raising awareness

Promoting compliance, including through targeted support and awareness-raising, alongside effective enforcement by regulators, is essential to ensure that legal provisions on psychosocial risks translate into safer and healthier working environments. This is typically pursued through a balanced regulatory mix in which enforcement is combined with education, guidance and capacity building. Such approaches reflect contemporary regulatory theory, including responsive regulation and strategic enforcement, which emphasise combining deterrence with support for compliance rather than relying on sanctions alone (Walters et al. 2021).

Labour inspectorates play a central role in this framework. Across jurisdictions, inspection systems typically combine proactive prevention activities – such as targeted campaigns, preventive visits, sectoral programmes and thematic inspections – with reactive enforcement functions, including responses to complaints, incident notifications and reported harms. This dual role is particularly important for psychosocial hazards, which require both anticipatory engagement with organizational risks and the capacity to intervene where legal obligations are not met.

In practice, many inspectorates operationalize this approach by integrating compliance checks with guidance during workplace inspections. Inspectors may verify adherence to legal requirements while also recommending improvements, providing tools or directing duty holders to relevant guidance. Regulatory contact thus functions simultaneously as a compliance mechanism and a capacity-building intervention, particularly for organizational risks that are less visible and more complex than traditional physical hazards.

Balancing advice, improvement support and enforcement requires the exercise of professional judgement by inspectors. In the context of psychosocial risks, this task is especially demanding, as it involves assessing work organization, workload allocation, supervision, leadership practices and organizational culture, as well as cumulative and interacting risk pathways. Inspectorate practice therefore influences whether psychosocial hazards are treated as substantive regulatory concerns alongside physical risks, with evidence suggesting that weaker regulatory signals can undermine deterrence and compliance (Popple et al. 2021).

These challenges are compounded by the scale and pervasiveness of psychosocial risks, which arise across all sectors and organizational sizes, placing sustained pressure on inspection systems with limited resources. This has reinforced the importance of strategic prioritisation, evidence-based targeting and complementary regulatory approaches, including guidance, enforceable undertakings and other mechanisms suited to addressing systemic and organizational sources of risk.

Research confirms that traditional inspection methods require adaptation for psychosocial risks. Studies from Europe show that labour inspection can stimulate improvements in psychosocial risk management, but that inspectors require specific training, diagnostic tools and sufficient time to address work-organization problems effectively (Toukas et al. 2015; Weissbrodt and Giauque 2017). In response, a number of inspectorates have developed targeted initiatives and specialized tools to strengthen regulatory capacity in this area.

In Europe, coordinated initiatives under the Senior Labour Inspectors' Committee (SLIC) have played an important role in shaping national approaches. Joint campaigns have supported the development of shared concepts, tools and methodologies for assessing work-related stress and organizational factors, including the *Guide for Assessing the Quality of Risk Assessments and Risk-Management Measures with Regard to the Prevention of Psychosocial Risks* (SLIC 2018). Several countries have built on this work within their national systems. In Denmark, the Working Environment Authority has developed dedicated inspection tools and thematic campaigns addressing stress, workload and offensive behaviours. [237] Sweden has similarly integrated psychosocial risks into labour inspection through targeted reviews of workload, working time arrangements and victimisation linked to provisions on the organizational and social work environment (Leka and Iavicoli 2020). In Spain, Technical Criteria[238]clarify inspection expectations regarding psychosocial

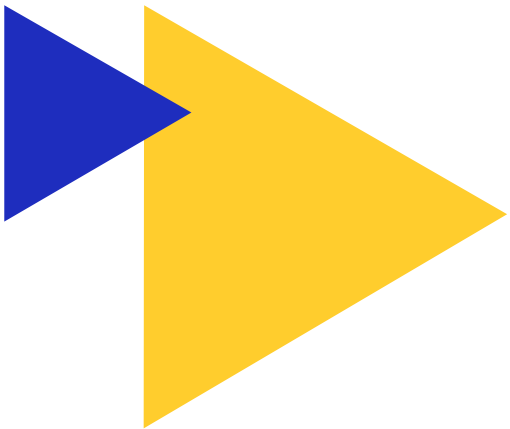
237 Danish Working Environment Authority (Arbejdstilsynet), Guidance and tools on the psychosocial working environment, (multiple sector guides publicly available).

238 Technical Criterias 69/2009, 87/2011 and 104/2021 of the Spanish Labor Inspectorate.

risk assessment and prevention, while in Estonia the Labour Inspectorate provides detailed public guidance on organizational and psychosocial risks.[239]

Outside Europe, a number of inspectorates have also developed specific approaches to psychosocial risk prevention. In Australia, recent reforms to the model Work Health and Safety framework have expanded incident notification requirements to include serious psychological harm and psychosocial risk outcomes, signalling increased regulatory attention to work-related mental health. In Canada, several provincial regulators provide inspectors with structured guidance and checklists on psychosocial hazards, including work-related stress and workload management.[240] In Chile, compliance with the CEAL-SM/SUSESO psychosocial risk assessment instrument, mandatory for organizations with 10 or more workers and in force since January 2023, is monitored through the national psychosocial risk surveillance system overseen by the Superintendencia de Seguridad Social (SUSESO) and the Dirección del Trabajo (DT).[241] In Brazil, psychosocial risk factors have recently been included in the requirements of NR-1 concerning OSH regulations and will be phased in through an initial educational period supported by guidance materials and tripartite monitoring. Labour inspection will begin assessing compliance from late May 2026, following an adaptation period for workplaces.[242] In Japan and the Republic of Korea, labour inspections include scrutiny of working time records and overtime limits as part of efforts to prevent overwork-related disorders (Kim et al. 2019; Yamauchi et al. 2017). In Kenya, Jordan and Lebanon, inspectorates have strengthened their capacity to address psychosocial risks through ILO-supported training, enabling inspectors to integrate organizational and psychosocial considerations into routine OSH inspections.

Alongside inspection activities, the development of **voluntary standards and guidance tools** plays an important role in promoting compliance and supporting employers and workers to identify and manage psychosocial risks. Across regions, OSH bodies, together with social partners, have produced general and sector-specific guidance, checklists and toolkits that translate legal requirements into practical, action-oriented approaches. These resources range from broad guidance on psychosocial risk assessment and preventive strategies to materials addressing specific issues, as well as emerging risks linked to digitalisation, new forms of work and changing work environments. These guidance tools help workplaces apply preventive measures adapted to different sectors, diverse forms of work and evolving challenges.[243] Targeted tools have also been developed for small and medium-sized enterprises (SMEs), helping to address practical barriers related to awareness, capacity and resources.



239 Estonian Labour Inspectorate, Risk Assessment Guidance (2021).

240 WorkSafeBC, Psychological Health and Safety (2023).

241 Protocolo de Vigilancia de Riesgos Psicosociales en el Trabajo.

242 Brazil, Ministry of Labor and Employment (MTE), "Inclusão de fatores de risco psicossociais no GRO começa em caráter educativo a partir de maio" (24 April 2025; updated 20 May 2025).

243 A non-exhaustive selection of guidance materials issued by national OSH bodies, social partners and related institutions is provided in the annex.

Voluntary standards on psychosocial risks

Voluntary standards and guidance instruments have increasingly been used to support the implementation of OSH legislation on psychosocial risks. Although non-binding, they provide structured methods for identifying, assessing and managing psychosocial hazards and help organizations operationalize general legal duties within existing OSH management systems.

At international level, ISO 45003:2021 offers guidance on managing psychosocial risks within OSH management systems, with an emphasis on organizational factors, worker participation and continuous improvement (ISO 2021). Several regional and national instruments play a similar role. In Europe, common concepts and approaches have been promoted through initiatives such as the Psychosocial Risk Management – European Framework (PRIMA-EF), which informed the development of PAS 1010:2011 in the United Kingdom. The United Kingdom Management Standards for Work-related Stress, issued by the Health and Safety Executive, provide practical tools for assessing and addressing key organizational stressors. In *Canada*, the National Standard of *Canada* for Psychological Health and Safety in the Workplace (CSA Z1003), introduced in 2013, was the first auditable national standard dedicated to this issue. Other countries have adopted guidance-based approaches, such as national guidance on work-related psychological safety and health in Australia, as well as sectoral tools addressing fatigue, bullying and workplace violence.

Research suggests that voluntary standards and guidance can strengthen preventive capacity and support more consistent organizational approaches to psychosocial risk management, particularly when used with meaningful worker participation and aligned with regulatory expectations (Leka et al. 2011; Leka et al. 2015). At the same time, research indicates that an emphasis on formalized procedures and documentation does not, on its own, ensure effective psychosocial risk management in practice (Bluff and Gunningham 2004; Hohnen et al. 2014). Evidence on the implementation and impact of some standards remains limited, underscoring the importance of using voluntary instruments as a complement to, rather than a replacement for, binding regulatory frameworks.

Awareness-raising campaigns play an important role in improving understanding of psychosocial risks and encouraging preventive action at the workplace level. Across regions, OSH authorities and social partners have implemented national and sectoral campaigns to inform employers, workers and the wider public about issues such as work-related stress, workload, work organization, mental health at work and emerging psychosocial risks. These initiatives typically aim to promote early recognition of risks, encourage dialogue and reinforce a preventive culture, often using a combination of public information materials, targeted communication strategies and workplace outreach. In many cases, campaigns are developed through tripartite or bipartite cooperation, which helps ensure credibility and relevance for different audiences. They frequently target specific groups – such as managers, OSH practitioners, workers' representatives or SMEs – and are rolled out through multiple channels, including dedicated websites, guidance materials, training events, social media, sector-specific resources and inspectorate outreach.

In Europe, EU-OSHA has played a central role in coordinated awareness-raising through its Healthy Workplaces Campaigns, including the 2014-2015 campaign on work-related stress, with a forthcoming campaign announced on psychosocial risks and mental health at work.[244]Several countries have complemented these initiatives with national campaigns. For example, France has conducted multi-year national awareness campaigns on psychosocial risks involving labour authorities, social security institutions and social partners, combining communication activities with workplace tools and training.[245] Belgium has implemented joint awareness initiatives on psychosocial risks and burn-out prevention through its Federal

244 Healthy Workplaces Campaigns and the upcoming Healthy Workplaces Campaign 2026-2028 which focuses on mental health and psychosocial risks at work.

245 INRS and Ministry of Labour campaigns on psychosocial risks.

Public Service Employment and the social partners, supporting dissemination of preventive approaches across sectors.[246]

Beyond Europe, awareness-raising on psychosocial risks and mental health at work is often pursued through national public campaigns or communication initiatives linked to prevention frameworks. For example, in Japan, the government has established an annual "Enlightenment Month" (November) to raise public awareness on preventing overwork-related harm (*karōshi*), accompanied by outreach activities and events.[247] In Chile, the national psychosocial risk protocol (CEAL-SM/SUSESO) explicitly include a "campaña de difusión y sensibilización" [awareness and outreach campaign] as part of its implementation process to encourage understanding and participation.[248] In the Pacific Alliance (Chile, Colombia, Mexico and Peru), a regional initiative is currently developing and implementing a communication campaign aimed at workers and employers to raise awareness and promote participatory management of psychosocial risks and mental health at work.[249]

246 FPS Employment & National Labour Council initiatives on psychosocial risks and burn-out.

247 Annual awareness month on preventing overwork-related harm *(karōshi)*, Act on Promoting Measures to Prevent Death and Injury from Overwork (Japan Law Translation), Article 5 (establishes November as the "Enlightenment Month").

248 Manual del método cuestionario CEAL-SM SUSESO, 2022.

249 The *International Seminar on psychosocial occupational risks* of the Pacific Alliance noted that the second stage "consists of the design and execution of a communication campaign aimed at workers and employers…".



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Part 4

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Workplace level prevention and management of psychosocial risks

Ensuring a safe and healthy working environment in all its dimensions requires that the psychosocial working environment be integrated into OSH management systems (OSH-MS). Embedding psychosocial considerations within OSH-MS supports a preventive safety and health culture in which all workplace hazards are addressed in a coherent and systematic manner. At the same time, the psychosocial working environment is shaped by decisions taken within broader organizational management systems, including human resources policies, operational planning, change management and work design. Effective prevention therefore depends on alignment between OSH management processes and these wider organizational practices, ensuring that psychosocial risks are addressed both within OSH governance and in everyday management decisions. Worker participation is central to this approach, as workers contribute essential knowledge about how work is experienced in practice.

The *ILO Guidelines on Occupational Safety and Health Management Systems* (ILO-OSH 2001) provide a foundation for managing risks arising from the working environment and are therefore applicable to psychosocial risks.[250] Within this framework, psychosocial risks should be integrated into all elements of the OSH management system – including policy, organization, planning and implementation, and evaluation and continual improvement – ensuring that the psychosocial working environment is treated as an integral component of effective OSH management rather than as a separate or parallel process (ILO 2001).

250 These guidelines are intended to be applied flexibly across diverse organisational contexts, with the aim of preventing work-related accidents and injury by minimising hazards so far as is reasonably practicable. Their principles emphasise proportionality, simplicity and integration of OSH into core business and work processes, recognising that many workplaces, particularly in MSMEs, informal economy settings and self-employment, operate under significant resource constraints and without formalised human resource or OSH functions.

The OSH **policy** expresses the organization's commitment to protecting workers' safety and health. It should be established by the employer in consultation with workers and their representatives and communicated to all workers, providing the basis for the organization's approach to managing occupational risks, including psychosocial risks.

Organizing within the OSH management system involves establishing leadership, responsibilities, accountability and authority for OSH, including responsibilities for identifying and controlling hazards and risks, including psychosocial risks. Arrangements should support supervision and promote cooperation and communication on OSH matters, ensure the participation of workers and their representatives, and provide the competence, training and resources needed for OSH.

Planning and implementation translate OSH policy commitments into action. Planning involves setting OSH objectives, defining responsibilities and allocating the resources needed to achieve them, based on a review of existing conditions. Implementation then puts these plans into practice by identifying hazards – including psychosocial hazards – assessing the associated risks and implementing preventive and protective measures to eliminate or control them (*given the importance and specific features of identifying psychosocial hazards, assessing associated risks and implementing preventive measures, the following section 3.1 elaborates on these matters*). Implementation also includes arrangements for the management of change to ensure that the safety and health implications of organizational, technological or work process changes are considered.

Evaluation and action for improvement involve monitoring and reviewing OSH performance to determine whether the system is functioning effectively and risks are being managed. This includes the use of appropriate indicators, the investigation of work-related injuries, ill health, diseases and incidents, and periodic audits and management reviews. Where deficiencies are identified, preventive and corrective actions should be implemented, supporting the continual improvement of the OSH management system and the prevention and control of hazards and risks, including psychosocial risks.

4.1. A spotlight on the management of psychosocial risks at work

Risk assessment is a core element of prevention within an OSH management system. It involves identifying hazards, evaluating associated risks and establishing priorities for preventive action. In the case of psychosocial risks, this requires examining the organizational conditions through which work is designed, organized and managed.

Identifying psychosocial hazards

Psychosocial hazards can arise from how jobs are designed, how work is organized and supervised, and how organizational policies and procedures influence daily operations. These aspects shape whether work is coherent, predictable and adequately resourced, or whether pressures such as excessive demands, role ambiguity, conflict or perceived unfairness may create harm.

Because psychosocial hazards are not always directly observable, evidence should be gathered from multiple complementary sources. Organizational documentation, job descriptions and management procedures may reveal how risks arise through their interaction, for example where insufficient staffing increases workload or where inadequate training leaves workers unable to meet expectations. This analysis should also consider factors such as discrimination, unequal power relations or gender norms that may influence task allocation, working time arrangements or supervisory practices.

Information from organizational records, human resources data and worker surveys can help identify patterns of workload pressure, role ambiguity or low perceived fairness. Self-report surveys are widely used because they are relatively easy to administer, although their interpretation requires caution since responses may be influenced by contextual factors or concerns about confidentiality. Recognized psychosocial risk assessment tools can be used to support this process,[251] alongside clear assurances of confidentiality and, where appropriate, independent administration. Survey data may also be complemented by indicators of outcomes such as work-related stress, burnout, mental ill-health or musculoskeletal symptoms.

Dialogue-based methods, including interviews, focus groups and worker–management discussions, provide additional insight into how work operates in practice. Combining organizational data with workers' experiences helps create a more reliable understanding of emerging psychosocial hazards.

Assessing and prioritizing psychosocial risks

Once hazards are identified, the next step is to evaluate the associated risks, considering both the likelihood of harm and the severity of its potential consequences. In assessing psychosocial risks, attention should also be given to the duration and frequency of exposure, the number of workers affected and the ways in which harm may develop or accumulate over time. Low numbers of complaints or reported incidents should therefore not be interpreted as evidence of low risk.

Research shows that psychosocial risks are frequently under-reported due to stigma, fear of negative consequences or uncertainty about what constitutes a reportable concern (Dollard et al. 2019; Klinefelter et al. 2021). For this reason, risk assessment should integrate multiple information sources, including organizational indicators, survey data, participatory dialogue with workers and the review of relevant organizational processes.

Examples of mechanism to encourage reporting

Workers should have accessible mechanisms to report psychosocial hazards that protect their privacy and confidentiality. These mechanisms should be proportionate to the size and structure of the organization and the level of risk, and may combine informal and formal channels. Examples include dedicated email addresses or telephone lines, confidential online or paper forms (including anonymous options), locked boxes for written reports, and clearly identified supervisors, managers or worker representatives available to receive reports, including those requiring urgent attention.

Awareness-raising about psychosocial hazards and clear communication about reporting and response procedures are essential. Information on reporting options, timely follow-up, involvement of workers and their representatives, and safeguards against victimization help build trust in reporting systems and support the early identification of psychosocial hazards.

Source: Adapted from Australian Government 2024.

Prioritization follows evaluation. Some hazards may require immediate action, such as violence and harassment, acute or sustained overload, or exposure to traumatic events. Others may call for longer-term improvements, including redesigning roles or workflows, revising performance management systems, strengthening communication structures, or enhancing participation and support mechanisms. A transparent and participatory prioritization process helps ensure that selected actions are legitimate, feasible and aligned with the organization's capacity for change.

251 A selection of these tools is provided in Annex 2.

Inclusive approaches to psychosocial risk management

Psychosocial hazards are unevenly distributed across the workforce, with certain groups facing higher exposure. Gendered patterns of work, occupational segregation, age, employment status, migration background, disability, and unequal power relations can influence both exposure to psychosocial hazards and workers' ability to raise concerns safely (Benach et al. 2014; Campos-Serna et al. 2013; ILO 2019; WHO and ILO 2022).

For example, women and younger workers are often overrepresented in roles with high emotional demands, low decision authority and insecure employment, while migrant, temporary and agency workers may face job insecurity and barriers to reporting (Eurofound 2017b; Nyberg et al. 2020; WHO and ILO 2022). These patterns are shaped by how work is organized and by prevailing social norms, and may result in unequal exposure to workload pressure, role ambiguity, unfair treatment, and violence or harassment over time (Nielsen and Einarsen 2018; Nyberg et al. 2020). Such sustained and uneven exposure reinforces the importance of reducing harmful exposure at its source, recognizing when exposure has occurred, and ensuring adequate recovery and supportive organisational responses for all workers (Kivimäki and Steptoe 2018; WHO and ILO 2022).

An inclusive psychosocial risk assessment process therefore considers not only what the hazards are, but also who is most exposed and whose voice may be missing from the assessment process. This can involve:

- disaggregating survey and organizational data, where appropriate (e.g. by gender, age or employment arrangement);
- ensuring the participation of all workers in surveys and dialogue processes, including those from minority or potentially vulnerable groups;
- examining how staffing models, shift allocation, performance systems and reporting lines may create unequal exposure to psychosocial hazards;
- identifying barriers to reporting, such as stigma, fear of retaliation, language barriers or lack of trust;
- integrating discrimination, as well as violence and harassment, into psychosocial risk assessment rather than addressing them separately (ILO 2020; WHO and ILO 2022).

Considering diversity in this way strengthens the accuracy and effectiveness of psychosocial risk management, helping ensure that preventive measures improve working conditions for all workers.



From assessment to action

Risk assessment should lead to concrete preventive action that improves how work is designed, managed and supported so that psychosocial risks are reduced at their source. In line with the hierarchy of controls, [252] priority should be given to measures that eliminate or reduce risks at their origin, while also recognizing the complementary role of measures that protect health and provide support where difficulties arise.

Preventive measures should therefore address the organizational determinants of work that can generate psychosocial risks by improving how work is designed, organized and managed. This includes strengthening workload management, job design, role clarity, work-time organization, staffing levels and participation mechanisms. Such collective, organization-level interventions correspond to the higher levels of the hierarchy of controls and form the foundation of effective prevention (Aust et al. 2023; Demerouti and Adaloudis 2024; Fox et al. 2021).

At the same time, a comprehensive approach also strengthens the conditions that enable workers to cope with demands and thrive in their roles. Building organizational resources such as fairness, worker voice, supportive leadership, social support, and opportunities for influence, learning, development and recovery helps protect and promote mental health and well-being at work. These resources can strengthen capacity, engagement and resilience at the level of teams and organizations, reducing the likelihood that everyday stressors escalate into harm.

Even where preventive measures are in place, some workers may experience difficulties related to psychosocial risks. In such situations, responses should be timely, non-stigmatizing and work-focused. Access to support services, temporary work adjustments, occupational health input and fair return-to-work processes can help stabilize the situation and support continued participation in work. Evidence indicates that participatory, work-focused return-to-work approaches are more effective than clinical treatment alone, particularly in cases of stress- or burnout-related conditions (Mikkelsen and Rosholm 2018; WHO and ILO 2022). These forms of support complement, but do not replace, preventive organizational measures.

Implementation measures should be monitored to assess their effectiveness, identify unintended consequences and ensure they remain relevant as work evolves. Regular review of psychosocial risk assessments and the preventive measures implemented helps ensure that preventive action remains effective and responsive to changes in work organization, processes and workforce conditions.

Where such review indicates emerging or persistent problems, organizations should revisit the psychosocial risk assessment to identify underlying causes, including organizational factors related to how work is designed, organized and managed. Based on this analysis, preventive measures may need to be adjusted, for example by revising procedures, adapting work organization or job design, strengthening supervisory practices, or improving participation and communication mechanisms. In this way, psychosocial risk management becomes a continuous process that supports ongoing improvement of the psychosocial working environment.

252 The hierarchy of control refers to a structured approach that prioritises measures to prevent and control hazards according to their effectiveness, giving precedence to eliminating the hazard, then substituting it with something less harmful, followed by minimising risks through engineering and organisational measures, then administrative controls, and only as a last resort relying on personal protective equipment. Applied to psychosocial risks, this logic has been further developed in the literature through a "psychosocial hierarchy of control", such as that proposed by LaMontagne et al. (2024). In this adaptation, the highest levels of control focus on eliminating working conditions that threaten safety, health and well-being, for example by reducing excessive workload or removing sources of harassment. This is followed by substituting safer and healthier work processes and practices, such as improving working time arrangements or job structures. The next level involves redesigning the work environment for safety, health and well-being, including improvements in supervision, support, communication and the broader social and organisational environment. Lower levels consist of administrative controls, including organisational strategies, policies, guidelines and communicative practices that promote healthy work. Finally, at the lowest level, are measures that encourage personal change, such as individual coping, resilience or stress-management interventions, which can be supportive but do not address risks at their source.

Examples of preventive measures to better design, manage and organize work

Collective preventive measures that address how work is designed, managed and organized should be prioritized in line with the hierarchy of controls. In practice, such measures often address several interacting psychosocial hazards simultaneously, as issues such as workload, role clarity, predictability, participation and support frequently arise from shared features of work design and organizational practice. Individual or reactive measures may still be necessary in some situations, but they should complement rather than replace these upstream interventions.

Examples of preventive measures in relation to the job and task design

Job demands

- Alternate periods of intense cognitive effort with lower-demand activities to avoid sustained concentration requirements
- Support cognitive work by implementing systems that reduce human error (e.g. use IT systems to capture important information and generate reminders)
- Limit unnecessary complexity in task sequencing, with clear steps and adequate time built in
- Provide additional support during periods of high demand (e.g. provide more workers, better equipment or outsource tasks)
- Plan the workforce so there are enough appropriately skilled staff, including training in advance to meet work demands
- Where high emotional demands are inherent to the role, build in structural buffers such as rotation away from high-demand interactions and recovery time between emotionally intense tasks
- Limit exposure to traumatic material and events to what is operationally necessary by designing work so as few workers as possible are exposed, for the shortest time
- Minimize unnecessary physical strain by applying ergonomic principles to task sequencing, load limits and posture requirements
- Rotate tasks to reduce prolonged exposure to physically demanding activities

Level of responsibility

- When high responsibility is a feature of a role, ensure this is made explicit and discussed with workers before they start the role

Task design

- Reduce unnecessary tasks, facilitate task rotation and provide alternatives to maintain engagement at work, avoiding tasks that are overly simple, repetitive or monotonous

Examples of preventive measures in relation to how work is managed and organized

Roles and expectations

- Provide clear job descriptions that define roles, responsibilities and performance requirements, carrying reviews of these and communicating any changes to the workers and those working alongside them
- Provide clear work instructions and expectations, explain why roles, responsibilities and tasks are allocated as they are, and ensure workers, particularly when they share the same tasks, understand who is doing what
- Design management structures with clear reporting lines, identifying immediate supervisor for each worker
- Provide clear guidelines for what to do when expectations do not align (e.g. between workers, workers and supervisors, or workers and clients)
- Encourage informal communication between managers and workers, and among workers, and implement information sharing systems so workers have access to the information they need to do their jobs

Job control or autonomy

- Match workers' level of autonomy and control in carrying out their tasks to their skills and experience
- Provide workers with appropriate autonomy, based on the nature of the job, to perform tasks effectively and safely (e.g. empower workers facing high emotional demands to exercise discretion, such as offering refunds to prevent customer aggression or distress)

Workload and work pace

- Adjust total workloads taking into account available resources and the number and capacity of workers
- Monitor working practices to detect harmful or discriminatory patterns in the assignment of workload and work pace, and introduce appropriate adjustments
- If work is machine or computer paced, design processes so the pace of work is adequate for workers, considering the need to change tasks, or pause the workflow to take breaks
- Match worker abilities to the amount of work and complexity to avoid work underload (e.g. running out of work or having work that is too easy)

Supervision and support

- Provide workers with the things they need to do their jobs properly and safely (e.g. the right tools, equipment, systems and resources) and ensure workers have sufficient access to them (e.g. they are conveniently located, and workers do not need to compete for access)
- Encourage supervisors to be empathetic in their leadership, establish open communication mechanisms (e.g. have an open-door policy) and safe spaces for workers to raise concerns early, helping workers when they are struggling
- Establish systems to ensure regular, fair, goal-focused and constructive feedback discussions occur between workers and supervisors to discuss work tasks, and any support or development needs (e.g. implement end of shift debriefs or require supervisors to do regular follow-ups)
- Encourage the development of positive working relationships and promote collaboration and cooperation (e.g. invest in team planning and building activities and encourage team discussions)

Examples of preventive measures in relation to the broader policies, practices and procedures that govern work

Employment arrangements

- Review and adjust employment agreements and entitlements, where feasible, to support more predictable and stable employment conditions
- Where possible, improve workers' access to social protection, such as paid sick leave and annual leave

Working time arrangements

- Establish hours of work guidelines to allow adequate rest and recovery (e.g. maximum shift length, maximum number of consecutive shifts, minimum break/rest periods, maximum night shifts, maximum overtime)
- Set up predictable rosters and shifts, and where possible, consult and provide advance notice of any changes impacting workers usual work hours or arrangements
- Allow reasonable adjustments in working hours and practices (e.g. hybrid or remote working arrangements) to facilitate workers' fulfilment of responsibilities outside of work, when compatible with the nature of work
- Provide clear information on the working hours and accessibility of workers (including remote workers) to facilitate disconnection from work

Organizational change management

- Plan changes to roles, tasks, objectives and supervisory arrangements to ensure they are reasonable and account for a period of change (e.g. adjusting performance targets whilst workers learn new roles)
- Provide mechanisms to guide workers and supervisors through change processes (e.g. provide information or feedback sessions to address any concerns)

Surveillance

- Consult workers when developing performance monitoring systems and procedures and adopt a clear policy to ensure that the purpose and level of monitoring are appropriate

Rewards, performance management, opportunities for development and recruitment processes

- Implement systems to support performance (e.g. training and mentoring) and provide opportunities for development (e.g. allow workers to take ownership of particular tasks)
- Establish effective communication and constructive feedback processes that ensure it is timely, specific, practical, fair and clearly relates to workers' performance
- Use fair, transparent and meaningful ways of providing recognition and rewards to reflect workers efforts (e.g. avoid only recognizing the workers doing high profile work; recognize teamwork and corporate contributions)
- Develop unbiased and transparent selection procedures that ensure promoted or hired workers have the right skills and experience for the position, including the level of autonomy the job will have

- Design fair and transparent performance management systems and processes (e.g. ensure performance measures relate to aspects of work within a worker's control and consult workers on performance expectations)
- When algorithms are used in workplace systems or processes, ensure transparency in automated decision-making and maintain human oversight to identify and address potential biases

An OSH policy and management system

- Establish and apply clear procedures to identify, assess and control psychosocial and environmental risks, using a range of assessment methods and appropriate indicators
- Monitor and review risk management measures regularly, particularly when there are changes in work organization or processes

Violence and harassment at work

- Establish and implement transparent and appropriate procedures to report incidents and deal with conflicts and problem solving, outlining expected behaviours and how this will be managed
- Encourage workers to report incidents and behaviours of concern and provide a range of accessible and user-friendly ways to make a report informally, formally and confidentially
- Ensure processes and systems for reporting and responding to incidents are widely communicated and regularly reviewed to ensure their fair and consistent application

Consultation and participation

- Implement information and consultation arrangements to regularly discuss the work, how it is done and any changes impacting workers
- Encourage workers to engage with consultation and raise any issues, concerns or suggestions
- Apply managerial practices that prioritise workers' psychosocial safety and health alongside organizational and productivity goals, supported by ethical and transparent leadership

Examples of preventive measures in relation to the physical work environment and equipment

- Implement good physical working conditions (e.g. noise, lighting, vibration, temperature, chemicals, adequate space) in accordance with relevant OSH guidance and standards
- Provide appropriate and well-maintained equipment, including training for new equipment and technology
- Minimize isolated work and exposure to extreme environmental conditions or unstable or traumatic environments and materials, and provision of security and support measures



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The way forward

The evidence reviewed in this report shows that the psychosocial working environment has a powerful influence on workers' physical and mental health and well-being, as well as on organizational performance. How jobs are designed, managed and organized, and the broader policies, practices and procedures that govern work, can lead to negative consequences – such as increased risk of illness, injury and reduced work ability – or to positive outcomes, including improved health, well-being and performance.

Translating this knowledge into consistent prevention requires coordinated efforts by governments, employers and workers and their organizations, OSH professionals and other relevant actors. These efforts need to focus on three interrelated areas: strengthening the knowledge base and monitoring systems; improving the coherence and clarity of policy and regulatory frameworks; and reinforcing workplace governance and leadership for psychosocial risk prevention. They must also keep pace with changes in the world of work that may influence the psychosocial working environment, including digitalisation, new forms of employment, demographic shifts and climate-related disruptions.

While **research** on the psychosocial working environment is increasing, greater methodological consistency and alignment remain necessary. Regular and comparable data on exposure to psychosocial risks and their related outcomes would improve understanding of prevalence, trends and the effectiveness of preventive efforts over time. However, these issues remain under-represented in many national OSH statistics and surveillance systems (EU-OSHA 2021; Niedhammer, Bertrais, and Witt 2021). Recent consultations with national OSH statistics focal points indicate growing recognition of these gaps. Several institutions have reported concrete plans to enhance OSH statistics in the coming years, including in relation to psychosocial risks and work-related mental health. Sustaining and expanding these efforts will be essential to strengthening the quality, comparability and policy relevance of national data systems.

Existing data sources – such as working-conditions surveys, labour force surveys, sickness absence and injury registers, labour inspection records and enterprise-level monitoring – contain valuable information but are often not designed or analysed in ways that adequately capture key aspects of the psychosocial working environment. More systematic integration of psychosocial indicators into OSH monitoring systems and national statistics – including through disaggregated data by sex, age, migration status, disability and other relevant characteristics – would improve tracking of both exposures and outcomes. This is particularly important given that psychosocial risks often develop gradually, accumulate over time and are shaped by organizational and social conditions.

Improved use of available data is also critical for understanding how psychosocial risks are distributed across the workforce. Evidence shows that exposure is not evenly

shared. Immigrants, persons with disabilities, ageing workers, young workers and those in precarious or informal employment may face higher exposure or more limited access to support (ILO 2019; Rugulies et al. 2023). Discrimination, unequal power relations and social norms can influence how work is organized and how risks arise in practice. Strengthened monitoring and analysis can therefore help identify differentiated patterns of exposure and inform more inclusive prevention strategies.

There are also important gaps in evaluation research. While many initiatives aim to improve the psychosocial working environment, there is still limited evidence on what works in different sectors, for different groups of workers and in different national contexts. Strengthening evaluation of policies and workplace measures would help identify effective approaches and support their wider adoption.

Policy and regulatory frameworks addressing psychosocial risks have evolved in recent years, yet approaches vary widely, resulting in differences in how psychosocial risks are addressed in practice. In some contexts, psychosocial risks have been addressed through general OSH legislation and employers' preventive duties; in others through more specific regulations, sectoral provisions, collective bargaining agreements, voluntary standards or combinations of these instruments. The extent to which different categories of workers and sectors are fully protected under OSH frameworks also varies.

The terminology used in laws and policies likewise differs. References to issues such as "mental health at work", "psychological safety" or "mentally healthy workplaces" often overlap with, but are not equivalent to, "psychosocial hazards or risks". When policy attention focuses primarily on outcomes rather than on the work-related causes, psychosocial risk prevention may receive less emphasis. By contrast, when psychosocial risks are more explicitly reflected in policy and regulatory frameworks, workplaces are more likely to conduct risk assessments and implement preventive measures as part of normal OSH management practice (Jain et al. 2022; Lerouge 2025).

Furthermore, in many contexts policy attention has focused primarily on acute and visible issues such as violence and harassment. While critical, these represent only one dimension of the psychosocial working environment. Greater attention to the structural and ongoing aspects of work organization – including workload, job control, predictability, participation and organizational justice – can encourage a more comprehensive preventive approach.

Strengthening the link between the policy and regulatory framework and workplace practice is essential. In addition to enforcement efforts, supporting employers through practical advice, guidance, tools and capacity-building measures – particularly for small and medium-sized enterprises – is essential to translating policy commitments into practice.

Finally, synergies between OSH authorities, public health bodies, employers' and workers' organizations, and sector-level institutions can reinforce preventive efforts. Recent expert consultations indicate that, even where legislative or policy measures exist, awareness and interpretation of their scope may differ within countries. Strengthening coordination, communication and shared understanding among institutions is therefore essential to ensure consistent implementation. Action beyond the individual workplace, including through employers' and workers' organizations, can help disseminate good practices and accelerate improvements across sectors.

At the **workplace level**, preventing psychosocial risks depends primarily on how work is designed, organized and managed. While not all psychosocial hazards can be fully eliminated, many risks can be reduced by adjusting core features of work organization – including workload distribution, job design, staffing, working time arrangements, supervision and opportunities for participation and development. These types of measures typically involve reviewing how tasks are allocated, how performance expectations are set and how work processes operate in practice. Such organizational measures can be complemented by initiatives that strengthen workers' skills and capacity to recognize psychosocial risks and respond appropriately where they arise.

Leadership and management commitment are central in this regard. Decisions taken by senior management and line managers shape the conditions under which work is performed and determine whether adequate resources, realistic expectations and supportive practices are in place. Recognizing this respon-

sibility helps ensure that psychosocial risk prevention is embedded in everyday management decisions and supported across functions within the organization.

Effective workplace practice therefore focuses on organizational measures that address psychosocial risks at their source, supported by participatory processes involving workers and their representatives. Cooperation between OSH professionals, human resources, management and workers helps ensure that preventive measures are adapted to the realities of work and sustained over time.

Given that psychosocial risks require preventive and multidisciplinary approaches, stronger links between research, policy and workplace practice remain essential. This includes continued attention to how policies are developed, implemented and evaluated, and to the broader social and organizational conditions that shape how work is designed, organized and managed in practice. Enhanced monitoring and surveillance, more systematic policy evaluation, and the development of practical and participatory approaches to improving work organization can support the uptake of effective practices and contribute to more consistently safe and healthy psychosocial working environments.





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Annexes

Annex 1. Key ILO publications and resources on psychosocial risks

- ***How to prevent and address violence and harassment at work: A training course for enterprises*** (2024): a course to help equip enterprises to effectively tackle workplace violence and harassment and is intended for Employer and Business Membership Organizations, globally, to deliver as a new training offering or service for their members.
- ***Preventing and addressing violence and harassment in the world of work through occupational safety and health measures*** (2024): a report that examines the widespread prevalence of violence and harassment at work and stresses the importance of comprehensive solutions, such as global conventions and protective protocols, to tackle these challenges.
- ***Ensuring Compliance with Legislation on Psychosocial Risks*** (2022): A training module oriented to labour inspectors, regulators and policymakers explaining the concept of psychosocial risks, steps towards effective psychosocial risk management and the role of compliance frameworks.
- ***Healthy and Safe Telework*** (2022): a technical brief providing evidence-based recommendations to protect and promote the physical and mental health, safety, and well-being of teleworkers.
- ***Mental Health at Work*** (2022): a policy brief from the ILO and WHO calling for global action to address mental health at work, outlining practical strategies to prevent psychosocial risks, promote mental well-being, and support workers with mental health conditions in all sectors.
- ***Violence and Harassment at Work: A Practical Guide for Employers*** (2022): a tool on Convention No. 190 and Recommendation No. 206 to provide insight into the definitions, guiding principles, and actions established under these instruments.
- ***Violence and harassment in the world of work: A guide on Convention No. 190 and Recommendation No. 206*** (2021): a resource explaining the principles and scope of Convention No. 190 and Recommendation No. 206, outlining the international standards and rights-based framework for preventing and eliminating violence and harassment across all sectors and forms of work.
- ***Managing Work-related Psychosocial Risks during the COVID-19 Pandemic*** (2020): a practical guide with key elements for assessing psychosocial risks and implementing preventive measures to protect workers' health and well-being, during lockdowns and return-to-work phases.

- ***Safe and Healthy Working Environments Free from Violence and Harassment*** (2020): a report exploring how violence and harassment in the world of work connects with OSH frameworks and psychosocial risk management.
- ***Workplace Stress: A Collective Challenge*** (2016): a report raising awareness of work-related stress, outlining its main causes and effects, and presenting frameworks to prevent and manage it.
- ***SOLVE Training Package: Integrating Health Promotion into Workplace OSH Policies*** (2012)**:** a training package focusing on the prevention of psychosocial risks and the promotion of health and well-being at work through policy design and action.
- ***Stress Prevention at Work Checkpoints: Practical Improvements for Stress Prevention in the Workplace*** (2012): a practical tool providing 50 checkpoints to identify, assess and prevent work-related stress through simple workplace improvements.
- Other resources on violence and harassment, including tools, guidelines and training materials, are available on the ILO's **Addressing violence and harassment at work** webpage.



Annex 2. A selection of tools for psychosocial hazards identification and risk assessment

Tool/ Instrument	Objective	Methodology
Job Content Questionnaire 2 (JCQ-2) (Karasek et al. 2025; Agbenyikey et al. 2025)[253]	Assesses psychosocial risk at the task, organizational and external-to-work levels. Multilevel development of the JCQ tool, with Demand, Control and Stability-Support operationalized at the task and the organizational level.	Self-report multiple-choice questionnaire where workers rate the frequency or degree of agreement on 4-point Likert scale. Data are analysed at individual/group/organizational level and can be used to identify priorities for improvement and to monitor change over time.
Method for Evaluating Psychosocial Factors (*Método de Evaluación de Factores Psicosociales FPSICO*) of the National Institute for Safety and Hygiene at Work (INSHT) version 4.1, 2022[254]	Tool to identify and evaluate psychosocial risk factors in workplaces. It assesses nine factors: working time, autonomy, workload, psychological demands, variety and content of work, participation/ supervision, interest in the worker/compensation, role at work, and relationships and social support.	Self-report questionnaire of 44 questions (89 items in total) with Likert-type response options. Responses are converted into scores for each dimension and processed with the FPSICO 4.1 software, which classifies risk into four levels (adequate, moderate, high and very high), allows comparison between groups or sites, and supports the planning and follow-up of preventive measures over time, including general recommendations for improvement based on the results obtained.
Modular Analysis of Stress Factors in Organizations (*Modulare Analyse der Belastungsfaktorenin Organisationen, MABO*), BAuA, 2022[255]	Modular and practice-oriented instrument for assessing work-related psychosocial hazards. It identifies stress factors across core domains such as job tasks and demands, communication and cooperation, leadership, work organization, working conditions, and employee attitudes, with the possibility to include organization-specific and optional thematic modules.	Self-report questionnaire with 81 core items rated on a 6-point Likert scale. Optional modules can be added depending on organizational needs. Includes a prior anonymous thematic consultation to integrate organization-specific topics. Results are analysed using norm-based comparisons from a multi-organization reference database and feed into a participatory process of feedback, measure development and follow-up evaluation.
iWorkHealth, 2021[256]	Online, company-administered psychosocial health assessment tool designed to help employers understand their workforce's well-being at work. It covers three domains: individual resilience (individual mental well-being), organisational resilience or workplace stressors (including factors such as job demands, job recognition and organisational culture), and well-being at work (including work-related stress, burnout and depressive symptoms).	Self-report questionnaire consisting of 44 items with Likert-type response options. Responses are aggregated and anonymized at company level. The tool generates company reports based on aggregated and anonymised responses which can be segmented by departments, providing an overview of factors and patterns influencing employees' mental well-being. Results are accompanied by recommended interventions to support employers in improving workplace psychosocial conditions.

253 Please note that the JCQ-2 is the new version of the Job Content Questionnaire (JCQ) (1985). Access can be requested at: https://www.jcqcenter.com/

254 Open access at: https://www.insst.es/documentacion/herramientas-de-prl/aip/fpsico-factores-psicosociales-metodo-evaluacion-version-4-1-2022

255 Available at: https://www.baua.de/DE/Angebote/Publikationen/Aufsaetze/artikel3256.html

256 This tool was developed by Singapores' Ministry of Manpower's Workplace Safety and Health (WSH) Institute, in partnership with the WSH Council, Institute of Mental Health, Changi General Hospital and Health Promotion Board. Available at: https://www.tal.sg/wshc/programmes/iworkhealth/overview

Tool/ Instrument	Objective	Methodology
Healthy Work Survey (HWS), 2021[257]	Assesses workplace psychosocial hazards and work organization factors, drawing on established items and scales from theory-based job stress models and national survey measures, such as job demands, control, social support, rewards, job insecurity and work-family conflict. It also covers emerging stressors, including scheduling practices, electronic monitoring, workplace bullying, emotional labour and procedural justice.	Self-report questionnaire with 55 core work-related items and optional sections. Responses generate automated individual or organizational reports. Organizational results are aggregated, benchmarked against national US data, classified into three risk levels, and used to "red flag" priority psychosocial hazards where group averages and high-risk worker percentages exceed national benchmarks. Results support awareness, organizational diagnosis, risk prioritization and follow-up interventions.
Copenhage Psychosocial Questionnaire (COPSOQ) (curent version III), 2019[258]	Designed as a cross-sector and cross-occupational tool for workplace psychosocial risk assessment and for organizational development. It covers a broad range of domains: demands at work, work organization and job contents, interpersonal relations and leadership, work-individual interface, social capital, offensive behaviours, health and well-being.	Self-report questionnaire available in a long version (148 itemscrvrfglr432 covering 45 dimensions), which can be adapted into medium and short versions. Workers rate the frequency or level of agreement regarding different aspects of work on Likert-type items. Item responses are converted to 0-100 scale scores and aggregated into dimension scores for each domain, allowing comparison over time and, where reference values are available, benchmarking to identify elevated psychosocial risks.
Effort-Reward Imbalance (ERI) Questionnaire, 2012[259]	Assesses elements of the work environment and workers' experiences and perceptions to evaluate imbalance between efforts and rewards, using three dimensions: Effort (e.g. demands, time pressure), Reward (e.g. promotions, respect, job security) and Over-commitment (e.g. personal pattern of excessive striving and difficulty disengaging from work).	Self-report questionnaire (long version 22 items, short version 16 items) using 4-point Likert scale. Item scores are combined into subscales for Effort, Reward and Over-commitment, and an Effort-Reward ratio is cal-culated (values >1 indicating more effort than reward). The ratio is typically interpreted as a continuous indicator of imbalance or grouped into categories (e.g. low/medium/high) for comparison between groups and over time.
Psychosocial Safety Climate Scale (PSC-12), 2010[260]	Assesses the organizational climate for psychological safety and health across four domains: management commitment, management priority, organizational communication and participation regarding psychosocial risk.	12-item self-report questionnaire on a 5-point Likert scale. Scores are averaged to produce a total PSC score, which can be classified into benchmark categories (e.g. high, medium, low, very low PSC) to indicate the level of psychosocial risk and priority for preventive action.
Australian Workplace Barometer (AWB), 2009[261]	National surveillance instrument designed to monitor psychosocial risk factors, psychosocial safety climate (PSC) and worker health and productivity in Australia. It contains 16 subscales that assess PSC, job demands, job resources, health outcomes and motivational outcomes.	Self-report questionnaire administered to workers using Likert-type scales from validated instruments (e.g. PSC-12, ERI components). Scale scores are calculated for key domains and can be benchmarked against national reference data from AWB waves to identify high-risk industries and occupations, estimate productivity costs, and track changes in psychosocial risks and health outcomes over time.

257 Open access to the free online asssessment tool available at: https://www.healthywork.org/what/assess/

258 This international instrument is available in 28 languages and counts with several validation for occupational groups and national contexts (including, but not limited to, the Peruvian version called CENSOPAS-COPSOQ). The English version is available for open access at: https://www.copsoq-network.org/li-cence-guidelines-and-questionnaire

259 Open access to long and short versions at: https://www.uniklinik-duesseldorf.de/fileadmin/Fuer-Patienten-und-Besucher/Kliniken-Zentren-Institute/Institute/Institut_fuer_Medizinische_Soziologie/Dateien/ERI/ERI_Psychometric-New.pdf

260 The PSC scale has been translated to 9 languages and can be found at: https://stresscafe.net/psc-tools/

261 See Dollard et al. (2014) for further details on the scales and their origins at: https://www.safeworkaustralia.gov.au/system/files/documents/1702/the-australian-workplace-barometer-report.pdf

Tool/ Instrument	Objective	Methodology
Work Organisation Assessment Questionnaire, 2006[262]	Developed as part of a risk assessment and risk reduction methodology for hazards inherent in the design and management of work. Assesses five broad categories: relationships with management, reward and recognition, workload, quality of relationships with colleagues, and quality of physical environment.	Self-report questionnaire of 28 items where workers rate each aspect of work on a 5-point Likert scale. Item scores are combined into subscale scores and an overall index of quality of work organization, useful as a means of identifying "hot spots" in organizations and informing subsequent discussions between managers and other employees over the exact nature of those problems and reasonable ways of dealing with them.
Demand Induced Strain Questionnaire (DISQ) (version 3.0), 2004[263]	Assesses multidimensional job demands and job resources, distinguishing six dimensions: cognitive, emotional and physical job demands, and cognitive, emotional and physical job resources. It is used to profile work environments and examine how specific types of resources may compensate for corresponding demands.	Self-report questionnaire of 31 items rated on a 5-point frequency Likert scale. Items are grouped into six subscales (three demand and three resource scales); mean scores are calculated for each subscale and used at group/organizational level to characterize demand-resource profiles and, in research, to test interactions and "matching" effects between demands, resources and outcomes.
Health & Safety Executive Management Standards Indicator Tool (HSE-MS IT), 2003[264]	Supports employers to apply the Management Standards for work-related stress by assessing six key areas of work design: demands, control, support, relationships, role and change, and by describing desirable organizational conditions ("good practice") for each area.	35-item self-report questionnaire on a 5-point Likert scale referring to experiences over the last six months. Responses are analysed with the HSE Management Standards Analysis Tool, which provides scores from 1 (poor) to 5 (desirable) for each domain and allows comparison over time and between groups.
The People at Work Risk Assessment Tool for Psychosocial Hazards (PAW), 2003[265]	Assesses psychosocial risk at the task, organizational and external-to-work levels. Multi-level development of the JCQ tool, with Demand, Control and Stability-Support operationalised at the task and the organizational level.	Self-report questionnaire of 47 items, measuring 13 common psychosocial hazards on a 7-point Likert scale. A range of behavioural experiences (bullying and violence), demographic questions, and outcome variables are also included. Allows comparison over time and between groups.
General Nordic Questionnaire (QPSNordic), 2000[266]	Assesses psychological, social and organizational working conditions, including job demands, role expectations, control, predictability and mastery of work, social interactions (including bullying and harassment), leadership, organizational climate and culture, work-private life interface, commitment and work motives, with the aim of supporting organizational development, documenting changes in working conditions and researching links between work, health and productivity.	Self-report multiple-choice questionnaire (full version of 129 items or short version QPSNordic 34+ with 37 items) where workers rate the frequency or degree of agreement on 5-point Likert scale. Data are analysed at group/organizational level (e.g. means, distributions), fed back to workers in survey feedback meetings, and used to identify priorities for improvement and to monitor change over time.

262 See Griffiths et al. (2006) for further details at: https://pmc.ncbi.nlm.nih.gov/articles/PMC2078053/
263 See Bova, De Jonge, and Guglielmi (2015) for further details at: https://onlinelibrary.wiley.com/doi/10.1002/smi.2550
264 Available for open access at: https://www.hse.gov.uk/stress/standards/downloads.htm
265 Available at: https://www.peopleatwork.gov.au/webcopy/peopleatwork
266 Open access to the user guide at: https://researchdata.se/en/catalogue/dataset/snd0837-4/1.0



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References

Adams, Byron G., M. Christina Meyers, and Lusanda Sekaja. 2020. "Positive Leadership: Relationships with Employee Inclusion, Discrimination, and Well-Being." *Applied Psychology* 69 (4). https://doi.org/10.1111/apps.12230.

ADP Research. 2025. "People at Work 2025: A Global Workforce View." *ADP Research*. https://www.adpresearch.com/research/people-at-work-2025-a-global-workforce-view/.

Adriaenssens, Jef, Véronique De Gucht, and Stan Maes. 2015. "Determinants and Prevalence of Burnout in Emergency Nurses: A Systematic Review of 25 Years of Research." *International Journal of Nursing Studies* 52 (2): 649–61. https://doi.org/10.1016/j.ijnurstu.2014.11.004.

Agbenyikey, Wilfred, Jian Li, Sung-Il Cho, Sarven S. McLinton, Maureen Dollard, Maren Formazin, Bongkyoo Choi, Irene Houtman, and Robert Karasek. 2025. "An International Comparative Reliability and Concurrent Validity Assessment of the Multi-Level Job Content Questionnaire (JCQ) 2.0." *International Journal of Environmental Research and Public Health* 22 (9): 1435. https://doi.org/10.3390/ijerph22091435.

Al-Malki, Mohammed, and Wang Juan. 2018. "Impact of Laissez-Faire Leadership on Role Ambiguity and Role Conflict: Implications for Job Performance." *International Journal of Management Science and Business Administration* 4 (1): 29–43. https://doi.org/10.18775/ijied.1849-7551-7020.2015.41.2003.

Alalhareth, Ahmed Saleh Mohammed, Hussain Abdullah Yahya Alghubari, Bader Salem Mohammad Alsharman, Ali Mohammed Abdullah Al Faraj, Hayan Mohammed Ali Alyami, Ibrahim Yahia Yahia Alsalem, Hamad Mahdi Al Bouzbedah, Abdullah Nasser Ali Al ajje, Hassan Rashed Hassan Alfuhayd, and Mohammed Ali Alswedan. 2024. "Chronic Occupational Stress and Health Outcomes: A Systematic Review of Recent Findings." *Journal of Ecohumanism* 3 (8): 909–17. https://doi.org/10.62754/joe.v3i8.4777.

Albrecht, Simon L., Camille R. Green, and Andrew Marty. 2021. "Meaningful Work, Job Resources, and Employee Engagement." *Sustainability* 13 (7): 4045. https://doi.org/10.3390/su13074045.

Allan, Blake A. 2017. "Task Significance and Meaningful Work: A Longitudinal Study." *Journal of Vocational Behavior* 102 (102): 174–82. https://doi.org/10.1016/j.jvb.2017.07.011.

Amiri, S., and S. Behnezhad. 2020. "Association between Job Strain and Sick Leave: A Systematic Review and Meta-Analysis of Prospective Cohort Studies." *Public Health* 185 (August): 235–42. https://doi.org/10.1016/j.puhe.2020.05.023.

Amoadu, Mustapha, Edward Wilson Ansah, and Jacob Owusu Sarfo. 2023. "Influence of Psychosocial Safety Climate on Occupational Health and Safety: A Scoping Review." *BMC Public Health* 23 (1). https://doi.org/10.1186/s12889-023-16246-x.

Andel, Ross, Michael Crowe, Elizabeth A. Hahn, James A. Mortimer, Nancy L. Pedersen, Laura Fratiglioni, Boo Johansson, and Margaret Gatz. 2011. "Work-Related Stress May Increase the Risk of Vascular Dementia." *Journal of the American Geriatrics Society* 60 (1): 60–67. https://doi.org/10.1111/j.1532-5415.2011.03777.x.

Antoni, Michael H., and Firdaus S. Dhabhar. 2019. "The Impact of Psychosocial Stress and Stress Management on Immune Responses in Patients with Cancer." *Cancer* 125 (9): 1417–31. https://doi.org/10.1002/cncr.31943.

Aon, and TELUS Health. 2023. "Asia Mental Health Index Report." Www.aon.com. 2023. https://www.aon.com/apac/insights/health/asia-mental-health-index-report.

APA (American Psychological Association). 2024. "Coping with Stress at Work." Apa.org. October 22, 2024. https://www.apa.org/topics/healthy-workplaces/work-stress.

Arends, Iris, Christophe Prinz, and Femke Abma. 2017. "Job Quality, Health and At-Work Productivity." *OECD Social, Employment and Migration Working Papers*, June. https://doi.org/10.1787/43ff6bdc-en.

Arlinghaus, Anna, and Friedhelm Nachreiner. 2016. "Unusual and Unsocial? Effects of Shift Work and Other Unusual Working Times on Social Participation." In *Social and Family Issues in Shift Work and Non Standard Working Hours*, 39–57. Springer, Cham. https://doi.org/10.1007/978-3-319-42286-2_3.

Arora, Vipra, and T Kamalanabhan. 2013. "Linking Supervisor and Coworker Support to Employee Innovative Behavior at Work: Role of Psychological Conditions." https://www.aabri.com/NO2013Manuscripts/NO13079.pdf.

Aust, Birgit, Jeppe Lykke Møller, Mads Nordentoft, Karen Bo Frydendall, Elizabeth Bengtsen, Andreas Brøgger Jensen, Anne Helene Garde, et al. 2023. "How Effective Are Organizational-Level Interventions in Improving the Psychosocial Work Environment, Health, and Retention of Workers? A Systematic Overview of Systematic Reviews." *Scandinavian Journal of Work, Environment & Health* 49 (5): 315–29. https://doi.org/10.5271/sjweh.4097.

Australian Government. 2024. "Federal Register of Legislation - Work Health and Safety (Managing Psychosocial Hazards at Work) Code of Practice 2024." Legislation.gov.au. 2024. https://www.legislation.gov.au/F2024L01380/latest/text.

Azagba, Sunday, and Mesbah F Sharaf. 2011. "The Effect of Job Stress on Smoking and Alcohol Consumption." *Health Economics Review* 1 (1). https://doi.org/10.1186/2191-1991-1-15.

Baiocco, Sara, Enrique Fernandez-Macías, Rani Uma, and Annarosa Pesole. 2022. "The Algorithmic Management of Work and Its Implications in Different Contexts." Seville: European Commission. https://publications.jrc.ec.europa.eu/repository/handle/JRC129749.

Bakker, Arnold B., and Evangelia Demerouti. 2017. "Job Demands–Resources Theory: Taking Stock and Looking Forward." *Journal of Occupational Health Psychology* 22 (3): 273–85. https://doi.org/10.1037/ocp0000056.

Baldwin, Susanna. 2006. *Organisational Justice*. Brighton: Institute for Employment Studies.

Ball, Kirstie. 2010. "Workplace Surveillance: An Overview." *Labor History* 51 (1): 87–106. https://doi.org/10.1080/00236561003654776.

———. 2021. "Electronic Monitoring and Surveillance in the Workplace." JRC Publications Repository. September 22, 2021. https://publications.jrc.ec.europa.eu/repository/handle/JRC125716.

Beehr, Terry. 1995. *Psychological Stress in the Workplace (Psychology Revivals)*. Routledge. https://doi.org/10.4324/9781315795560.

Bell, Ruth. 2017. "Psychosocial Pathways and Health Outcomes: Informing Action on Health Inequalities." Public Health England & UCL Institute of Health Equity. https://assets.publishing.service.gov.uk/media/5a74d3e440f0b65f613228d7/Psychosocial_pathways_and_health_equity.pdf.

Benach, J., A. Vives, M. Amable, C. Vanroelen, G. Tarafa, and C. Muntaner. 2014. "Precarious Employment: Understanding an Emerging Social Determinant of Health." *Annual Review of Public Health* 35 (1): 229–53. https://doi.org/10.1146/annurev-publhealth-032013-182500.

Bernal, Dinora, Javier Campos-Serna, Aurelio Tobias, Sergio Vargas-Prada, Fernando G. Benavides, and Consol Serra. 2015. "Work-Related Psychosocial Risk Factors and Musculoskeletal Disorders in Hospital Nurses and Nursing Aides: A Systematic Review and Meta-Analysis." *International Journal of Nursing Studies* 52 (2): 635–48. https://doi.org/10.1016/j.ijnurstu.2014.11.003.

Bezzina, Aaron, Emma Austin, Ha Nguyen, and Carole James. 2023. "Workplace Psychosocial Factors and Their Association with Musculoskeletal Disorders: A Systematic Review of Longitudinal Studies." *AAOHN Journal* 71 (12). https://doi.org/10.1177/21650799231193578.

Bierens de Haan, Barthold. 2020. "Humanitarian Action and Armed Conflict: Coping with Stress." *Icrc.org*. https://www.icrc.org/en/publication/0576-humanitarian-action-and-armed-conflict-coping-stress.

Billings, Jo, Nicholas Zhan, Helen Nicholls, Paul Burton, Maya Zosmer, Idit Albert, Nick Grey, et al. 2023. "Post-Incident Psychosocial Interventions after a Traumatic Incident in the Workplace: A Systematic Review of Current Research Evidence and Clinical Guidance." *European Journal of Psychotraumatology* 14 (2). https://doi.org/10.1080/20008066.2023.2281751.

Bittle, Steven, Sarah Waters, China Mills, and Loïc Lerouge. 2025. "Work-Related Suicide: An International Social Justice Perspective." *Crime, Law and Social Change* 83 (1). https://doi.org/10.1007/s10611-025-10237-2.

Bluff, Elizabeth, and Neil Gunningham. 2026. "Principle, Process, Performance or What? New Approaches to OHS Standards Setting." *The Australian National University*, 12–42. https://researchportalplus.anu.edu.au/en/publications/principle-process-performance-or-what-new-approaches-to-ohs-stand.

Boini, Stephanie, Eve Bourgkard, Jean Ferrières, and Yolande Esquirol. 2022. "What Do We Know about the Effect of Night-Shift Work on Cardiovascular Risk Factors? An Umbrella Review." *Frontiers in Public Health* 10 (November). https://doi.org/10.3389/fpubh.2022.1034195.

Boot, Cécile RL, Anthony D LaMontagne, and Ida EH Madsen. 2024. "Fifty Years of Research on Psychosocial Working Conditions and Health: From Promise to Practice." *Scandinavian Journal of Work Environment & Health* 50 (August): 395–405. https://doi.org/10.5271/sjweh.4180.

Bova, Nicoletta, Jan De Jonge, and Dina Guglielmi. 2013. "The Demand-Induced Strain Compensation Questionnaire: A Cross-National Validation Study." *Stress and Health* 31 (3): 236–44. https://doi.org/10.1002/smi.2550.

Bowling, Nathan A., and Cristina Kirkendall. 2012. "Workload: A Review of Causes, Consequences, and Potential Interventions." *Contemporary Occupational Health Psychology* 2 (March): 221–38. https://doi.org/10.1002/9781119942849.ch13.

Brborović, Hana, Qëndresë Daka, Kushtrim Dakaj, and Ognjen Brborović. 2017. "Antecedents and Associations of Sickness Presenteeism and Sickness Absenteeism in Nurses: A Systematic Review." *International Journal of Nursing Practice* 23 (6): e12598. https://doi.org/10.1111/ijn.12598.

Bronkhorst, Babette, Lars Tummers, Bram Steijn, and Dominique Vijverberg. 2015. "Organizational Climate and Employee Mental Health Outcomes." *Health Care Management Review* 40 (3): 254–71. https://doi.org/10.1097/hmr.0000000000000026.

Bruk-Lee, Valentina, and Paul E. Spector. 2006. "The Social Stressors-Counterproductive Work Behaviors Link: Are Conflicts with Supervisors and Coworkers the Same?" *Journal of Occupational Health Psychology* 11 (2): 145–56. https://doi.org/10.1037/1076-8998.11.2.145.

Burr, Hermann, Hanne Berthelsen, Salvador Moncada, Matthias Nübling, Emilie Dupret, Yucel Demiral, John Oudyk, et al. 2019. "The Third Version of the Copenhagen Psychosocial Questionnaire." *Safety and Health at Work* 10 (4): 482–503. https://doi.org/10.1016/j.shaw.2019.10.002.

Buruck, Gabriele, Anne Tomaschek, Johannes Wendsche, Elke Ochsmann, and Denise Dörfel. 2019. "Psychosocial Areas of Worklife and Chronic Low Back Pain: A Systematic Review and Meta-Analysis." *BMC Musculoskeletal Disorders* 20 (1). https://doi.org/10.1186/s12891-019-2826-3.

Butterworth, Peter, Kim Kiely, and Perri Timmins. 2017. "The Role of Psychosocial Work Factors in the Decision to Retire Early." Canberra: Safe Work Australia. https://www.safeworkaustralia.gov.au/system/files/documents/1712/role-of-psychosocial-work-factors-in-the-decision-to-retire-early.pdf.

Cai, Chenxi, Ben Vandermeer, Rshmi Khurana, Kara Nerenberg, Robin Featherstone, Meghan Sebastianski, and Margie H. Davenport. 2019. "The Impact of Occupational Shift Work and Working Hours during Pregnancy on Health Outcomes: A Systematic Review and Meta-Analysis." *American Journal of Obstetrics and Gynecology* 221 (6). https://doi.org/10.1016/j.ajog.2019.06.051.

Campos-Serna, Javier, Elena Ronda-Pérez, Lucia Artazcoz, Bente E Moen, and Fernando G Benavides. 2013. "Gender Inequalities in Occupational Health Related to the Unequal Distribution of Working and Employment Conditions: A Systematic Review." *International Journal for Equity in Health* 12 (1): 57. https://doi.org/10.1186/1475-9276-12-57.

Cappelli, Peter, and Martin J. Conyon. 2018. "What Do Performance Appraisals Do?" *ILR Review* 71 (1): 88–116.

Casalini, Sara. 2023. "The Essential Role of Supervision: Fostering Growth, Excellence, and Accountability." *Journal of Educational Sciences Research* 13 (1). https://doi.org/10.22521/JESR.2023.13.1.09.

Cavanaugh, Marcie A., Wendy R. Boswell, Mark V. Roehling, and John W. Boudreau. 2000. "An Empirical Examination of Self-Reported Work Stress among U.S. Managers." *Journal of Applied Psychology* 85 (1): 65–74. https://doi.org/10.1037/0021-9010.85.1.65.

Chiang, Flora F. T., and Thomas A. Birtch. 2012. "The Performance Implications of Financial and Non-Financial Rewards: An Asian Nordic Comparison." *Journal of Management Studies* 49 (3): 538–70. https://doi.org/10.1111/j.1467-6486.2011.01018.x.

Chida, Yoichi, Mark Hamer, Jane Wardle, and Andrew Steptoe. 2008. "Do Stress-Related Psychosocial Factors Contribute to Cancer Incidence and Survival?" *Nature Clinical Practice Oncology* 5 (8): 466–75. https://doi.org/10.1038/ncponc1134.

Chirico, Francesco, Tarja Heponiemi, Milena Pavlova, Salvatore Zaffina, and Nicola Magnavita. 2019. "Psychosocial Risk Prevention in a Global Occupational Health Perspective. A Descriptive Analysis." *International Journal of Environmental Research and Public Health* 16 (14): 2470. https://doi.org/10.3390/ijerph16142470.

Christian, Michael S., Jill C. Bradley, J. Craig Wallace, and Michael J. Burke. 2009. "Workplace Safety: A Meta-Analysis of the Roles of Person and Situation Factors." *Journal of Applied Psychology* 94 (5): 1103–27. https://doi.org/10.1037/a0016172.

CIPD. 2025. "Good Work Index." CIPD. June 9, 2025. https://www.cipd.org/uk/knowledge/reports/goodwork/.

Colquitt, Jason A., Brent A. Scott, Jessica B. Rodell, David M. Long, Cindy P. Zapata, Donald E. Conlon, and Michael J. Wesson. 2013. "Justice at the Millennium, a Decade Later: A Meta-Analytic Test of Social Exchange and Affect-Based Perspectives." *Journal of Applied Psychology* 98 (2): 199–236. https://doi.org/10.1037/a0031757.

Conseil national du travail. 2023. "Risques Psychosociaux Dont Le Burn-Out." Cnt-Nar.be. 2023. https://cnt-nar.be/fr/dossiers-thematiques/risques-psychosociaux-dont-le-burn-out.

Countouris, Nicola, and Valerio De Stefano. 2019. "New Trade Union Strategies for New Forms of Employment." Brussels: ETUC. https://www.etuc.org/sites/default/files/publication/file/2019-04/2019_new%20trade%20union%20strategies%20for%20new%20forms%20of%20employment_0.pdf.

Cox, Thomas, and Amanda Griffiths. 2010. "Work-Related Stress: A Theoretical Perspective." In *Occupational Health Psychology*, edited by Stavroula Leka and Jonathan Houdmont, 31–56. Malden, Ma: Wiley-Blackwell.

Cox, Tom. 1993. *Stress Research and Stress Management: Putting Theory to Work*. London: Health and Safety Executive.

———. 2000. *Organisational Interventions for Work Stress: A Risk Management Approach*. Sudbury: Health and Safety Executive (HSE) Books.

Cox, Tom, and Sue Cox. 1993. "Occupational Health: Control and Monitoring of Psychosocial and Organisational Hazards at Work." *Journal of the Royal Society of Health* 113 (4): 201–5. https://doi.org/10.1177/146642409311300411.

Cropanzano, Russell, David E. Bowen, and Stephen W. Gilliland. 2007. "The Management of Organizational Justice." *Academy of Management Perspectives* 21 (4): 34–48. https://doi.org/10.5465/amp.2007.27895338.

CSA Group (Canadian Standards Association), and Bureau de normalisation du Québec (BNQ). 2013. "CAN/CSA-Z1003-13/BNQ 9700-803/2013 (R2022), Psychological Health and Safety in the Workplace." CSA Group. 2013. https://www.csagroup.org/article/can-csa-z1003-13-bnq-9700-803-2013-r2022-psychological-health-and-safety-in-the-workplace/.

De Crom, Nellie, and S. Rothmann. 2018. "Demands–Abilities Fit, Work Beliefs, Meaningful Work and Engagement in Nature-Based Jobs." *SA Journal of Industrial Psychology* 44 (March). https://doi.org/10.4102/sajip.v44i0.1496.

De Witte, Hans. 2005. "Job Insecurity: Review of the International Literature on Definitions, Prevalence, Antecedents and Consequences." *SA Journal of Industrial Psychology* 31 (4). https://doi.org/10.4102/sajip.v31i4.200.

Dediu, Vlad, Stavroula Leka, and Aditya Jain. 2018. "Job Demands, Job Resources and Innovative Work Behaviour: A European Union Study." *European Journal of Work and Organizational Psychology* 27 (3): 310–23. https://doi.org/10.1080/1359432x.2018.1444604.

Demerouti, Evangelia, and Niels Adaloudis. 2024. "Addressing Burnout in Organisations: A Literature Review." *Econstor.eu*. https://hdl.handle.net/10419/302237.

Demerouti, Evangelia, Arnold B. Bakker, Friedhelm Nachreiner, and Wilmar B. Schaufeli. 2001. "The Job Resources Model of Burnout." *Journal of Applied Psychology* 86 (3): 499–512. https://doi.org/10.1037/0021-9010.86.3.499.

Derdowski, Lukasz Andrzej, and Gro Ellen Mathisen. 2023. "Psychosocial Factors and Safety in High-Risk Industries: A Systematic Literature Review." *Safety Science* 157: 105948. https://doi.org/10.1016/j.ssci.2022.105948.

Descatha, Alexis, Grace Sembajwe, Frank Pega, Yuka Ujita, Michael Baer, Fabio Boccuni, Cristina Di Tecco, et al. 2020. "The Effect of Exposure to Long Working Hours on Stroke: A Systematic Review and Meta-Analysis from the WHO/ILO Joint Estimates of the Work-Related Burden of Disease and Injury." *Environment International* 142 (1): 105746. https://doi.org/10.1016/j.envint.2020.105746.

Dhanani, Lindsay Y., Matthew L. LaPalme, and Dana L. Joseph. 2021. "How Prevalent Is Workplace Mistreatment? A Meta-Analytic Investigation." *Journal of Organizational Behavior* 42 (8). https://doi.org/10.1002/job.2534.

Dobrotić, Ivana, and Sonja Blum. 2019. "Inclusiveness of Parental-Leave Benefits in Twenty-One European Countries: Measuring Social and Gender Inequalities in Leave Eligibility." *Social Politics: International Studies in Gender, State & Society* 27 (3). https://doi.org/10.1093/sp/jxz023.

Dollard, Maureen F, Christian Dormann, Michelle R. Tuckey, and Jordi Escartín. 2017. "Psychosocial Safety Climate (PSC) and Enacted PSC for Workplace Bullying and Psychological Health Problem Reduction." *European Journal of Work and Organizational Psychology* 26 (6): 844–57. https://doi.org/10.1080/1359432x.2017.1380626.

Dollard, Maureen F., May Loh, Harry Becher, Daniel Neser, Sophie Richter, Amy Zadow, Ali Afsharian, and Rachael Potter. 2024. "PSC as an Organisational Level Determinant of Working Time Lost and Expenditure Following Workplace Injuries and Illnesses." *Safety Science* 177 (September): 106602–2. https://doi.org/10.1016/j.ssci.2024.106602.

Dollard, Maureen F., and Arnold B. Bakker. 2010. "Psychosocial Safety Climate as a Precursor to Conducive Work Environments, Psychological Health Problems, and Employee Engagement." *Journal of Occupational and Organizational Psychology* 83 (3): 579–99. https://doi.org/10.1348/096317909X470690.

Dollard, Maureen F., Christian Dormann, and Mohd Awang Idris. 2019. "Psychosocial Safety Climate: A New Work Stress Theory and Implications for Method." In *Psychosocial Safety Climate*, edited by Maureen F. Dollard, Christian Dormann, and Mohd Awang Idris, 3–30. Springer, Cham. https://doi.org/10.1007/978-3-030-20319-1_1.

Dollard, Maureen, Tessa Bailey, Sarven McLinton, Penelope Richards, Wes McTernan, Anne Taylor, and Stephanie Bond. 2012. "The Australian Workplace Barometer: Report on Psychosocial Safety Climateand Worker Health in Australia." Safe Work Australia. https://www.safeworkaustralia.gov.au/system/files/documents/1702/the-australian-workplace-barometer-report.pdf.

Dragano, N., and L. Schneider. 2011. "Psychosoziale Arbeitsbelastungen Als Prädiktoren Der Krankheitsbedingten Frühberentung: Ein Beitrag Zur Beurteilung Des Rehabilitationsbedarfs." *Die Rehabilitation* 50 (01): 28–36. https://doi.org/10.1055/s-0030-1270431.

Duchaine, Caroline S., Karine Aubé, Mahée Gilbert-Ouimet, Michel Vézina, Ruth Ndjaboué, Victoria Massamba, Denis Talbot, et al. 2020. "Psychosocial Stressors at Work and the Risk of Sickness Absence due to a Diagnosed Mental Disorder." *JAMA Psychiatry* 77 (8): 842–51. https://doi.org/10.1001/jamapsychiatry.2020.0322.

Eddy, Pennie, Eleanor H. Wertheim, Matthew W. Hale, and Bradley J. Wright. 2023. "A Systematic Review and Revised Meta-Analysis of the Effort-Reward Imbalance Model of Workplace Stress and Hypothalamic-Pituitary-Adrenal Axis Measures of Stress." *Psychosomatic Medicine* 85 (5): 450. https://doi.org/10.1097/PSY.0000000000001155.

Edmondson, Amy C., and Zhike Lei. 2014. "Psychological Safety: The History, Renaissance, and Future of an Interpersonal Construct." *Annual Review of Organizational Psychology and Organizational Behavior* 1 (1): 23–43. https://doi.org/10.1146/annurev-orgpsych-031413-091305.

Edwards, Jeffrey, Robert Caplan, and Van Harrison. 2025. "Person-Environment Fit Theory: Conceptual Foundations, Empirical Evidence, and Directions for Future Research." *Research Explorer the University of Manchester*, 28–67. https://research.manchester.ac.uk/en/publications/person-environment-fit-theory-conceptual-foundations-empirical-ev/.

Efimov, Ilona, Volker Harth, and Stefanie Mache. 2020. "Health-Oriented Self- and Employee Leadership in Virtual Teams: A Qualitative Study with Virtual Leaders." *International Journal of Environmental Research and Public Health* 17 (18): 6519. https://doi.org/10.3390/ijerph17186519.

Egan, M., C. Bambra, S. Thomas, M. Petticrew, M. Whitehead, and H. Thomson. 2007. "The Psychosocial and Health Effects of Workplace Reorganisation. 1. A Systematic Review of Organisational-Level Interventions That Aim to Increase Employee Control." *Journal of Epidemiology & Community Health* 61 (11): 945–54. https://doi.org/10.1136/jech.2006.054965.

Eguchi, Hisashi, Kazuhiro Watanabe, Norito Kawakami, Emiko Ando, Kotaro Imamura, Asuka Sakuraya, Natsu Sasaki, et al. 2023. "Work-Related Psychosocial Factors and Inflammatory Markers: A Systematic Review and Meta-Analysis." *Journal of Psychosomatic Research* 170 (July): 111349. https://doi.org/10.1016/j.jpsychores.2023.111349.

Einarsen, Kari, Reidar J. Mykletun, Ståle Valvatna Einarsen, Anders Skogstad, and Denise Salin. 2017. "Ethical Infrastructure and Successful Handling of Workplace Bullying." *Nordic Journal of Working Life Studies* 7 (1). https://doi.org/10.18291/njwls.v7i1.81398.

Einarsen, Ståle, Helge Hoel, Cary L Cooper, and Dieter Zapf. 2020. *Bullying and Harassment in the Workplace*. CRC Press.

Ellen Pinkos Cobb. 2022. *Managing Psychosocial Hazards and Work-Related Stress in Today's Work Environment*. 1st ed. New York: Routledge. https://doi.org/10.4324/9781003187349.

Elovainio, Marko, Mika Kivimäki, and Jussi Vahtera. 2002. "Organizational Justice: Evidence of a New Psychosocial Predictor of Health." *American Journal of Public Health* 92 (1): 105–8. https://doi.org/10.2105/ajph.92.1.105.

EU-OSHA (European Agency for Safety and Health at Work). 2021. "Improving Compliance with Occupational Safety and Health Regulations: An Overarching Review European Risk Observatory Literature Review European Agency for Safety and Health at Work." European Agency for Safety and Health at Work. https://osha.europa.eu/sites/default/files/Improving_compliance_OSH_regulatons_lit%20review.pdf.

———. 2025a. "First Findings of the Fourth European Survey of Enterprises on New and Emerging Risks (ESENER 2024)." *EU-OSHA*. European Agency for Safety and Health at Work. https://osha.europa.eu/en/publications/first-findings-fourth-european-survey-enterprises-new-and-emerging-risks-esener-2024.

———. 2025b. "Denmark: Psychosocial Risk Prevention – Strategies and Legislation." *EU-OSHA*. European Agency for Safety and Health at Work. https://osha.europa.eu/en/publications/denmark-psychosocial-risk-prevention-strategies-and-legislation.

———. 2025c. "OSH Pulse 2025 - Occupational Safety and Health in the Era of Climate and Digital Change." EU-OSHA. European Agency for Safety and Health at Work. September 23, 2025. https://osha.europa.eu/en/publications/osh-pulse-2025-occupational-safety-and-health-era-climate-and-digital-change.

Eurofound (European Foundation for the Improvement of Living and Working Conditions). 2017a. "Exploring Self-Employment in the European Union." *Europa.eu*. Luxembourg: Publications Office of the European Union. https://doi.org/10.2806/651917.

———. 2017b. "Sixth European Working Conditions Survey – Overview Report (2017 Update)." *Europa.eu*. Luxembourg: Publications Office of the European Union. https://doi.org/10.2806/422172.

———. 2022. "Working Conditions in the Time of COVID-19: Implications for the Future Working Conditions and Sustainable Work European Working Conditions Telephone Survey 2021." Luxembourg: Publications Office of the European Union. https://assets.eurofound.europa.eu/f/279033/30357e5fd8/ef22012en.pdf.

———. 2025a. "EWCS 2024 Data Explorer." Eurofound. September 1, 2025. https://www.eurofound.europa.eu/en/surveys-and-data/surveys/european-working-conditions-survey/ewcs-2024/dashboard.

———. 2025b. "European Working Conditions Survey 2024: First Findings." Publications Office of the EU. November 10, 2025. https://doi.org/10.2806/9436104.

Eurofound, and ILO (International Labour Organization). 2019. "Working Conditions in a Global Perspective." Publications Office of the European Union, Luxembourg, and International Labour Organization, Geneva. https://doi.org/10.2806/870542.

European Commission. 2011. *Report on the Implementation of the European Social Partners' Framework Agreement on Work-Related Stress*. Vol. Commission Staff Working Paper, SEC (2011) 241 final. Brussels. https://www.europarl.europa.eu/RegData/docs_autres_institutions/commission_europeenne/sec/2011/0241/COM_SEC%282011%290241_EN.pdf.

European Social Partners. 2008. "Implementation of the European Autonomous Framework Agreement on Work-Related Stress." https://www.businesseurope.eu/wp-content/uploads/2025/02/2009-01163-E-e7e-1.pdf.

Ferrie, Jane E., Hugo Westerlund, Marianna Virtanen, Jussi Vahtera, and Mika Kivimäki. 2008. "Flexible Labor Markets and Employee Health." *SJWEH Supplements*, no. 6: 98–110. https://www.sjweh.fi/article/1257.

Finnish Centre for Pensions. 2025. "Statistical Yearbook of Pensioners in Finland 2024." *Julkari.fi*. Finnish Centre for Pensions. https://www.julkari.fi/items/459fde62-78e9-413a-a5ef-4c79330c633f.

Fishta, Alba, and Eva-Maria Backé. 2015. "Psychosocial Stress at Work and Cardiovascular Diseases: An Overview of Systematic Reviews." *International Archives of Occupational and Environmental Health* 88 (8): 997–1014. https://doi.org/10.1007/s00420-015-1019-0.

Fond, Guillaume, Anderson Loundou, Nora Hamdani, Wahid Boukouaci, Aroldo Dargel, José Oliveira, Matthieu Roger, Ryad Tamouza, Marion Leboyer, and Laurent Boyer. 2014. "Anxiety and Depression Comorbidities in Irritable Bowel Syndrome (IBS): A Systematic Review and Meta-Analysis." *European Archives of Psychiatry and Clinical Neuroscience* 264 (8): 651–60. https://doi.org/10.1007/s00406-014-0502-z.

Fox, Kimberly E., Sydney T. Johnson, Lisa F. Berkman, Marjaana Sianoja, Yenee Soh, Laura D. Kubzansky, and Erin L. Kelly. 2021. "Organisational- and Group-Level Workplace Interventions and Their Effect on Multiple Domains of Worker Well-Being: A Systematic Review." *Work & Stress* 36 (1): 1–30.

French, J.R., and R.D. Caplan. 1972. “Organizational Stress and Individual Strain.” *The Failure of Success* 30 (66): 61–77.

Fryers, Tom. 2006. “Work, Identity and Health.” *Clinical Practice and Epidemiology in Mental Health* 2 (1): 12. https://doi.org/10.1186/1745-0179-2-12.

Gallup. 2022. “The Economic Cost of Poor Employee Mental Health.” *Gallup.com*. https://www.gallup.com/workplace/404174/economic-cost-poor-employee-mental-health.aspx.

———. 2023. “State of the Global Workplace: The Voice of the World’s Employees.” Gallup.

———. 2025a. “Indicator Employee Engagement.” Gallup.com. 2025. https://www.gallup.com/394373/indicator-employee-engagement.aspx.

———. 2025b. “State of the Global Workplace: Understanding Employees, Informing Leaders.” *Gallup*. https://www.gallup.com/workplace/349484/state-of-the-global-workplace.aspx.

Gammarano, Rosina. 2020. “Measuring Job Quality: Difficult but Necessary.” ILOSTAT. January 27, 2020. https://ilostat.ilo.org/measuring-job-quality-difficult-but-necessary/.

Gerhardt, Christin, Norbert K. Semmer, Sabine Sauter, Alexandra Walker, Nathal de Wijn, Wolfgang Kälin, Maria U. Kottwitz, Bernd Kersten, Benjamin Ulrich, and Achim Elfering. 2021. “How Are Social Stressors at Work Related to Well-Being and Health? A Systematic Review and Meta-Analysis.” *BMC Public Health* 21 (1). https://doi.org/10.1186/s12889-021-10894-7.

Gilbreath, Brad. 2005. “Creating Healthy Workplaces: The Supervisor’s Role.” *International Review of Industrial and Organizational Psychology 2004*, January, 93–118. https://doi.org/10.1002/0470013311.ch3.

Glaser, Ronald, and Janice K. Kiecolt-Glaser. 2005. “Stress-Induced Immune Dysfunction: Implications for Health.” *Nature Reviews Immunology* 5 (3): 243–51. https://doi.org/10.1038/nri1571.

Greenberg, Jerald. 1987. “A Taxonomy of Organizational Justice Theories.” *The Academy of Management Review* 12 (1): 9–22. https://doi.org/10.2307/257990.

Greiner, Birgit A, and Ella Arensman. 2022. “The Role of Work in Suicidal Behavior – Uncovering Priorities for Research and Prevention.” *Scandinavian Journal of Work, Environment & Health* 48 (6): 419–24. https://doi.org/10.5271/sjweh.4051.

Griffiths, Amanda, Tom Cox, M Karanika, S Khan, and J-M Tomás. 2006. “Work Design and Management in the Manufacturing Sector: Development and Validation of the Work Organisation Assessment Questionnaire.” *Occupational and Environmental Medicine* 63 (10): 669–75. https://doi.org/10.1136/oem.2005.023671.

Guisolan, S.C., M. Ambrogi, A. Meeussen, F. Althaus, and G. Eperon. 2022. “Health and Security Risks of Humanitarian Aid Workers during Field Missions: Experience of the International Red Cross.” *Travel Medicine and Infectious Disease* 46 (March): 102275. https://doi.org/10.1016/j.tmaid.2022.102275.

Guldenmund, F.W. 2000. “The Nature of Safety Culture: A Review of Theory and Research.” *Safety Science* 34 (1-3): 215–57. https://doi.org/10.1016/s0925-7535(00)00014-x.

Gupta, Roma Kumari, and Chandrabhan M Tembhurnekar. 2024. “AI-Driven Performance Appraisal Systems: A Critical Literature Review of Emerging Issues and Challenges.” *ShodhKosh Journal of Visual and Performing Arts* 5 (7). https://doi.org/10.29121/shodhkosh.v5.i7.2024.3481.

Hackman, J. Richard. 1980. “Work Redesign and Motivation.” *Professional Psychology* 11 (3): 445–55. https://doi.org/10.1037/0735-7028.11.3.445.

Hackman, J. Richard, and Greg R. Oldham. 1976. “Motivation through the Design of Work: Test of a Theory.” *Organizational Behavior and Human Performance* 16 (2): 250–79. https://doi.org/10.1016/0030-5073(76)90016-7.

Hakanen, Jari J., and Arnold B. Bakker. 2017. “Born and Bred to Burn Out: A Life-Course View and Reflections on Job Burnout.” *Journal of Occupational Health Psychology* 22 (3): 354–64. https://doi.org/10.1037/ocp0000053.

Halbesleben, Jonathon R. B. 2010. “A Meta-Analysis of Work Engagement: Relationships with Burnout, Demands, Resources, and Consequences.” In *Work Engagement: A Handbook of Essential Theory and Research*, edited by Arnold B. Bakker. Psychology Press.

Hall, Nicole A., Adam T. Everson, Madison R. Billingsley, and Mary Beth Miller. 2021. “Moral Injury, Mental Health, and Behavioral Health Outcomes: A Systematic Review of the Literature.” *Clinical Psychology & Psychotherapy* 29 (1). https://doi.org/10.1002/cpp.2607.

Harvey, Samuel B., Matthew Modini, Sadhbh Joyce, Josie S. Milligan-Saville, Leona Tan, Arnstein Mykletun, Richard A. Bryant, Helen Christensen, and Philip B. Mitchell. 2017. “Can Work Make You Mentally Ill? A Systematic

Meta-Review of Work-Related Risk Factors for Common Mental Health Problems." *Occupational and Environmental Medicine* 74 (4): 301–10.

Hassard, Juliet, Kevin R. H. Teoh, Gintare Visockaite, Philip Dewe, and Tom Cox. 2018. "The Cost of Work-Related Stress to Society: A Systematic Review." *Journal of Occupational Health Psychology* 23 (1): 1–17. https://doi.org/10.1037/ocp0000069.

Hassard, Juliet, Kevin Teoh, Tom Cox, Philip Dewe, Marlen Cosmar, Robert Gründler, and Danny Flemming. 2014. "Calculating the Costs of Work-Related Stress and Psychosocial Risks – a Literature Review." *European Agency for Safety and Health at Work* https://doi.org/10.2802/20493.

Haveraaen, Lise Aasen, Lisebet Skeie Skarpaas, and Randi Wågø Aas. 2017. "Job Demands and Decision Control Predicted Return to Work: The Rapid-RTW Cohort Study." *BMC Public Health* 17 (1). https://doi.org/10.1186/s12889-016-3942-8.

Heery, Edmund, and Mike Noon. 2008. *A Dictionary of Human Resource Management*. Oxford University Press. https://doi.org/10.1093/acref/9780199298761.001.0001.

Heikkilä, Katarina, S. T. Nyberg, T. Theorell, E. I. Fransson, L. Alfredsson, J. B. Bjorner, S. Bonenfant, et al. 2013. "Work Stress and Risk of Cancer: Meta-Analysis of 5700 Incident Cancer Events in 116 000 European Men and Women." *BMJ* 346 (feb07 1): f165–65. https://doi.org/10.1136/bmj.f165.

Heikkilä, Katriina, Eleonor I. Fransson, Solja T. Nyberg, Marie Zins, Hugo Westerlund, Peter Westerholm, Marianna Virtanen, et al. 2013. "Job Strain and Health-Related Lifestyle: Findings from an Individual-Participant Meta-Analysis of 118 000 Working Adults." *American Journal of Public Health* 103 (11): 2090–97. https://doi.org/10.2105/ajph.2012.301090.

Herrera-Ballesteros, Juan, Carlos Javier de las Heras-Rosas, Pedro Mota Veiga, and Felipe Sampaio Rodrigues. 2025. "How Do Flexible Working Time Policies, Telework from Home and Work Fatigue Impact on the Work-Life Balance?" *International Journal of Organizational Analysis* 33 (12): 153–73. https://doi.org/10.1108/ijoa-03-2025-5353.

Hesketh, I., and C. L. Cooper. 2014. "Leaveism at Work." *Occupational Medicine* 64 (3): 146–47. https://doi.org/10.1093/occmed/kqu025.

Hintsa, T., A. Kouvonen, M. McCann, M. Jokela, M. Elovainio, and P. Demakakos. 2015. "Higher Effort–Reward Imbalance and Lower Job Control Predict Exit from the Labour Market at the Age of 61 Years or Younger: Evidence from the English Longitudinal Study of Ageing." *J Epidemiol Community Health* 69 (6): 543–49. https://doi.org/10.1136/jech-2014-205148.

Hohnen, Pernille, Peter Hasle, Anne Helbo Jespersen, and Christian Uhrenholdt Madsen. 2014. "Hard Work in Soft Regulation: A Discussion of the Social Mechanisms in OHS Management Standards and Possible Dilemmas in the Regulation of Psychosocial Work Environment." *Nordic Journal of Working Life Studies* 4 (3): 13. https://doi.org/10.19154/njwls.v4i3.4177.

Huettermann, Hendrik, and Heike Bruch. 2019. "Mutual Gains? Health-Related HRM, Collective Well-Being and Organizational Performance." *Journal of Management Studies* 56 (6): 1045–72. https://doi.org/10.1111/joms.12446.

Hünefeld, Lena, Sophie-Charlotte Meyer, Serife Erol, and Elke Ahlers. 2025. "Work Intensity: Identification and Analysis of Key Determinants." *International Journal of Workplace Health Management* 18 (2). https://doi.org/10.1108/ijwhm-12-2023-0177.

Ikramullah, Malik, Jan-Willem Van Prooijen, Muhammad Zahid Iqbal, and Faqir Sajjad Ul-Hassan. 2016. "Effectiveness of Performance Appraisal." *Personnel Review* 45 (2): 334–52. https://doi.org/10.1108/pr-07-2014-0164.

ILO (International Labour Organization). 1986. *Psychosocial Factors at Work: Recognition and Control: Report of the Joint ILO/WHO Committee on Occupational Health, Ninth Session*. Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma992480113402676.

———. 2001. "Guidelines on Occupational Safety and Health Management Systems (ILO-OSH 2001)." Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/1k2j8bg/alma994988793002676.

———. 2008. Resolution concerning the measurement of working time. International Conference of Labour Statisticians 18th Session. https://www.ilo.org/resource/resolution-concerning-measurement-working-time-0.

———. 2012a. "Encyclopaedia of Occupational Health and Safety," 4: Section V. Psychosocial and Organizational Factors. https://www.ilo.org/publications/encyclopaedia-occupational-health-and-safety.

———. 2012b. "From Precarious Work to Decent Work: Outcome Document to the Workers' Symposium on Policies and Regulations to Combat Precarious Employment." Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma994692543402676.

———. 2013. “Working Conditions Laws 2012: a global review.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma994838083402676.

———. 2016a. “Non-Standard Employment around the World: Understanding Challenges, Shaping Prospects – Overview.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995219144602676.

———. 2016b. “Workplace Stress: A Collective Challenge.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995218738602676.

———. 2018. *Statistics on work relationships*. ICLS/20/2018/2. https://labordoc.ilo.org/permalink/41ILO_INST/1k2j8bg/alma994999776802676.

———. 2019. “Work for a Brighter Future.” *Www.ilo.org*. Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995271389102676.

———. 2020. “Safe and Healthy Working Environments Free from Violence and Harassment.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/1k2j8bg/alma995088291202676.

———. 2022a. “Module 14 – Ensuring Compliance with Legislation on Psychosocial Risks.” In *ILO Curriculum on Building Modern and Effective Labour Inspection Systems*. Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995650072602676.

———. 2022b. “Working Time and Work-Life Balance around the World.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/1k2j8bg/alma995221493502676.

———. 2022c. “Care at Work: Investing in Care Leave and Services for a More Gender Equal World of Work.” Geneva: International Labour Office. https://doi.org/10.54394/AQOF1491.

———. 2022d. “Transforming Enterprises through Diversity and Inclusion.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995180193102676.

———. 2022e. Resolution on the Inclusion of a Safe and Healthy Working Environment in the ILO’s Framework of Fundamental Principles and Rights at Work.” International Labour Conference. 110th Session. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995192191602676.

———. 2023a. Resolution to amend the 19th ICLS resolution concerning statistics of work, employment and labour underutilization. ICLS/21/2023/RES.II https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995334393302676.

———. 2023b. “The International Standard Classification of Occupations (ISCO-08) Companion Guide.” Geneva: International Labour Office. https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995330791802676.

———. 2024a. Resolution Concerning Decent Work and the Care Economy. International Labour Conference. 112th Session. https://labordoc.ilo.org/permalink/41ILO_INST/1k2j8bg/alma995376393002676.

———. 2024b. “Tackling Work-Related Violence and Harassment: What Data from Three Pilot Surveys in West Africa Reveal.” ILOSTAT. November 25, 2024. https://ilostat.ilo.org/blog/tackling-work-related-violence-and-harassment-what-data-from-three-pilot-surveys-in-west-africa-reveal/.

———. 2025a. “Transnational Company Agreements.” ILO Cross-Border Social Dialogue. 2025. https://cbsd.ilo.org/category/transnational-agreements/transnational-company-agreements.

———. 2025b. “World Employment and Social Outlook: Trends 2025.” Geneva: International Labour Office. https://doi.org/10.54394/izln1673.

———. 2025c. “Revolutionizing Health and Safety: The Role of AI and Digitalization at Work.” Geneva: International Labour Office. https://doi.org/10.54394/KNZE0733.

———. 2025d. “Issue Paper: OSH Statistics Supporting the Review and Update of the International OSH Statistical Standards at the 22nd ICLS in 2028.” https://labordoc.ilo.org/permalink/41ILO_INST/56f4ou/alma995681570602676.

———. 2026. “Employment and Social Trends 2026.” Geneva: International Labour Office. https://doi.org/10.54394/ahrt2681.

ILO and Lloyd’s Register Foundation. 2022. “Experiences of Violence and Harassment at Work: A Global First Survey.” Geneva: International Labour Office. https://doi.org/10.54394/IOAX8567.

INSST (Instituto Nacional de Seguridad y Salud en el Trabajo). 2012. *NTP 0926 Factores Psicosociales: Metodología de Evaluación*. *Portal INSST*. https://www.insst.es/documentacion/colecciones-tecnicas/ntp-notas-tecnicas-de-prevencion/27-serie-ntp-numeros-926-a-960-ano-2012/nota-tecnica-de-prevencion-ntp-926.

ISO (International Organization for Standardization). 2021. "Occupational Health and Safety Management — Psychological Health and Safety at Work — Guidelines for Managing Psychosocial Risks." Iso.org. 2021. https://www.iso.org/obp/ui/#iso:std:iso:45003:ed-1:v1:en.

Jack, Michelle Marion, Marian Baird, and Elizabeth Hill. 2025. "The Impact of Firm-Level Work and Family Initiatives on Organizational Outcomes, Employees and Their Families: An Umbrella Review." *Community, Work & Family*, October, 1–60. https://doi.org/10.1080/13668803.2025.2563055.

Jain, Aditya, Luis D. Torres, Kevin Teoh, and Stavroula Leka. 2022. "The Impact of National Legislation on Psychosocial Risks on Organisational Action Plans, Psychosocial Working Conditions, and Employee Work-Related Stress in Europe." *Social Science & Medicine* 302 (June): 114987. https://doi.org/10.1016/j.socscimed.2022.114987.

Jamieson, Nikki, Lindsay B Carey, Anthony Jamieson, and Myfanwy Maple. 2023. "Examining the Association between Moral Injury and Suicidal Behavior in Military Populations: A Systematic Review." *Journal of Religion & Health* 62 (August): 3904–25. https://doi.org/10.1007/s10943-023-01885-6.

Jespersen, Anne Helbo, Peter Hasle, and Klaus T. Nielsen. 2016. "The Wicked Character of Psychosocial Risks: Implications for Regulation." *Nordic Journal of Working Life Studies* 6 (3): 23–42. https://doi.org/10.19154/njwls.v6i3.5526.

Jimmieson, Nerina L., Adele J. Bergin, Prashant Bordia, and Michelle K. Tucker. 2021. "Supervisor Strategies and Resources Needed for Managing Employee Stress: A Qualitative Analysis." *Safety Science* 136. https://doi.org/10.1016/j.ssci.2020.105149.

Johnson, J V, and E M Hall. 1988. "Job Strain, Work Place Social Support, and Cardiovascular Disease: A Cross-Sectional Study of a Random Sample of the Swedish Working Population." *American Journal of Public Health* 78 (10): 1336–42. https://doi.org/10.2105/ajph.78.10.1336.

K. Lindström, A.-L Elo, A Skogstad, M. Dallner, F. Gamberale, V. Hottinen, S. Knardahl, and E. Ørhede. 2000. "User's Guide for the QPSNordic. General Nordic Questionnaire for the Psychological and Social Factors at Work." The National Research Centre for the Working Environment. Nordic Council of Ministers. https://nfa.elsevierpure.com/en/publications/users-guide-for-the-qpsnordic-general-nordic-questionnaire-for-th/.

Kahn, W. A. 1990. "Psychological Conditions of Personal Engagement and Disengagement at Work." *Academy of Management Journal* 33 (4): 692–724. https://www.jstor.org/stable/256287.

Kalleberg, Arne L. 2008. "The Mismatched Worker: When People Don't Fit Their Jobs." *Academy of Management Perspectives* 22 (1): 24–40. https://doi.org/10.5465/AMP.2008.31217510.

Karasek, Robert A. 1979. "Job Demands, Job Decision Latitude, and Mental Strain: Implications for Job Redesign." *Administrative Science Quarterly* 24 (2): 285–308. https://doi.org/10.2307/2392498.

Karasek, Robert, Maureen Dollard, Sung-Il Cho, Per-Olof Östergren, and Irene Houtman. 2025. "The Multi-Level Job Content Questionnaire 2.0 (JCQ 2.0) and the Associationalist Demand–Control (ADC) Theory for a Sustainable Global Economy." https://doi.org/10.20944/preprints202507.1172.v1.

Karasek, Robert, and Töres Theorell. 1990. *Healthy Work: Stress, Productivity, and the Reconstruction of Working Life*. New York (N.Y): Basic Books.

Karkkola, Petri, Matti Kuittinen, and Taina Hintsa. 2019. "Role Clarity, Role Conflict, and Vitality at Work: The Role of the Basic Needs." *Scandinavian Journal of Psychology* 60 (5): 456–63. https://doi.org/10.1111/sjop.12550.

Kieselbach, Thomas, Elisabeth Armgarth, Sebastiano Bagnara, Anna-Liisa Elo, Steve Jefferys, Catelijne Joling, Karl Kuhn, et al. 2010. *Health in Restructuring (HIRES) : Recommendations, National Responses and Policy Issues in the EU ; Consisting of Health in Restructuring: Innovative Approaches and Policy Recommendations (HIRES), Dissemination and Consultation of the HIRES Recommendations in 13 EU Countries (HIRES Plus)*. Munich Hampp.

Kim, Inah, Min Ji Koo, Hye-Eun Lee, Yong Lim Won, and Jaechul Song. 2019. "Overwork-Related Disorders and Recent Improvement of National Policy in South Korea." *Journal of Occupational Health* 61 (4). https://doi.org/10.1002/1348-9585.12060.

Kim, Joungsue, Ria Kwon, Hyunok Yun, Ga-Young Lim, Kyung-Sook Woo, and Inah Kim. 2024. "The Association between Long Working Hours, Shift Work, and Suicidal Ideation: A Systematic Review and Meta-Analyses." *Scandinavian Journal of Work, Environment & Health* 50 (7): 503–18. https://doi.org/10.5271/sjweh.4182.

Kivimäki, Mika, and Andrew Steptoe. 2018. "Effects of Stress on the Development and Progression of Cardiovascular Disease." *Nature Reviews Cardiology* 15 (4): 215–29. https://doi.org/10.1038/nrcardio.2017.189.

Kivimäki, Mika, Marianna Virtanen, Ichiro Kawachi, Solja T. Nyberg, Lars Alfredsson, G. David Batty, Jakob B. Bjorner, et al. 2015. "Long Working Hours, Socioeconomic Status, and the Risk of Incident Type 2 Diabetes: A Meta-Analysis of Published and Unpublished Data from 222 120 Individuals." *The Lancet Diabetes & Endocrinology* 3 (1): 27–34. https://doi.org/10.1016/S2213-8587(14)70178-0.

Klinefelter, Zachary, Robert R. Sinclair, Thomas W. Britt, Gargi Sawhney, Kristen Jennings Black, and Alec Munc. 2020. "Psychosocial Safety Climate and Stigma: Reporting Stress-Related Concerns at Work." *Stress and Health* 37 (3). https://doi.org/10.1002/smi.3010.

Knardahl, Stein, Håkon A. Johannessen, Tom Sterud, Mikko Härmä, Reiner Rugulies, Jorma Seitsamo, and Vilhelm Borg. 2017. "The Contribution from Psychological, Social, and Organizational Work Factors to Risk of Disability Retirement: A Systematic Review with Meta-Analyses." *BMC Public Health* 17 (1). https://doi.org/10.1186/s12889-017-4059-4.

Konturek, Peter C., T. Brzozowski, and S. J. Konturek. 2011. "Stress and the Gut: Pathophysiology, Clinical Consequences, Diagnostic Approach and Treatment Options." *Journal of Physiology and Pharmacology: An Official Journal of the Polish Physiological Society* 62 (6): 591–99. https://pubmed.ncbi.nlm.nih.gov/22314561/.

Kop, Jean-Luc, Virginie Althaus, Nadja Formet-Robert, and Vincent Grosjean. 2016. "Systematic Comparative Content Analysis of 17 Psychosocial Work Environment Questionnaires Using a New Taxonomy." *International Journal of Occupational and Environmental Health* 22 (2): 128–41. https://doi.org/10.1080/10773525.2016.1185214.

Koranyi, Isa, Johanna Jonsson, Torkel Rönnblad, Leo Stockfelt, and Theo Bodin. 2018. "Precarious Employment and Occupational Accidents and Injuries – a Systematic Review." *Scandinavian Journal of Work, Environment & Health* 44 (4): 341–50. https://doi.org/10.5271/sjweh.3720.

Korunka, Christian. 2017. "Challenges for Job Design." In *Job Demands in a Changing World of Work*, edited by Christian Korunka and Bettina Kubicek, 131–51. Springer, Cham. https://doi.org/10.1007/978-3-319-54678-0_8.

Kossek, Ellen Ernst, Shaun Pichler, Todd Bodner, and Leslie B. Hammer. 2011. "Workplace Social Support and Work-Family Conflict: A Meta-Analysis Clarifying the Influence of General and Work-Family-Specific Supervisor and Organizational Support." *Personnel Psychology* 64 (2): 289–313.

Kotera, Yasuhiro, and Katia Correa Vione. 2020. "Psychological Impacts of the New Ways of Working (NWW): A Systematic Review." *International Journal of Environmental Research and Public Health* 17 (14): 5080. https://doi.org/10.3390/ijerph17145080.

Kouvonen, Anne, Minna Mänty, Tea Lallukka, Eero Lahelma, and Ossi Rahkonen. 2016. "Changes in Psychosocial and Physical Working Conditions and Common Mental Disorders." *The European Journal of Public Health* 26 (3): 458–63. https://doi.org/10.1093/eurpub/ckw019.

Kubicek, Bettina, Matea Paškvan, and Johanna Bunner. 2017. "The Bright and Dark Sides of Job Autonomy." In *Job Demands in a Changing World of Work*, 45–63. Springer, Cham. https://doi.org/10.1007/978-3-319-54678-0_4.

Kucharczyk, Erica R., Kevin Morgan, and Andrew P. Hall. 2012. "The Occupational Impact of Sleep Quality and Insomnia Symptoms." *Sleep Medicine Reviews* 16 (6): 547–59. https://doi.org/10.1016/j.smrv.2012.01.005.

Kyung, MinJung, Soo-Jeong Lee, Caroline Dancu, and OiSaeng Hong. 2023. "Underreporting of Workers' Injuries or Illnesses and Contributing Factors: A Systematic Review." *BMC Public Health* 23 (1). https://doi.org/10.1186/s12889-023-15487-0.

Lambreghts, Charlotte, Sofie Vandenbroeck, Kaat Goorts, and Lode Godderis. 2023. "Return-To-Work Interventions for Sick-Listed Employees with Burnout: A Systematic Review." *Occupational and Environmental Medicine* 80 (9): 538–44. https://doi.org/10.1136/oemed-2023-108867.

LaMontagne, Anthony D, Angela Martin, Kathryn M Page, Nicola J Reavley, Andrew J Noblet, Allison J Milner, Tessa Keegel, and Peter M Smith. 2014. "Workplace Mental Health: Developing an Integrated Intervention Approach." *BMC Psychiatry* 14 (1). https://doi.org/10.1186/1471-244x-14-131.

LaMontagne, Anthony D., Maria Åberg, Sandra Blomqvist, Nick Glozier, Birgit A. Greiner, Jorgen Gullestrup, Samuel B. Harvey, et al. 2024. "Work-Related Suicide: Evolving Understandings of Etiology & Intervention." *American Journal of Industrial Medicine* 67 (8): 679–95. https://doi.org/10.1002/ajim.23624.

Landsbergis, Paul A., Joseph G. Grzywacz, and Anthony D. LaMontagne. 2012. "Work Organization, Job Insecurity, and Occupational Health Disparities." *American Journal of Industrial Medicine* 57 (5): 495–515. https://doi.org/10.1002/ajim.22126.

Landsbergis, Paul A., Peter L. Schnall, Karen L. Belkić, Dean Baker, Joseph Schwartz, and Thomas G. Pickering. 2001. "Work Stressors and Cardiovascular Disease." *Work (Reading, Mass.)* 17 (3): 191–208. https://pubmed.ncbi.nlm.nih.gov/12441599/.

Lawler, Edward E., George S. Benson, and Michael McDermott. 2012. "What Makes Performance Appraisals Effective?" *Compensation & Benefits Review* 44 (4): 191–200. https://journals.sagepub.com/doi/abs/10.1177/0886368712462331.

Leach, Desmond J., Toby D. Wall, and Paul R. Jackson. 2003. "The Effect of Empowerment on Job Knowledge: An Empirical Test Involving Operators of Complex Technology." *Journal of Occupational and Organizational Psychology* 76 (1): 27–52. https://doi.org/10.1348/096317903321208871.

Leake, Gillian, Albert Amankwaa, and Irene Elisabeth. 2025. "Workplace Mistreatment: A Systematic Review of Interventions and Future Research Agenda." *Journal of Business Ethics*, June. https://doi.org/10.1007/s10551-025-06058-x.

Lee, Christina Y, Katherine Moawad, and Grace W Pien. 2025. "Impact of Night Shift Work on Women's Fertility, Pregnancy and Menopause." *Frontiers in Sleep* 4. https://doi.org/10.3389/frsle.2025.1545258.

Leiter, Michael P., William Zanaletti, and Piergiorgio Argentero. 2009. "Occupational Risk Perception, Safety Training, and Injury Prevention: Testing a Model in the Italian Printing Industry." *Journal of Occupational Health Psychology* 14 (1): 1–10. https://doi.org/10.1037/1076-8998.14.1.1.

Leka, Stavroula, and Tom Cox, eds. 2008. *The European Framework for Psychosocial Risk Management: PRIMA-EF. Institute of Work, Health and Organisations.* http://www.prima-ef.org/uploads/1/1/0/2/11022736/prima-ef_ebook.pdf.

Leka, Stavroula, Amanda Griffiths, and Tom Cox. 2003. "Work Organisation and Stress: Systematic Problem Approaches for Employers, Managers and Trade Union Representatives." *Who.int*. Geneva: World Health Organization. https://doi.org/9241590475.

Leka, Stavroula, and Sergio Iavicoli. 2020. "Peer Review on 'Legislation and Practical Management of Psychosocial Risks at Work.'" *A Critical Evaluation of the EU Policy Context, Stockholm (Sweden), 3-4 October 2019: Thematic Discussion Paper*. Luxembourg: Publications Office of the European Union. https://doi.org/10.2767/021888.

Leka, Stavroula, and Aditya Jain. 2024. "Conceptualising Work-Related Psychosocial Risks: Current State of the Art and Implications for Research, Policy and Practice, Report 2024.09." Brussels: ETUI. https://doi.org/10.2139/ssrn.5114626.

Leka, Stavroula, Aditya Jain, Tom Cox, and Evelyn Kortum. 2011. "The Development of the European Framework for Psychosocial Risk Management: PRIMA-EF." *Journal of Occupational Health* 53 (2): 137–43. https://doi.org/10.1539/joh.o10010.

Leka, Stavroula, Aditya Jain, Sergio Iavicoli, and Cristina Di Tecco. 2015. "An Evaluation of the Policy Context on Psychosocial Risks and Mental Health in the Workplace in the European Union: Achievements, Challenges, and the Future." *BioMed Research International* 2015 (1): 1–18. https://doi.org/10.1155/2015/213089.

Leka, Stavroula, Aditya Jain, and Loïc Lerouge. 2017. "Work-Related Psychosocial Risks: Key Definitions and an Overview of the Policy Context in Europe." In *Psychosocial Risks in Labour and Social Security Law. Aligning Perspectives on Health, Safety and Well-Being*, 1–12. Springer, Cham. https://doi.org/10.1007/978-3-319-63065-6_1.

Lerouge, Loïc. 2017. *Psychosocial Risks in Labour and Social Security Law. Aligning Perspectives on Health, Safety and Well-Being*. Springer, Cham. https://doi.org/10.1007/978-3-319-63065-6.

———. 2024. "Droit de La Santé Au Travail Dans La Fonction Publique Hospitalière et Qualité Des Soins." *AJFP. Actualité Juridique Fonctions Publiques*, p. 499-504. https://doi.org/10.2471/BLT.19.020919.

———. 2025. "The Concepts of 'Mental Health in the Workplace' and 'Psychosocial Risks': A Clarification from a Legal Perspective." *European Labour Law Journal* 16 (3). https://doi.org/10.1177/20319525251336018.

Lerouge, Loïc, and Francisco Trujillo Pons. 2022. "Contribution to the Study on the 'Right to Disconnect' from Work. Are France and Spain Examples for Other Countries and EU Law?" *European Labour Law Journal* 13 (3): 450–65. https://doi.org/10.1177/20319525221105102.

LeWine, Howard E. 2024. "Understanding the Stress Response." Harvard Health. Harvard Health Publishing. April 3, 2024. https://www.health.harvard.edu/staying-healthy/understanding-the-stress-response.

Li, Jian, Frank Pega, Yuka Ujita, Chantal Brisson, Els Clays, Alexis Descatha, Marco M. Ferrario, et al. 2020. "The Effect of Exposure to Long Working Hours on Ischaemic Heart Disease: A Systematic Review and Meta-Analysis from the WHO/ILO Joint Estimates of the Work-Related Burden of Disease and Injury." *Environment International* 142 (142): 105739. https://doi.org/10.1016/j.envint.2020.105739.

Li, Jian, Min Zhang, Adrian Loerbroks, Peter Angerer, and Johannes Siegrist. 2014. "Work Stress and the Risk of Recurrent Coronary Heart Disease Events: A Systematic Review and Meta-Analysis." *International Journal of Occupational Medicine and Environmental Health*, November. https://doi.org/10.2478/s13382-014-0303-7.

Li, Ruoxuan, and Meilin Yao. 2022. "What Promotes Teachers' Turnover Intention? Evidence from a Meta-Analysis." *Educational Research Review* 37: 100477. https://doi.org/10.1016/j.edurev.2022.100477.

Li, Xiaomei, and Thitinant Wareewanich. 2024. "A Casual Model to Understand Psychological Safety Affecting Employee Creativity: Role of Knowledge Sharing, Knowledge Hiding and Organizational Safety Climate." *International Journal of Religion* 5 (11): 5074–92. https://doi.org/10.61707/eztzn198.

Lillberg, K., Pia K. Verkasalo, Jaakko Kaprio, Lyly Teppo, Hans Helenius, and Markku Koskenvuo. 2003. "Stressful Life Events and Risk of Breast Cancer in 10,808 Women: A Cohort Study." *American Journal of Epidemiology* 157 (5): 415–23. https://doi.org/10.1093/aje/kwg002.

Lissitsa, Sabina, and Svetlana Chachashvili-Bolotin. 2020. "It's Not What You Know but Where You Come From: Cognitive Skills, Job Autonomy and Latent Discrimination of Ethnic Minorities in Israel." *International Review of Education*, February. https://doi.org/10.1007/s11159-020-09825-2.

Llorens, Clara, Ramon Alós, Ernest Cano, Ariadna Font, Pere Jódar, Vicente López, Albert Navarro, Amat Sánchez, Mireia Utzet, and Salvador Moncada. 2009. "Psychosocial Risk Exposures and Labour Management Practices. An Exploratory Approach." *Scandinavian Journal of Public Health* 38 (3_suppl): 125–36. https://doi.org/10.1177/1403494809354363.

Madsen, Ida E H, Nidhi Gupta, Esben Budtz-Jørgensen, Jens Peter Bonde, Elisabeth Framke, Esben Meulengracht Flachs, Sesilje Bondo Petersen, Annemette Coop Svane-Petersen, Andreas Holtermann, and Reiner Rugulies. 2018. "Physical Work Demands and Psychosocial Working Conditions as Predictors of Musculoskeletal Pain: A Cohort Study Comparing Self-Reported and Job Exposure Matrix Measurements." *Occupational and Environmental Medicine* 75 (10): 752–58. https://doi.org/10.1136/oemed-2018-105151.

Magill, Rowan. 2024. "Work-Related Suicide: Examining the Role of Work Factors in Suicide." *WorkSafe*. WorkSafe New Zealand. https://www.worksafe.govt.nz/research/work-related-suicide-examining-the-role-of-work-factors-in-suicide/.

Makarevičienė, Alina, Madeline Nightingale, Greta Skubiejūtė, Emily Hutton, Vaida Gineikytė-Kanclerė, and Deimantė Kazlauskaitė. 2023. "Minimum Health and Safety Requirements for the Protection of Mental Health in the Workplace." Luxembourg: European Parliament. https://www.europarl.europa.eu/RegData/etudes/STUD/2023/740078/IPOL_STU(2023)740078_EN.pdf.

Maslach, Christina, Wilmar B. Schaufeli, and Michael P. Leiter. 2001. "Job Burnout." *Annual Review of Psychology* 52 (1): 397–422. https://doi.org/10.1146/annurev.psych.52.1.397.

Masoodi, M, Nour Abdelaal, Stephanie Tran, Yuan Stevens, Sam, and Andrey Karim Bardeesy. 2021. "Workplace Surveillance and Remote Work Exploring the Impacts and Implications amidst Covid-19 in Canada." https://dais.ca/wp-content/uploads/2023/11/WorkplaceSurveillanceandRemoteWork.pdf.

Mateescu, Alexandra, and Aiha Nguyen. 2019. "Algorithmic Management in the Workplace." https://datasociety.net/wp-content/uploads/2019/02/DS_Algorithmic_Management_Explainer.pdf.

Maulik, Pallab K. 2017. "Workplace Stress: A Neglected Aspect of Mental Health Wellbeing." *The Indian Journal of Medical Research* 146 (4): 441–44. https://doi.org/10.4103/ijmr.IJMR_1298_17.

Mauno, Saija, Mari Herttalampi, Jaana Minkkinen, Taru Feldt, and Bettina Kubicek. 2022. "Is Work Intensification Bad for Employees? A Review of Outcomes for Employees over the Last Two Decades." *Work & Stress* 37 (1): 1–26. https://doi.org/10.1080/02678373.2022.2080778.

McEwen, Bruce S., and Eliot Stellar. 1993. "Stress and the Individual." *Archives of Internal Medicine* 153 (18): 2093. https://doi.org/10.1001/archinte.1993.00410180039004.

Mercer. 2019. "Performance Transformation in the Future of Work Four Truths and Three Predictions Based on Insights from Mercer's 2019 Global Performance Management Study." https://www.mercer.com/content/dam/mercer/attachments/private/us-2019-performance-transformation-in-the-future-of-work.pdf.

Meyer, John D. 2013. "Race-Based Job Discrimination, Disparities in Job Control, and Their Joint Effects on Health." *American Journal of Industrial Medicine* 57 (5): 587–95. https://doi.org/10.1002/ajim.22255.

MHLW (Ministry of Health, Labour and Welfare). 2025. "Japan Recognizes Record Number of Deaths and Health Disorders Related to Overwork in Fiscal 2024." *Nippon.com*. https://www.nippon.com/en/japan-data/h02473/.

MHRC (Mental Health Research Center). 2021. "Psychological Health & Safety in Canadian Workplaces." 2021. https://static1.squarespace.com/static/5f31a311d93d0f2e28aaf04a/t/61e59ce735bb7b247057299d/1642437865230/Long+Form+EN+Final+-+MHRC+PHS+Report.pd.

Microsoft. 2025. "Microsoft 2025 Annual Work Trend Index." *Microsoft 2025 Annual Work Trend Index*. https://news.microsoft.com/annual-work-trend-index-2025/.

Mikkelsen, Mai Bjørnskov, and Michael Rosholm. 2018. "Systematic Review and Meta-Analysis of Interventions Aimed at Enhancing Return to Work for Sick-Listed Workers with Common Mental Disorders, Stress-Related Disorders, Somatoform Disorders and Personality Disorders." *Occupational and Environmental Medicine* 75 (9): 675–86. https://doi.org/10.1136/oemed-2018-105073.

Milanez, Anna, Annikka Lemmens, and Carla Ruggiu. 2025. "Algorithmic Management in the Workplace: New Evidence from an OECD Employer Survey." *OECD Artificial Intelligence Papers* 31. https://doi.org/10.1787/287c13c4-en.

Milner, Allison, Katrina Witt, Anthony D LaMontagne, and Isabelle Niedhammer. 2018. "Psychosocial Job Stressors and Suicidality: A Meta-Analysis and Systematic Review." *Occupational and Environmental Medicine* 75 (4): 245–53. https://doi.org/10.1136/oemed-2017-104531.

Ministro del Trabajo, and OISS (Organización Iberoamericana de Seguridad Social). 2022. "Tercera Encuesta Nacional de Condiciones de SST En Colombia." https://www.researchgate.net/publication/362455968_TERCERA_ENCUESTA_NACIONAL_DE_CONDICIONES_DE_SST_EN_COLOMBIA.

Miraglia, Mariella, and Gary Johns. 2016. "Going to Work Ill: A Meta-Analysis of the Correlates of Presenteeism and a Dual-Path Model." *Journal of Occupational Health Psychology* 21 (3): 261–83. https://doi.org/10.1037/ocp0000015.

Misra, Joya, Alexandra Kuvaeva, Kerryann O'meara, Dawn Kiyoe Culpepper, and Audrey Jaeger. 2021. "Gendered and Racialized Perceptions of Faculty Workloads." *Gender & Society* 35 (3): 358–94. https://doi.org/10.1177/08912432211001387.

Montani, Francesco, François Courcy, and Christian Vandenberghe. 2017. "Innovating under Stress: The Role of Commitment and Leader-Member Exchange." *Journal of Business Research* 77 (August): 1–13. https://doi.org/10.1016/j.jbusres.2017.03.024.

Moore, Phoebe V. 2017. "The Quantified Self in Precarity: Work, Technology and What Counts." Routledge. https://doi.org/10.4324/9781315561523.

Morgeson, Frederick P., and Stephen E. Humphrey. 2006. "The Work Design Questionnaire (WDQ): Developing and Validating a Comprehensive Measure for Assessing Job Design and the Nature of Work." *Journal of Applied Psychology* 91 (6): 1321–39. https://doi.org/10.1037/0021-9010.91.6.1321.

Mujtaba, Bahaudin, and S Shuaib. 2010. "An Equitable Total Rewards Approach to Pay for Performance Management." *Journal of Management Policy and Practice* 11. https://scholars.nova.edu/en/publications/an-equitable-total-rewards-approach-to-pay-for-performance-manage-2/.

Nahrgang, Jennifer D., Frederick P. Morgeson, and David A. Hofmann. 2011. "Safety at Work: A Meta-Analytic Investigation of the Link between Job Demands, Job Resources, Burnout, Engagement, and Safety Outcomes." *Journal of Applied Psychology* 96 (1): 71–94. https://doi.org/10.1037/a0021484.

National Mental Health Commission. n.d. "What Is a Mentally Healthy Workplace?" Mentally Healthy Workplaces. https://beta.mentallyhealthyworkplaces.gov.au/explore-modules/foundations-mentally-healthy-workplace/setting-success/what-mentally-healthy.

Neupane, Subas, Jukka Takala and Alexis Descatha. 2026. "Work-related psychosocial risk factors of mental and cardiovascular health, prevalence and the attributable fraction: From a translational review to evaluation of the burden". Internal report prepared for the International Labour Organization (ILO), Geneva (unpublished).

Nie, Tiezheng, Min Tian, Mingyang Cai, and Yan Qiao. 2023. "Job Autonomy and Work Meaning: Drivers of Employee Job-Crafting Behaviors in the VUCA Times." *Behavioural Sciences* 13 (6): 493–93. https://doi.org/10.3390/bs13060493.

Niedhammer, Isabelle, Sandrine Bertrais, and Katrina Witt. 2021. "Psychosocial Work Exposures and Health Outcomes: A Meta-Review of 72 Literature Reviews with Meta-Analysis." *Scandinavian Journal of Work, Environment & Health* 47 (7). https://doi.org/10.5271/sjweh.3968.

Nielsen, Karina, and Jo Yarker. 2022. "Employees' Experience of Supervisor Behaviour – a Support or a Hindrance on Their Return-To-Work Journey with a CMD? A Qualitative Study." *Work & Stress* 37 (4): 1–22. https://doi.org/10.1080/02678373.2022.2145622.

Nielsen, Morten Birkeland, and Ståle Valvatne Einarsen. 2018. "What We Know, What We Do Not Know, and What We Should and Could Have Known about Workplace Bullying: An Overview of the Literature and Agenda for Future Research." *Aggression and Violent Behavior* 42 (42): 71–83. https://doi.org/10.1016/j.avb.2018.06.007.

Nigam, Jeannie A.S., R. Michael Barker, Thomas R. Cunningham, Naomi G. Swanson, and Lewis Casey Chosewood. 2023. "Vital Signs: Health Worker–Perceived Working Conditions and Symptoms of Poor Mental Health — Quality of Worklife Survey, United States, 2018–2022." *MMWR. Morbidity and Mortality Weekly Report* 72 (44): 1197–1205. https://doi.org/10.15585/mmwr.mm7244e1.

Nilsson, Tove, Abid Lashari, Per Gustavsson, Mikko Härmä, Carolina Bigert, Theo Bodin, Laura Maclachlan, Annika Lindahl Norberg, and Emma Brulin. 2025. "Night and Shift Work and Incidence of Physician-Diagnosed Sleep Disorders in Nursing Staff: A Prospective Cohort Study." *International Journal of Nursing Studies* 164 (April): 105017. https://doi.org/10.1016/j.ijnurstu.2025.105017.

Nyberg, Anna, Göran Kecklund, Linda Magnusson Hanson, and Kristiina Rajaleid. 2020. "Workplace Violence and Health in Human Service Industries: A Systematic Review of Prospective and Longitudinal Studies." *Occupational and Environmental Medicine* 78 (2): 69–81.https://doi.org/10.1136/oemed-2020-106450.

Ohly, Sandra, and Charlotte Fritz. 2009. "Work Characteristics, Challenge Appraisal, Creativity, and Proactive Behavior: A Multi-Level Study." *Journal of Organizational Behavior* 31 (4): 543–65. https://doi.org/10.1002/job.633.

Özkan, Ahmet Hakan. 2022. "Organizational Justice Perceptions and Turnover Intention: A Meta-Analytic Review." *Kybernetes* 52 (8): 2886–99. https://doi.org/10.1108/k-01-2022-0119.

Palumbo, Rocco, Giulia Flamini, Luca Gnan, Massimiliano Matteo Pellegrini, Damiano Petrolo, and Mohammad Fakhar Manesh. 2021. "Disentangling the Implications of Teleworking on Work–Life Balance: A Serial Mediation Analysis through Motivation and Satisfaction." *Journal of Organizational Effectiveness: People and Performance* ahead-of-print (ahead-of-print). https://doi.org/10.1108/joepp-08-2020-0156.

Park, Jongsoo. 2023. "The Role of Organizational Efforts in Mitigating the Adverse Effects of Workplace Mistreatment on Attitudinal Responses." *Sustainability* 15 (3): 1800. https://doi.org/10.3390/su15031800.

Parker, Sharon K., and Karina Jorritsma. 2020. "Good Work Design for All: Multiple Pathways to Making a Difference." *European Journal of Work and Organizational Psychology* 30 (3): 1–13. https://doi.org/10.1080/1359432x.2020.1860121.

Pavlo Saik, Vitaliy Tsopa, Serhii Cheberyachko, Oleg Deryugin, Svitlana Sokurenko, Iryna Suima, and Vasyl Lozynskyi. 2024. "Improving the Process of Managing Psychosocial Risks in Organizations." *Risk Management and Healthcare Policy* Volume 17 (December): 2997–3016. https://doi.org/10.2147/rmhp.s488263.

Pega, Frank, Bálint Náfrádi, Natalie C. Momen, Yuka Ujita, Kai N. Streicher, Annette M. Prüss-Üstün, Alexis Descatha, et al. 2021. "Global, Regional, and National Burdens of Ischemic Heart Disease and Stroke Attributable to Exposure to Long Working Hours for 194 Countries, 2000–2016: A Systematic Analysis from the WHO/ILO Joint Estimates of the Work-Related Burden of Disease and Injury." *Environment International* 154 (106595): 106595. https://doi.org/10.1016/j.envint.2021.106595.

Peiró, Jose M, and Isabel Rodríguez. 2008. "Work Stress, Leadership and Organizational Health." *Papeles Del Psicólogo - Psychologist Papers* 29 (1): 88–99.

Peña-Gralle, Ana Paula B, Denis Talbot, Caroline S Duchaine, Mathilde Lavigne-Robichaud, Xavier Trudel, Karine Aubé, Matthias Gralle, Mahée Gilbert-Ouimet, Alain Milot, and Chantal Brisson. 2021. "Job Strain and Effort-Reward Imbalance as Risk Factors for Type 2 Diabetes Mellitus: A Systematic Review and Meta-Analysis of Prospective Studies." *Scandinavian Journal of Work, Environment & Health* 48 (1). https://doi.org/10.5271/sjweh.3987.

Penconek, Tatiana, Kaitlyn Tate, Andrea Bernardes, Sarah Lee, Simone P.M. Micaroni, Alexandre P. Balsanelli, Andre A. de Moura, and Greta G. Cummings. 2021. "Determinants of Nurse Manager Job Satisfaction: A Systematic Review." *International Journal of Nursing Studies* 118 (1): 103906. https://doi.org/10.1016/j.ijnurstu.2021.103906.

Pero, Angelica Salvi del, Peter Wyckoff, and Ann Vourc'h. 2022. "Using Artificial Intelligence in the Workplace: What Are the Main Ethical Risks?" *OECD Social, Employment and Migration Working Papers* 273. https://doi.org/10.1787/840a2d9f-en.

Popple, Sam, Kïrsten A Way, Richard Johnstone, Richard Croucher, and Peta Miller. 2021. "A Comparative Analysis of Inspector Responses to Complaints about Psychosocial and Physical Hazards." *Regulation & Governance* 17 (1): 234–49. https://doi.org/10.1111/rego.12447.

Potter, R. E., M. Y. Loh, D. Neser, A. Friebel, M. F. Dollard, S. Leka, A. Jain and L. Lerouge. Forthcoming. "International trends in national psychosocial policy coverage, implementation effects and worker outcomes: Comparative results from the National Policy Index for Worker Mental". Report, University of Adelaide.

Proper, Karin I., Daniëlla van de Langenberg, Wendy Rodenburg, Roel C.H. Vermeulen, Allard J. van der Beek, Harry van Steeg, and Linda W.M. van Kerkhof. 2016. "The Relationship between Shift Work and Metabolic Risk Factors." *American Journal of Preventive Medicine* 50 (5): e147–57. https://doi.org/10.1016/j.amepre.2015.11.013.

Quinlan, Michael, and Philip Bohle. 2009. "Overstretched and Unreciprocated Commitment: Reviewing Research on the Occupational Health and Safety Effects of Downsizing and Job Insecurity." *International Journal of Health Services* 39 (1): 1–44. https://doi.org/10.2307/45131124.

Rahman, Mantaka, Mozahidul Islam Zahid, Habiba Kabir, Imtiaz Abdullah, Tamal Saha, Ummul Khair Alam, Afroza Tamanna Shimu, Mohammad Nur Uddin, and Ritesh Ghimire. 2025. "Sleep Problems and Associated Risk Factors among Physicians in Bangladesh: A Protocol for Systematic Review and Meta-Analysis of Observational Cross-Sectional Studies." *BMJ Open* 15 (12): e112676. https://doi.org/10.1136/bmjopen-2025-112676.

Randstad. 2025. "Workmonitor 2025 Global Report." *Randstad.com*. Randstad NV. https://workforceinsights.randstad.com/en/download-the-workmonitor-2025-report.

Reh, Susan, and Susanne Scheibe. 2025. "Emotional Job Demands Diminish Employees' Sympathy over 9 Years, but Only for Those with a Low Learning Goal Orientation." *Work, Aging and Retirement* 11 (3): 294–306. https://doi.org/10.1093/workar/waae022.

Rodgers, Justin, Adolfo G. Cuevas, David R. Williams, Ichiro Kawachi, and S.V. Subramanian. 2021. "The Relative Contributions of Behavioral, Biological, and Psychological Risk Factors in the Association between Psychosocial Stress and All-Cause Mortality among Middle- and Older-Aged Adults in the USA." *GeroScience* 43 (2): 655–72. https://doi.org/10.1007/s11357-020-00319-5.

Rugulies, Reiner, Birgit Aust, Birgit A Greiner, Ella Arensman, Norito Kawakami, Anthony D LaMontagne, and Ida E H Madsen. 2023. "Work-Related Causes of Mental Health Conditions and Interventions for Their Improvement in Workplaces." *The Lancet* 402 (10410): 1368–81. https://doi.org/10.1016/s0140-6736(23)00869-3.

S. Attiq, S Wahid, N Javaid, M Kanwal, and H.J Shah. 2017. "The Impact of Employees' Core Self-Evaluation Personality Trait, Management Support, Co-Worker Support on Job Satisfaction, and Innovative Work Behaviour." *Pakistan Journal of Psychological Research* 32 (1): 247–71.

Sales, Stephen M. 1970. "Some Effects of Role Overload and Role Underload." *Organizational Behavior and Human Performance* 5 (6): 592–608. https://doi.org/10.1016/0030-5073(70)90042-5.

Salvagioni, Denise Albieri Jodas, Francine Nesello Melanda, Arthur Eumann Mesas, Alberto Durán González, Flávia Lopes Gabani, and Selma Maffei de Andrade. 2017. "Physical, Psychological and Occupational Consequences of Job Burnout: A Systematic Review of Prospective Studies." *PLOS ONE* 12 (10). https://doi.org/10.1371/journal.pone.0185781.

Schaufeli, Wilmar B. 2017. "Applying the Job Demands-Resources Model: A 'How To' Guide to Measuring and Tackling Work Engagement and Burnout." *Organizational Dynamics* 46 (2): 120–32. https://doi.org/10.1016/j.orgdyn.2017.04.008.

Schaufeli, Wilmar B., Arnold B. Bakker, and Marisa Salanova. 2006. "The Measurement of Work Engagement with a Short Questionnaire: A Cross-National Study." *Educational and Psychological Measurement* 66 (4): 701–16. https://doi.org/10.1177/0013164405282471.

Schein, Edgar H. 2010. *Organizational Culture and Leadership*. 5th ed. Hoboken, New Jersey Wiley.

Schmitt, Antje, Deanne N. Den Hartog, and Frank D. Belschak. 2015. "Is Outcome Responsibility at Work Emotionally Exhausting? Investigating Employee Proactivity as a Moderator." *Journal of Occupational Health Psychology* 20 (4): 491–500. https://doi.org/10.1037/a0039011.

Schoellbauer, Julia, Sabine Sonnentag, Roman Prem, and Christian Korunka. 2021. "I'd Rather Know What to Expect…Work Unpredictability as Contemporary Work Stressor with Detrimental Implications for Employees' Daily Wellbeing." *Work & Stress* 36 (3): 1–18. https://doi.org/10.1080/02678373.2021.1976881.

Schulte, Paul A, Jessica M K Streit, Fatima Sheriff, George Delclos, Sarah A Felknor, Sara L Tamers, Sherry Fendinger, James Grosch, and Robert Sala. 2020. "Potential Scenarios and Hazards in the Work of the Future: A Systematic Review of the Peer-Reviewed and Gray Literatures." *Annals of Work Exposures and Health* 64 (8): 786–816. https://doi.org/10.1093/annweh/wxaa051.

Senior Labour Inspectors' Committee (SLIC). 2018. "Guide for Assessing the Quality of Risk Assessments and Risk-Management Measures with Regard to the Prevention of Psychosocial Risks." *https://circabc.europa.eu/ui/group/fea534f4-2590-4490-bca6-504782b47c79/library/22e5a918-47d6-4646-93f3-ebd341f6c571*

Seo, Hyun Jeong, Eun-jung Hyun, and Young-Geun Yoon. 2024. "The Impact of Physical Hazards on Workers' Job Satisfaction in the Construction Industry: A Case Study of Korea." *Behavioral Sciences* 14 (12): 1197–97. https://doi.org/10.3390/bs14121197.

Shaw, James B., and Jeff A. Weekley. 1985. "The Effects of Objective Work-Load Variations of Psychological Strain and Post-Work-Load Performance." *Journal of Management* 11 (1): 87–98. https://doi.org/10.1177/014920638501100108.

Siegrist, Johannes. 1996. "Adverse Health Effects of High-Effort/Low-Reward Conditions." *Journal of Occupational Health Psychology* 1 (1): 27–41. https://doi.org/10.1037/1076-8998.1.1.27.

Skakon, Janne, Karina Nielsen, Vilhelm Borg, and Jaime Guzman. 2010. "Are Leaders' Well-Being, Behaviours and Style Associated with the Affective Well-Being of Their Employees? A Systematic Review of Three Decades of Research." *Work & Stress* 24 (2): 107–39. https://doi.org/10.1080/02678373.2010.495262.

Soile Juutinen, Kirsi Sjöblom, Maureen F Dollard, and Anne Mäkikangas. 2023. "Psychosocial Safety Climate: Measurement and Relationship with Well-Being in a Four-Wave Longitudinal Study during Remote Work." *Scandinavian Journal of Psychology* 64 (4): 504–11. https://doi.org/10.1111/sjop.12917.

Soltanzadeh, Ahmad, Meysam Eyvazlou, Mostafa Mohammad-ghasemi, Mansoureh Sadeghi-Yarandi, Monireh Rahimkhani, Neda Ghasemi, Mojdeh Bonyadi, and Mohsen Sadeghi-Yarandi. 2024. "Investigating the Relationship between Shift Work Schedule and Blood and Metabolic Parameters: A 10-Years Retrospective Cohort Study." *Scientific Reports* 14 (1). https://doi.org/10.1038/s41598-024-68378-8.

Stansfeld, Stephen, and Bridget Candy. 2006. "Psychosocial Work Environment and Mental Health—a Meta-Analytic Review." *Scandinavian Journal of Work, Environment & Health* 32 (6): 443–62. https://doi.org/10.5271/sjweh.1050.

Statistics Canada. 2023. "Work-Related Stress Most Often Caused by Heavy Workloads and Work-Life Balance." https://www150.statcan.gc.ca/n1/en/daily-quotidien/230619/dq230619c-eng.pdf?st=Brj0ccqa.

Sultan-Taïeb, Hélène, Tania Villeneuve, Jean-François Chastang, and Isabelle Niedhammer. 2024. "The Costs of Cardiovascular Diseases and Depression Attributable to Psychosocial Work Exposures in the European Union." *ETUI*. ETUI. https://doi.org/10.2139/ssrn.4519658.

Sun, Chenjunyan, Carol K.H. Hon, Kïrsten A. Way, Nerina L. Jimmieson, and Bo Xia. 2022. "The Relationship between Psychosocial Hazards and Mental Health in the Construction Industry: A Meta-Analysis." *Safety Science* 145 (1): 105485. https://doi.org/10.1016/j.ssci.2021.105485.

Superintendencia de Seguridad Social. 2024. "Riesgo Psicosocial Laboral En Chile Resultados de La Aplicación Del Cuestionario CEAL-SM/SUSESO En 2024." Superintendencia de Seguridad Social. https://www.suseso.cl/607/articles-758966_archivo_01.pdf.

Sverke, Magnus, Lena Låstad, Johnny Hellgren, Anne Richter, and Katharina Näswall. 2019. "A Meta-Analysis of Job Insecurity and Employee Performance: Testing Temporal Aspects, Rating Source, Welfare Regime, and Union Density as Moderators." *International Journal of Environmental Research and Public Health* 16 (14): 2536. https://doi.org/10.3390/ijerph16142536.

Taibi, Yacine, Yannick A. Metzler, Silja Bellingrath, and Andreas Müller. 2021. "A Systematic Overview on the Risk Effects of Psychosocial Work Characteristics on Musculoskeletal Disorders, Absenteeism, and Workplace Accidents." *Applied Ergonomics* 95 (September): 103434. https://doi.org/10.1016/j.apergo.2021.103434.

Tanjung, Kamilia, Prakash K. C., Saila Kyrönlahti, Marcel Goldberg, Clas-Håkan Nygård, and Subas Neupane. 2025. "Associations of Psychosocial and Physical Work Demands with All-Cause Mortality: A Pooled Analysis of Prospective Cohort Studies." *International Journal of Epidemiology* 54 (3). https://doi.org/10.1093/ije/dyaf045.

Taylor, Adrian H., and Lisa Dorn. 2006. "Stress, Fatigue, Health, and Risk of Road Traffic Accidents among Professional Drivers: The Contribution of Physical Inactivity." *Annual Review of Public Health* 27 (1): 371–91. https://doi.org/10.1146/annurev.publhealth.27.021405.102117.

The Lancet. 2026. "No Health without Peace." *The Lancet* 407 (10523): 1. https://doi.org/10.1016/s0140-6736(25)02596-6.

Thibodeau, Pari Shah, Aela Nash, Jennifer C Greenfield, and Jennifer L Bellamy. 2023. "The Association of Moral Injury and Healthcare Clinicians' Wellbeing: A Systematic Review." *International Journal of Environmental Research and Public Health* 20 (13): 6300–6300. https://doi.org/10.3390/ijerph20136300.

Thompson, Rebecca J., Stephanie C. Payne, and Aaron B. Taylor. 2014. "Applicant Attraction to Flexible Work Arrangements: Separating the Influence of Flextime and Flexplace." *Journal of Occupational and Organizational Psychology* 88 (4): 726–49. https://doi.org/10.1111/joop.12095.

Tiesman, Hope M., Srinivas Konda, Dan Hartley, Cammie Chaumont Menéndez, Marilyn Ridenour, and Scott Hendricks. 2015. "Suicide in U.S. Workplaces, 2003–2010: A Comparison with Non-Workplace Suicides." *American Journal of Preventive Medicine* 48 (6): 674–82. https://doi.org/10.1016/j.amepre.2014.12.011.

Tiwari, Mansi, Garima Mathur, and Suvijna Awasthi. 2018. "Gender-Based Discrimination Faced by Females at Workplace: A Perceptual Study of Working Females." *Journal of Entrepreneurship Education* 17 (3): 1–7.

Toukas, Dimitrios, Miltiadis Delichas, Chryssoula Toufekoula, and Anastasia Spyrouli. 2015. "The Role of Labour Inspectorates in Tackling the Psychosocial Risks at Work in Europe: Problems and Perspectives." *Safety and Health at Work* 6 (4): 263–67. https://doi.org/10.1016/j.shaw.2015.06.001.

Tummers, Lars G., and Arnold B. Bakker. 2021. "Leadership and Job Demands-Resources Theory: A Systematic Review." *Frontiers in Psychology* 12 (September): 1–13. https://doi.org/10.3389/fpsyg.2021.722080.

U.S. Bureau of Labor Statistics. 2025. "American Time Use Survey — 2024 Results." https://www.bls.gov/news.release/pdf/atus.pdf. Van der Doef, Margot, and Stan Maes. 1999. "The Job Demand-Control (-Support) Model and Psychological Well-Being: A Review of 20 Years of Empirical Research." *Work & Stress* 13 (2): 87–114. https://doi.org/10.1080/026783799296084.

Vander Elst, Tinne, Hans De Witte, and Nele De Cuyper. 2014. "The Job Insecurity Scale: A Psychometric Evaluation across Five European Countries." *European Journal of Work and Organizational Psychology* 23 (3): 364–80. https://doi.org/10.1080/1359432x.2012.745989.

Vesper, Denise, Michael J Zickar, Liam O'Brien, Rory O'Neill, Maureen F Dollard, Kevin Flynn, Keaton A Fletcher, et al. 2025. "Occupational Health and Labor Unions." *Occupational Health Science* 9 (June): 835–64. https://doi.org/10.1007/s41542-025-00233-2.

Virtanen, Marianna, Jane E Ferrie, David Gimeno, Jussi Vahtera, Marko Elovainio, Archana Singh-Manoux, Michael G Marmot, and Mika Kivimäki. 2009. "Long Working Hours and Sleep Disturbances: The Whitehall II Prospective Cohort Study." *Sleep* 32 (6): 737–45. https://doi.org/10.1093/sleep/32.6.737.

Vollrath, Margarete E., and Svenn Torgersen. 2008. "Personality Types and Risky Health Behaviors in Norwegian Students." *Scandinavian Journal of Psychology* 49 (3): 287–92. https://doi.org/10.1111/j.1467-9450.2008.00631.x.

Walters, David, Richard Johnstone, Hans Jørgen Limborg, Elizabeth Bluff, and Ulrik Gensby. 2021. "Securing Compliance: Some Lessons for EU Strategy on Occupational Health and Safety." Brussels: ETUI aisbl. https://doi.org/10.2139/ssrn.3873809.

Walton, Gregory M. 2014. "The New Science of Wise Psychological Interventions." *Current Directions in Psychological Science* 23 (1): 73–82. https://doi.org/10.1177/0963721413512856.

Wang, Hui-Xin, Maria Wahlberg, Anita Karp, Bengt Winblad, and Laura Fratiglioni. 2012. "Psychosocial Stress at Work Is Associated with Increased Dementia Risk in Late Life." *Alzheimer's & Dementia* 8 (2): 114–20. https://doi.org/10.1016/j.jalz.2011.03.001.

Warr, P. 1987. "Work, Unemployment, and Mental Health." Psycnet.apa.org. 1987. https://psycnet.apa.org/record/1987-98751-000.

Weissbrodt, Rafaël, and David Giauque. 2017. "Labour Inspections and the Prevention of Psychosocial Risks at Work: A Realist Synthesis." *Safety Science* 100 (December): 110–24. https://doi.org/10.1016/j.ssci.2017.02.012.

White, Marc, Shannon Wagner, Izabela Z. Schultz, Susan M. Bradley, Vernita Hsu, Lisa McGuire, and Werner Schultz. 2013. "Modifiable Workplace Risk Factors Contributing to Workplace Absence across Health Conditions: A Stakeholder-Centered Best-Evidence Synthesis of Systematic Reviews." *IOS Press* 45 (4): 475–92. https://doi.org/10.3233/WOR-131628.

WHO (World Health Organization). 2019. "Burn-out an 'Occupational Phenomenon': International Classification of Diseases." World Health Organization. World Health Organization. May 28, 2019. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases.

———. 2022a. "World Mental Health Report: Transforming Mental Health for All." *Www.who.int*. Geneva: World Health Organization. https://www.who.int/publications/i/item/9789240049338.

———. 2022b. "Guidelines on Mental Health at Work." *Www.who.int*. Geneva: World Health Organization. https://www.who.int/publications/i/item/9789240053052.

WHO and ILO. 2021. "WHO/ILO Joint Estimates of the Work-Related Burden of Disease and Injury, 2000-2016: Global Monitoring Report." Geneva: World Health Organization and the International Labour Organization. https://www.who.int/publications/i/item/9789240034945

———. 2022. "Mental Health at Work: Policy Brief." *Www.who.int*. Geneva: World Health Organization and International Labour Organization. https://www.who.int/publications/i/item/9789240057944.

Wieclaw, Joanna, Esben Agerbo, Preben Bo Mortensen, Hermann Burr, Finn Tuchsen, and Jens Peter Bonde. 2008. "Psychosocial Working Conditions and the Risk of Depression and Anxiety Disorders in the Danish Workforce." *BMC Public Health* 8 (1). https://doi.org/10.1186/1471-2458-8-280.

Wiegand, Douglas M, Peter Y Chen, Joseph J Hurrell, Steve Jex, Akinori Nakata, Jeannie A Nigam, Michelle Robertson, and Lois E Tetrick. 2012. "A Consensus Method for Updating Psychosocial Measures Used in NIOSH Health Hazard Evaluations." *Journal of Occupational and Environmental Medicine* 54 (3): 350–55. https://doi.org/10.1097/jom.0b013e3182440a04.

Williams, Geoffrey C., Hallgeir Halvari, Christopher P. Niemiec, Øystein Sørebø, Anja H. Olafsen, and Cathrine Westbye. 2014. "Managerial Support for Basic Psychological Needs, Somatic Symptom Burden and Work-Related Correlates: A Self-Determination Theory Perspective." *Work & Stress* 28 (4): 404–19. https://doi.org/10.1080/02678373.2014.971920.

Williamson, Victoria, Sharon A.M. Stevelink, and Neil Greenberg. 2018. "Occupational Moral Injury and Mental Health: Systematic Review and Meta-Analysis." *The British Journal of Psychiatry* 212 (6): 339–46. https://doi.org/10.1192/bjp.2018.55.

WorkSafe Queensland. 2023. “Managing the Risk of Psychosocial Hazards at Work Code of Practice 2022.” Www.worksafe.qld.gov.au. April 1, 2023. https://www.worksafe.qld.gov.au/laws-and-compliance/codes-of-practice/managing-the-risk-of-psychosocial-hazards-at-work-code-of-practice-2022.

Xu, Tianwei, Linda L. Magnusson Hanson, Theis Lange, Liis Starkopf, Hugo Westerlund, Ida E. H. Madsen, Reiner Rugulies, et al. 2017. “Workplace Bullying and Violence as Risk Factors for Type 2 Diabetes: A Multicohort Study and Meta-Analysis.” *Diabetologia* 61 (1): 75–83. https://doi.org/10.1007/s00125-017-4480-3.

Xue, Yuanxin, Jillian Lopes, Kimberly Ritchie, Andrea M. D’Alessandro, Laura Banfield, Randi E. McCabe, Alexandra Heber, Ruth A. Lanius, and Margaret C. McKinnon. 2022. “Potential Circumstances Associated with Moral Injury and Moral Distress in Healthcare Workers and Public Safety Personnel across the Globe during COVID-19: A Scoping Review.” *Frontiers in Psychiatry* 13 (June). https://doi.org/10.3389/fpsyt.2022.863232.

Yamauchi, Takashi, Toru Yoshikawa, Masahiro Takamoto, Takeshi Sasaki, Shun Matsumoto, Kotaro Kayashima, Tadashi Takeshima, and Masaya Takahashi. 2017. “Overwork-Related Disorders in Japan: Recent Trends and Development of a National Policy to Promote Preventive Measures.” *Industrial Health* 55 (3): 293–302. https://doi.org/10.2486/indhealth.2016-0198.

Yang, Tingting, Yan Qiao, Siyun Xiang, Wenzhen Li, Yong Gan, and Yongchun Chen. 2018. “Work Stress and the Risk of Cancer: A Meta-Analysis of Observational Studies.” *International Journal of Cancer* 144 (10): 2390–2400. https://doi.org/10.1002/ijc.31955.

Yong, Lee C., Jia Li, and Geoffrey M. Calvert. 2017. “Sleep-Related Problems in the US Working Population: Prevalence and Association with Shiftwork Status.” *Occupational and Environmental Medicine* 74 (2): 93–104. https://doi.org/10.1136/oemed-2016-103638.

Yu, Ying-Fen, Yi-Ya Chang, and Shu-Hung Chang. 2024. “Exploring Health Promotion Behaviors, Occupational Burnout, and Sleep Disturbances in Traditional Industry Workers.” *Healthcare* 13 (1): 51. https://doi.org/10.3390/healthcare13010051.

Zadow, Amy Jane, Maureen F. Dollard, Christian Dormann, and Paul Landsbergis. 2021. “Predicting New Major Depression Symptoms from Long Working Hours, Psychosocial Safety Climate and Work Engagement: A Population-Based Cohort Study.” *BMJ Open* 11 (6): e044133. https://doi.org/10.1136/bmjopen-2020-044133.

Zadow, Amy, May Young Loh, Maureen Frances Dollard, Gro Ellen Mathisen, and Bella Yantcheva. 2023. “Psychosocial Safety Climate as a Predictor of Work Engagement, Creativity, Innovation, and Work Performance: A Case Study of Software Engineers.” *Frontiers in Psychology* 14 (April). https://doi.org/10.3389/fpsyg.2023.1082283.

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